CHAPTER 11.0 RADIOACTIVE WASTE MANAGEMENT

11.1 SOURCE TERMS

A discussion of the fission product inventory in the reactor core and the diffusion to the fuel pellet/cladding gap is presented in chapter 15. Models used in the preparation of the source terms are presented in references 1 and 2 and are based on operating plant data where available.

Two source terms are presented. The first is a conservative design base that utilizes a conventional fuel clad defect model. This design model serves as a basis for system and shielding requirements and calculations of the maximum offsite doses resulting from credible accidents.

The second source term is a realistic model used to predict expected long-term average concentrations of radionuclides in the primary and secondary fluid stream and an average plant's releases over its lifetime. This realistic model, based on available measured nuclide concentrations during normal operation, was formulated as a standard for the American National Standard Source Term Specifications, ANSI N237,[3] and is the source term model used in NUREG-0017.[4]

Recommended design bases for the liquid waste volumes originating from leakage or drainage of nuclear steam supply system (NSSS) components are presented in reference 1.

11.1.1 REACTOR COOLANT AND SECONDARY SIDE ACTIVITY

11.1.1.1 Design Basis Model

The parameters used in the calculation of the reactor coolant fission and corrosion product specific activities together with the pertinent information concerning the expected coolant cleanup flowrate, demineralizer effectiveness, and volume control tank noble gas stripping behavior are summarized in table 11.1-1. No purge is assumed in the volume control tank to the gaseous waste processing system. Calculated reactor coolant radionuclide concentrations are presented in table 11.1-2.

The time dependent fission product core inventories that serve as a source term to the reactor coolant system are calculated by the ORIGEN code[2] using a data library based on ENDF/B-IV.[5]

The specific activity for major nuclides in the pressurizer steam and liquid phases is given in table 11.1-3. The pressurizer liquid specific activity is assumed to be the same as that of the reactor coolant. Table 11.1-3 lists the nuclides that are the major contributors to total source strength.

The pressurizer nitrogen-16 activity calculations are based on an insurge to the pressurizer following a 10-percent step-load power decrease. It is assumed that the incoming reactor coolant mixes only with the pressurizer liquid below the first baffle (109 ft^3) and that the nitrogen-16 concentration is corrected for decay during transit through the surge line. With these assumptions, the pressurizer nitrogen-16 activity is found to be 12.9 μCi/g.

Pressurizer steam phase radiogas concentrations in table 11.1-3 are based on the stripping of radiogases from the continuous 2-gal/min pressurizer spray and the subsequent buildup of these radiogases in the steam space. The buildup time is assumed to be 1 effective full-power year. The radiogases are assumed to be completely stripped from the spray, with the exception of krypton-85, where a stripping fraction of 0.9 is used.

Pressurizer steam phase iodine concentrations in table 11.1-3 are obtained from the liquid phase nuclide activities and measured values of the partition coefficient for iodine-131. A partition coefficient of 100 is used and is assumed to apply to all radioiodines.

The specific activity for major nuclides in the volume control tank liquid and vapor phases is given in table 11.1-4.

The radiation sources in the volume control tank are based on a nominal operating level in the tank of 200 ft^3 in the liquid phase and 200 ft^3 in the vapor phase and on the stripping fractions given in table 11.1-1, assuming no volume control tank purge.

The design basis steam generator secondary side fission and corrosion product concentrations for the steam generator blowdown processing system are given in subsection 10.4.8. Design basis iodine concentrations in the steam generator secondary side based on Technical Specification maximum activity of 0.1 μCi/g dose equivalent iodine-131 are given in table 11.1-5.

11.1.1.2 Realistic Model

The parameters used to describe the realistic model are given in table 11.1-6 together with the range of values utilized by ANSI N237-1976.

Corrections have been made according to the ANSI N237-1976 standard formulas. Operation of a Westinghouse gaseous waste management system is assumed. The Y parameter is interpreted as equal to the stripping fraction and is calculated using the formula in table 11.1-1. These stripping fractions apply to the VEGP gaseous waste management system. Stripping fractions (Y parameter) are listed in table 11.1-7.

Specific activities in the primary coolant, steam generator water, and steam generator steam, based on the parameters of table 11.1-6, are also given in table 11.1-7.

Regulatory Guide 1.112, appendix B, recommends input parameters needed to execute the gaseous and liquid effluents (GALE) computer code[4] for pressurized water reactors. These values are listed in table 11.1-8.

11.1.2 TRITIUM PRODUCTION AND RELEASE TO THE REACTOR COOLANT

There are two principal contributors to tritium production within the NSSS: the ternary fission source and the dissolved boron in the reactor coolant. Additional contributions are made by lithium-6, lithium-7, and deuterium in the reactor water and by tritium produced in the B_4C section of the control rods. *(HISTORICAL): Tritium production from various sources is shown in table 11.1-9.*

Additional background information on tritium production is given in reference 1.

11.1.3 SPENT RESIN VOLUMES

The expected volume of spent demineralizer resin supplied to the solid waste management system from demineralizers is presented in table 11.1-10. The demineralizer resin source strengths are given in section 12.2. The information is based on plant experience as further outlined in reference 1.

11.1.4 REFERENCES

1. "Source Term Data for Westinghouse Pressurized Water Reactors," WCAP-8253, Revision 1, February 1976.

2. Bell, M. J., "ORIGEN - The ORNL Isotope Generation and Depletion Code," ORNL-4628, Oak Ridge National Laboratory, Oak Ridge, Tennessee, May 1973.

3. American National Standard Source Term Specification, ANSI N237-1976/AN8-18.1, approved May 11, 1976.

4. U.S. Nuclear Regulatory Commission, "Calculation of Releases of Radioactive Materials in Gaseous and Liquid Effluents from Pressurized Water Reactors," NUREG-0017 Office of Standard Development, April 1976.

5. "ORIGEN Yields and Cross Sections - Nuclear Transmutation and Decay Data From ENDF/B-IV," RSIC-DLC-38, Oak Ridge National Laboratory, Radiation Shielding Information Center, Oak Ridge, Tennessee.

TABLE 11.1-1 (SHEET 1 OF 3)

PARAMETERS USED IN THE CALCULATION OF DESIGN BASIS FISSION AND CORROSION PRODUCT ACTIVITIES

Ultimate core thermal power (MWt)	3636
Reactor coolant liquid volume (ft^3)	11,500
Reactor coolant full-power average temperature (°F)	590
Purification flowrate, normal (gal/min)[(a)]	75
Effective cation demineralizer flow (gal/min)[(a)]	7.5
Volume control tank	
Vapor volume (ft^3)	200
Liquid volume (ft^3)	200
Pressure (psig)	15
Temperature (°F)	115
Nuclide release coefficients (the product of the failed fuel fraction and the fission product escape rate coefficient)	
Failed fuel fractions:	
Equivalent fraction of core power produced by fuel rods containing small cladding defects	0.01
Fission product escape rate coefficients during full-power operation (s^{-1}):	
Kr and Xe isotopes	6.5×10^{-8}
Br, Rb, I, and Cs isotopes	1.3×10^{-8}
Mo, Tc, and Ag isotopes	2.0×10^{-9}
Te isotopes	1.0×10^{-9}
Sr and Ba isotopes	1.0×10^{-11}
Y, Zr, Nb, Ru, Rh, La, Ce, and Pr isotopes	1.6×10^{-12}
Chemical and volume control system mixed bed demineralizers	
Resin volume (ft^3)	30
Demineralizer isotopic decontamination factors[(b)]:	
Kr and Xe isotopes	1
Br and I isotopes	10
Sr and Ba isotopes	10
Other isotopes	1

TABLE 11.1-1 (SHEET 2 OF 3)

Chemical and volume control system cation bed demineralizer	
Resin volume (ft^3)	20
Demineralizer isotopic decontamination factors[(b)]:	
Kr and Xe isotopes	1
Sr and Ba isotopes	1
Rb-86, Cs-134, and Cs-137	10
Rb-88, Rb-89, Cs-136, and Cs-138	1
Other isotopes	1

Volume control tank noble gas stripping fractions (based on no purge to the gaseous waste management system)

Isotopic	Stripping Fraction[(c)]
Kr-83m	0.73
Kr-85m	0.53
Kr-85	5.3×10^{-5}
Kr-87	0.80
Kr-88	0.64
Kr-89	0.99
Xe-131m	0.012
Xe-133m	0.061
Xe-133	0.027
Xe-135m	0.93
Xe-135	0.27
Xe-137	0.98
Xe-138	0.94

Pressurizer volumes (ft^3)

Liquid volume	1080
Vapor volume	720

Initial boron concentrations (ppm)

First cycle	900
Second cycle	1070
Third cycle	1110
Equilibrium cycle	1110

Operation times (effective full-power hours)

First cycle	9240
Second cycle	6570
Third cycle	6700
Equilibrium cycle	7200

TABLE 11.1-1 (SHEET 3 OF 3)

a. Flow calculated at designed charging pump discharge (2250 psia and 130°F). The 75 gpm flow rate bounds operation at higher letdown flow rates.

b. For all isotopes, except the isotopes of Kr, Xe, Br, I, Rb, Cs, Sr, and Ba, a removal decontamination factor of 10 is assumed. This is done to account for removal mechanisms other than ion exchange, such as plateout, etc.

c. The nuclide stripping fractions are calculated using the following equation:

$$\psi = 1 - \frac{KQ}{KQ + \lambda (KL + V) + P}$$

where:

ψ = nuclide volume control tank stripping fraction.

$$K = \frac{RT}{MH}$$

$$R = \text{gas constant} = 45.59 \ \frac{\text{atm/cm}^3}{\text{gmol/°R}}$$

T = nominal volume control tank temperature (°R).

M = molecular weight of water = 18.0 g/g mol.

H = Henry's Law constant equal to 3.02×10^4 atm/mol fraction for krypton and 2.04×10^4 atm/mol fraction for xenon, at 115°F.

Q = letdown or purification flowrate (g/s).

λ = nuclide decay constant (s^{-1}).

L = volume control tank liquid mass (g).

V = volume control tank vapor volume (cm^3).

P = volume control tank purge rate to the gaseous waste management system, at volume control tank conditions (cm^3/s).

TABLE 11.1-2 (SHEET 1 OF 2)

REACTOR COOLANT DESIGN BASIS FISSION AND CORROSION PRODUCT SPECIFIC ACTIVITY

Nuclide	Activity (μCi/g)	Nuclide	Activity (μCi/g)
Kr-83m	4.6×10^{-1}	Cs-134	2.3
Kr-85m	2.0	Cs-136	2.9
Kr-85	7.3	Cs-137	1.5
Kr-87	1.3	Ba-137m	1.4
Kr-88	3.6	Cs-138	9.6×10^{-1}
Kr-89	1.1×10^{-1}	H-3	3.5 (maximum)
Xe-131m	2.2	Cr-51	5.5×10^{-3}
Xe-133m	1.7×10^{1}	Mn-54	4.4×10^{-4}
Xe-133	2.7×10^{2}	Mn-56	2.0×10^{-2}
Xe-135m	4.8×10^{-1}	Fe-55	2.0×10^{-3}
Xe-135	7.2	Fe-59	5.2×10^{-4}
Xe-137	1.7×10^{-1}	Co-58	1.5×10^{-2}
Xe-138	6.4×10^{-1}	Co-60	1.9×10^{-3}
Br-83	9.5×10^{-2}	Sr-89	4.3×10^{-3}
Br-84	4.7×10^{-2}	Sr-90	1.2×10^{-4}
Br-85	6.0×10^{-3}	Sr-91	6.2×10^{-3}
I-127[(a)]	6.2×10^{-11}	Sr-92	1.3×10^{-3}
I-129	4.3×10^{-8}	Y-90	3.4×10^{-5}
I-130	2.1×10^{-2}	Y-91m	3.3×10^{-3}
I-131	2.8	Y-91	5.7×10^{-4}
I-132	2.8	Y-92	1.2×10^{-3}
I-133	4.2	Y-93	3.8×10^{-4}
I-134	5.7×10^{-1}	Zr-95	6.5×10^{-4}
I-135	2.3	Nb-95	6.5×10^{-4}
Rb-86	2.2×10^{-2}	Mo-99	7.5×10^{-1}
Rb-88	4.8	Tc-99m	6.9×10^{-1}
Rb-89	2.1×10^{-1}	Ru-103	5.7×10^{-4}
Ru-106	1.4×10^{-4}	Te-131	1.2×10^{-2}
Rh-103m	5.7×10^{-4}	Te-132	2.9×10^{-1}
Rh-106	1.4×10^{-4}	Te-134	3.0×10^{-2}
Ag-110m	1.4×10^{-3}	Ba-140	4.2×10^{-3}
Te-125m	2.8×10^{-4}	La-140	1.4×10^{-3}
Te-127m	2.9×10^{-3}	Ce-141	6.3×10^{-4}
Te-127	1.2×10^{-2}	Ce-143	5.0×10^{-4}
Te-129m	1.9×10^{-2}	Ce-144	3.9×10^{-4}
Te-129	1.8×10^{-2}	Pr-143	6.3×10^{-4}
Te-131m	2.6×10^{-2}	Pr-144	3.9×10^{-4}

TABLE 11.1-2 (SHEET 2 OF 2)

a. Grams of iodine-127 per gram of coolant.

The above concentrations are based on the assumptions listed below:

1. Reactor coolant mass = 2.3×10^8 g.
2. Operation with small defects in the cladding of fuel rods generating 1 percent of the core-rated power.
3. Reactor coolant purification or letdown rate = 75 gal/min at 130°F and 2250 psia.
4. No volume control tank purge.

TABLE 11.1-3

PRESSURIZER ACTIVITIES

Liquid Phase Specific Activity (1080-ft^3 liquid phase)

Nuclide	Activity (μCi/g)
Kr-88	3.7
Kr-89	1.1×10^{-1}
Xe-133	2.7×10^{2}
Xe-135	7.3
I-132	2.8
I-133	4.2
I-135	2.3
Rb-88	4.8
Cs-134	2.3
Cs-136	2.9
Cs-138	9.6×10^{-1}

Steam Phase Specific Activity (720-ft^3 steam phase)

Nuclide	Activity (μCi/g)
Kr-83m	2.0×10^{-2}
Kr-85m	2.2×10^{-1}
Kr-85	9.3×10^{2}
Kr-87	3.8×10^{-2}
Kr-88	2.4×10^{-1}
Kr-89	1.3×10^{-4}
Xe-131m	1.5×10^{1}
Xe-133m	2.2×10^{1}
Xe-133	8.1×10^{2}
Xe-135m	2.9×10^{-3}
Xe-135	1.6
Xe-137	2.6×10^{-4}
Xe-138	3.6×10^{-3}
I-129	4.3×10^{-10}
I-130	2.1×10^{-4}
I-131	2.8×10^{-2}
I-132	2.8×10^{-2}
I-133	4.2×10^{-2}
I-134	5.7×10^{-3}
I-135	2.3×10^{-2}

TABLE 11.1-4

VOLUME CONTROL TANK ACTIVITIES[a]

Liquid Phase Specific Activity (200-ft^3 liquid phase)

Nuclide	Activity (μCi/g)
Kr-88	1.3
Xe-133	2.6×10^{2}
Xe-135	5.3
Rb-88	4.8
Cs-134	2.3
Cs-136	2.9
Cs-138	9.6×10^{-1}

Vapor Phase Specific Activity (200-ft^3 steam phase)

Nuclide	Activity (μCi/g)
Kr-83m	3.7
Kr-85m	2.1×10^{1}
Kr-85	1.5×10^{2}
Kr-87	5.5
Kr-88	3.0
Kr-89	2.4×10^{-2}
Xe-131m	3.2×10^{1}
Xe-133m	2.5×10^{2}
Xe-133	3.9×10^{3}
Xe-135m	8.5
Xe-135	9.5×10^{1}
Xe-137	4.8×10^{-2}
Xe-138	6.2×10^{-1}

a. Based on parameters given in table 11.1-1.

TABLE 11.1-5

DESIGN BASIS IODINE CONCENTRATIONS IN SECONDARY SIDE WATER AT TECHNICAL SPECIFICATION LIMITS

Isotope	Concentration (μCi/g)
I-131	8.0×10^{-2}
I-132	2.8×10^{-2}
I-133	9.0×10^{-2}
I-134	2.0×10^{-3}
I-135	3.3×10^{-2}

TABLE 11.1-6

PARAMETERS USED TO DESCRIBE THE REACTOR SYSTEM - REALISTIC BASIS

Parameter	Symbol	Units	Value	ANSI N237 Range Maximum	ANSI N237 Range Minimum
Thermal power	P	MWt	3636	3800	3000
Steam flowrate	FS	lb/h	1.5×10^7	1.7×10^7	1.3×10^7
Weight of water in reactor coolant system	WP	lb	5.6×10^5	6.0×10^5	5.0×10^5
Weight of water in all steam generators	WS	lb	4.7×10^5	5.0×10^5	4.0×10^5
Reactor coolant letdown flow (purification)	FD	lb/h	3.8×10^4	4.2×10^4	3.2×10^4
Reactor coolant letdown flow (yearly average for boron control)	FB	lb/h	100	1.0×10^3	2.5×10
Steam generator blowdown flow (total)	FBD	lb/h	1.8×10^5	1.0×10^5	5.0×10^4
Fraction of radioactivity in blowdown steam that is not returned to the secondary coolant system	NBD	-	1.0	1.0	0.9
Flow through the purification system cation demineralizer	FA	lb/h	3.8×10^3	7.5×10^4	0.0
Ratio of condensate demineralizer flowrate to the total stream flowrate	NC	-	0.0	0.01	0.0
Ratio of the total amount of noble gases routed to gaseous radwaste from the purification system to the total amount routed from the primary coolant system (not including the boron recycle system)	Y	-	See table 11.1-7	0.01	0.0
Primary-to-secondary leak rate	-	lb/day	100	-	100

TABLE 11.1-7 (SHEET 1 OF 3)

SPECIFIC ACTIVITIES IN PRINCIPAL FLUID STREAMS - REALISTIC BASIS

Normal Plant Operation Source Terms
(based on ANSI N237)

Volume Control Tank Purge of 1.736×10^{2} cm^3/s

Group I - Noble Gases

Nuclide	Y Parameter	Reactor Coolant Activity (μCi/g)	Steam Gen. Liq. Activity (μCi/g)	Steam Gen. Steam Activity (μCi/g)
Kr-83m	7.8×10^{-1}	2.1×10^{-2}	Nil	5.7×10^{-9}
Kr-85m	6.5×10^{-1}	9.6×10^{-2}	Nil	2.7×10^{-8}
Kr-85	4.3×10^{-1}	4.8×10^{-3}	Nil	1.3×10^{-9}
Kr-87	8.2×10^{-1}	6.2×10^{-2}	Nil	1.6×10^{-8}
Kr-88	7.2×10^{-1}	1.9×10^{-1}	Nil	5.1×10^{-8}
Kr-89	9.9×10^{-1}	5.7×10^{-3}	Nil	1.6×10^{-9}
Xe-131m	3.5×10^{-1}	1.5×10^{-2}	Nil	4.1×10^{-9}
Xe-133m	3.7×10^{-1}	8.6×10^{-2}	Nil	2.4×10^{-8}
Xe-133	3.5×10^{-1}	4.1×10^{-1}	Nil	1.1×10^{-7}
Xe-135m	9.3×10^{-1}	1.4×10^{-2}	Nil	4.0×10^{-9}
Xe-135	4.7×10^{-1}	2.8×10^{-1}	Nil	7.6×10^{-8}
Xe-137	9.8×10^{-1}	1.0×10^{-2}	Nil	2.8×10^{-9}
Xe-138	9.4×10^{-1}	4.9×10^{-2}	Nil	1.3×10^{-8}

Group II - Halogens

Nuclide	Reactor Coolant Activity (μCi/g)	Steam Gen. Liq. Activity (μCi/g)	Steam Gen. Steam Activity (μCi/g)
Br-83	5.4×10^{-3}	1.2×10^{-7}	1.2×10^{-9}
Br-84	3.0×10^{-3}	2.0×10^{-8}	2.0×10^{-10}
Br-85	3.4×10^{-4}	2.2×10^{-10}	2.2×10^{-12}
I-130	2.3×10^{-3}	1.1×10^{-7}	1.1×10^{-9}
I-131	2.8×10^{-1}	2.2×10^{-5}	2.2×10^{-7}
I-132	1.1×10^{-1}	9.1×10^{-6}	9.1×10^{-8}
I-133	4.0×10^{-1}	2.4×10^{-5}	2.4×10^{-7}
I-134	5.3×10^{-2}	5.5×10^{-8}	5.5×10^{-9}
I-135	2.1×10^{-1}	8.3×10^{-6}	8.3×10^{-8}

TABLE 11.1-7 (SHEET 2 OF 3)

Group III - Rubidium and Cesium

Nuclide	Reactor Coolant Activity (μCi/g)	Steam Gen. Liq. Activity (μCi/g)	Steam Gen. Steam Activity (μCi/g)
Rb-86	8.9×10^{-5}	7.4×10^{-9}	7.4×10^{-12}
Rb-88	2.3×10^{-1}	8.3×10^{-7}	8.3×10^{-10}
Cs-134	2.6×10^{-2}	1.7×10^{-6}	1.7×10^{-9}
Cs-136	1.4×10^{-2}	9.5×10^{-7}	9.5×10^{-10}
Cs-137	1.9×10^{-2}	1.4×10^{-6}	1.4×10^{-9}
Ba-137m	1.8×10^{-2}	1.3×10^{-6}	1.3×10^{-9}

Group IV - Nitrogen-16

Nuclide	Reactor Coolant Activity (μCi/g)	Steam Gen. Liq. Activity (μCi/g)	Steam Gen. Steam Activity (μCi/g)
N-16	4.0×10^{1}	9.5×10^{-7}	9.5×10^{-7}

Group V - Tritium

Nuclide	Reactor Coolant Activity (μCi/g)	Steam Gen. Liq. Activity (μCi/g)	Steam Gen. Steam Activity (μCi/g)
H-3	1.0	1.0×10^{-3}	1.0×10^{-3}

TABLE 11.1-7 (SHEET 3 OF 3)

Group VI - Miscellaneous Isotopes

Nuclide	Reactor Coolant Activity (μCi/g)	Steam Gen. Liq. Activity (μCi/g)	Steam Gen. Steam Activity (μCi/g)
Cr-51	1.9×10^{-3}	1.4×10^{-7}	1.4×10^{-10}
Mn-54	3.1×10^{-4}	3.4×10^{-8}	3.4×10^{-11}
Fe-55	1.6×10^{-3}	1.2×10^{-8}	1.2×10^{-10}
Fe-59	1.0×10^{-3}	8.7×10^{-8}	8.7×10^{-11}
Co-58	1.6×10^{-2}	1.2×10^{-6}	1.2×10^{-9}
Co-60	2.0×10^{-3}	1.5×10^{-7}	1.5×10^{-10}
Sr-89	3.6×10^{-4}	3.5×10^{-8}	3.5×10^{-11}
Sr-90	1.0×10^{-5}	8.4×10^{-10}	8.4×10^{-13}
Sr-91	7.0×10^{-4}	3.1×10^{-8}	3.1×10^{-11}
Y-90	1.2×10^{-6}	4.8×10^{-10}	4.8×10^{-13}
Y-91m	4.1×10^{-4}	2.9×10^{-8}	2.9×10^{-11}
Y-91	6.5×10^{-5}	5.2×10^{-9}	5.2×10^{-12}
Y-93	3.7×10^{-5}	2.0×10^{-9}	2.0×10^{-12}
Zr-95	6.1×10^{-5}	5.2×10^{-9}	5.2×10^{-12}
Nb-95	5.1×10^{-5}	5.2×10^{-9}	5.2×10^{-12}
Mo-99	8.7×10^{-2}	7.2×10^{-6}	7.2×10^{-9}
Tc-99m	5.2×10^{-2}	1.8×10^{-5}	1.8×10^{-8}
Ru-103	4.6×10^{-5}	3.5×10^{-9}	3.5×10^{-12}
Ru-106	1.0×10^{-5}	8.5×10^{-10}	8.5×10^{-13}
Rh-103m	5.1×10^{-5}	1.9×10^{-8}	1.9×10^{-11}
Rh-106	1.1×10^{-5}	5.4×10^{-9}	5.4×10^{-12}
Te-125m	2.9×10^{-5}	1.6×10^{-9}	1.6×10^{-12}
Te-127m	2.8×10^{-4}	1.5×10^{-8}	1.5×10^{-11}
Te-127	9.2×10^{-4}	1.0×10^{-7}	1.0×10^{-10}
Te-129m	1.4×10^{-3}	1.1×10^{-7}	1.1×10^{-10}
Te-129	1.8×10^{-3}	5.6×10^{-7}	5.6×10^{-10}
Te-131m	2.6×10^{-3}	1.6×10^{-7}	1.6×10^{-10}
Te-131	1.2×10^{-3}	5.1×10^{-7}	5.1×10^{-10}
Te-132	2.8×10^{-2}	1.8×10^{-6}	1.8×10^{-9}
Ba-140	2.2×10^{-4}	1.7×10^{-8}	1.7×10^{-11}
La-140	1.6×10^{-4}	2.2×10^{-8}	2.2×10^{-11}
Ce-141	7.1×10^{-5}	5.3×10^{-9}	5.3×10^{-12}
Ce-143	4.2×10^{-5}	2.7×10^{-9}	2.7×10^{-12}
Ce-144	3.3×10^{-5}	3.4×10^{-9}	3.4×10^{-12}
Pr-143	5.1×10^{-5}	3.7×10^{-9}	3.7×10^{-12}
Pr-144	3.8×10^{-5}	2.1×10^{-8}	3.1×10^{-11}
Np-239	1.2×10^{-3}	7.5×10^{-8}	7.5×10^{-11}

TABLE 11.1-8 (SHEET 1 OF 3)

PARAMETERS SPECIFIED BY REGULATORY GUIDE 1.112 APPENDIX B
(INPUT PARAMETERS FOR THE GALE COMPUTER CODE)

Description	Value
Thermal power level (MWt)	3626
Mass of primary coolant (lb)	5.1×10^5
Primary system letdown rate (gal/min)	75[(a)]
Letdown cation demineralizer flowrate (gal/min)	7.5
Number of steam generators	4
Total steam flow (lb/h)	16.25×10^6
Mass of steam in each steam generator (lb)	6.4×10^3
Mass of liquid in each steam generator (lb)	1.11×10^5
Total mass of secondary coolant	2.022×10^6
Total blowdown rate (lb/h)	1.8×10^5
Condensate demineralizer regeneration time	0.0
Condensate demineralizer flow fraction	0.0
Maximum radwaste dilution flow (gal/min)	15.0×10^3
Shim Bleed	
Shim bleed flowrate (gal/day)	1.7×10^3
Decontamination factor for I	10^6
Decontamination factor for Cs and Rb	1.6×10^4
Decontamination factor for others	10^8
Collection time (day)	22.4
Process and discharge time (day)	0.31
Fraction discharged	1.0
Equipment Drains	
Equipment drains flowrate (gal/day)	300
Fraction of reactor coolant activity	1.0
Decontamination factor for I	10^5
Decontamination factor for Cs and Rb	8×10^3
Decontamination factor for others	1×10^7
Collection time (day)	22.4
Process and discharge time (day)	2.1
Fraction discharged	0.25
Clean Waste	
Clean waste input flowrate (gal/day)	7.13×10^2
Fraction of reactor coolant activity	0.051
Decontamination factor for I	10^4
Decontamination factor for Cs and Rb	4×10^3

TABLE 11.1-8 (SHEET 2 OF 3)

Description	Value
Decontamination factor for others	1×10^6
Collection time (day)	5.6
Process and discharge time (day)	0.03
Fraction discharged	1.0
Dirty Waste	
Dirty waste input flowrate (gal/day)	2.047×10^3
Fraction of reactor coolant activity	0.02
Decontamination factor for I	10^4
Decontamination factor for Cs and Rb	4×10^3
Decontamination factor for others	1×10^6
Collection time (day)	1.95
Process and discharge time (day)	0.19
Fraction discharged	1.0
Blowdown Waste	
Blowdown fraction processed	1.0
Decontamination factor for I	10^3
Decontamination factor for Cs and Rb	10^2
Decontamination factor for others	10^3
Collection time	0.0
Process and discharge time	0.0
Fraction discharged	0.1
Regenerant flowrate	
Decontamination factor for I	N/A
Decontamination factor for Cs and Rb	N/A
Decontamination factor for others	N/A
Collection time	N/A
Process and discharge time	N/A
Fraction discharged	N/A

TABLE 11.1-8 (SHEET 3 OF 3)

Description	Value
Gaseous Waste System	
Holdup time for xenon (day)	90
Holdup time for krypton (day)	90
Fill time of decay tanks for gas stripper	0.0
Gas waste system: HEPA?	Yes
Auxiliary building: charcoal?	Yes
Auxiliary building: HEPA?	Yes
Containment volume (ft^3)	2.75×10^6
Containment atmosphere cleanup rate (ft^3/min)	30×10^3
Containment shutdown purge: charcoal?, HEPA?	Yes, yes
Number purge per year	4
Containment normal purge rate (ft^3/min); charcoal?, HEPA?	5000; yes, yes
Fraction of iodine released from blowdown tank vent	0
Fraction of iodine released from main condenser air ejector	1.0
Detergent waste decontamination factor	1.0

a. The primary system letdown rate of 75 gpm results in effluent releases and doses which bound those for higher letdown rates to the maximum design flow of 130 gpm.

TABLE 11.1-9

TRITIUM PRODUCTION[a] (HISTORICAL)

Tritium Source	*Total Produced (Ci/cycle)*	*Release Expected to Reactor Coolant (Ci/cycle)*
Ternary fissions		
Initial cycle	*14,000*	*1400*
Equilibrium cycle	*10,900*	*1090*
Burnable poison rods		
Initial cycle	*1950*	*195*
Control rods	*70*	*70*
Coolant (soluble boron)		
Initial cycle	*388*	*388*
Equilibrium cycle	*285*	*285*
Coolant (lithium, deuterium)		
Initial cycle	*141*	*141*
Equilibrium cycle	*109*	*109*
Total initial cycle	*16,500*	*2130*
Total equilibrium cycle tabs; reset	*11,400*	*1560*

a. The following parameters were used:

Power level, 3565 MWt.
Release fraction from fuel, 10 percent.
Release fraction from burnable poison rods, 10 percent.
Weight of boron-10 in burnable poison rods, 6160 g.
Initial cycle boron, 900 ppm.
Equilibrium cycle boron, 1110 ppm.
Lithium concentration (99.9 atom-percent lithium-7), 2.2 ppm.
Initial cycle operating time, 9240 effective full-power h.
Equilibrium cycle operating time, 7200 effective full-power h.
Ag-In-Cd tip length in hybrid B_4C/Ag-In-Cd control rods, 40 in.
Release fraction from control rods to the coolant, 100 percent.
Production in control rods is based on continuous daily load follow (12, 3, 6, 3 cycle).
During baseload full-power operation, the production would be negligible.

TABLE 11.1-10

EXPECTED SPENT RESIN VOLUME

Item	Number	Resin Volume Each (ft^3)	Replacement Frequency (number/year)	Annual Average Spent Resin Volume (ft^3)
CVCS mixed bed demineralizers	2 per unit	30	2 beds/year (per unit)	60
CVCS cation bed demineralizer	1 per unit	20	1 bed/4 years (per unit)	5
Recycle evaporator feed demineralizers	2 for 2 units	30	1 bed/year (for both units)	30
Recycle evaporator condensate demin-eralizer	1 for 2 units	30	1 bed/5 years (for both units)	6
Thermal regeneration demineralizers	5 per unit	90	1 bed/year (per unit)	90
Steam generator blowdown mixed bed demineralizer	2 per unit	75	2 beds/year (per unit)	150
Waste monitor tank demineralizer	1 per unit	30	5 beds/6 years (per unit)	25
Spent fuel pool demineralizer	1 per unit	30	4 beds/3 years (per unit)	40

11.2 LIQUID WASTE MANAGEMENT SYSTEMS

The liquid waste management systems include all systems that may be used to process for disposal liquids containing radioactive material. These include:

A. Steam generator blowdown processing system (subsection 10.4.8).

B. Turbine building floor drain system (subsection 9.3.3).

C. Liquid waste processing system (LWPS) (section 11.2).

This section primarily addresses the LWPS. The other systems are also addressed in subsection 11.2.3, which discusses the expected releases from all liquid waste management systems.

The LWPS is designed to control, collect, process, handle, store, and dispose of liquid radioactive waste generated as the result of normal operation, including anticipated operational occurrences.

The LWPS services both units; most equipment is not shared but is associated with either Unit 1 or Unit 2. The shared portions of the LWPS are identified in subsection 11.2.2.

11.2.1 DESIGN BASES

The conformance of the LWPS design with the criteria of Regulatory Guide 1.143 is discussed in section 1.9.

11.2.1.1 Capacity

The projected flows of various liquid waste streams to the LWPS are specified in table 11.2.1-1. The LWPS provides adequate capacity to meet the anticipated processing requirements of the plant.

The LWPS design can accept equipment malfunctions without affecting the capability of the system to handle both anticipated liquid waste flows and possible surge load due to excessive leakage. Surge capacity of individual tanks is discussed in paragraph 11.2.2.6.2.

Portions of the LWPS may become unavailable as a result of the malfunctions listed in paragraphs 11.2.1.1.1 and 11.2.1.1.2.

Ample surge capacity of the system and the low load factor of the processing equipment permit the system to accommodate waste until failures can be repaired and normal plant operation resumed. In addition, the LWPS is designed to accommodate the anticipated operational occurrences described in paragraphs 11.2.1.1.3 through 11.2.1.1.5.

As an additional measure of assurance that liquid wastes can be processed during all plant conditions, piping connections have been installed in the LWPS process stream to allow use of a portable filtration system and the portable demineralizer system, which are both located in the radwaste processing facility.

11.2.1.1.1 Pump Failure

Where operation is not essential and surge capacity is available, a single pump is provided. Two reactor coolant drain tank (RCDT) pumps are provided because the relative inaccessibility of the containment during plant operation would hinder maintenance. Pump repair and replacement is facilitated by using three standard pump designs for the eight applications in the LWPS. To protect the pumps from damage due to loss of suction, each pump is interlocked to stop on a low level condition in the tank feeding the pump.

11.2.1.1.2 Filter, Strainer, or Demineralizer Plugging

Instrumentation is provided to give local indication of the pressure drop across all filters, strainers, and demineralizers. Periodic checks of the pressure drops provide indication of the equipment fouling, thus permitting corrective action to be taken before an excessive pressure drop is reached.

11.2.1.1.3 High Leakage Rate

The system is designed to handle a 1-gal/min primary coolant system leak in addition to the expected leakage during normal operation. Operation of the system is the same as for normal operation except that the load on the system is increased.

The extra leakage may be directed to the waste holdup tank or the floor drain tank. The floor drain tank contents are not necessarily processed; they may be discharged without treatment if sample analysis indicates that the water quality is acceptable for discharge.

11.2.1.1.4 High Use of Decontamination Water

Should large quantities of water be used to decontaminate areas or equipment, the load on the LWPS will be increased. However, the LWPS is designed to handle a 6-gal/min continuous and indefinite input to the waste holdup and/or floor drain tanks. If the flow to the floor drain tank can be discharged without processing, the overall LWPS capacity is increased. Laundry and hot shower wastes are in addition to the 6-gal/min capacity.

The 6-gal/min process capacity is equivalent to almost 5000 gal/day above what is normally sent to the waste holdup and floor drain tanks.

11.2.1.1.5 Refueling

During refueling the load on the LWPS is expected to increase, but operation is the same as for normal plant operation, and there is no significant effect on the performance capability of the LWPS.

11.2.1.2 Controlled Release of Radioactivity

The LWPS provides the capability to reduce the amounts of radioactive nuclides released in the liquid wastes through the use of demineralization and recycling of clean tritiated water for reuse in the reactor plant when possible and time delay for decay of short-lived nuclides.

The assumed equipment decontamination factors are included in table 11.2.1-2. The estimated LWPS isotopic inventories, concentrations, and annual average flowrates that will be processed in the LWPS or discharged to the environment during normal operation are estimated in table 11.2.1-3.

Before any liquid radioactive waste is discharged, it is pumped to a monitor tank. A sample of the monitor tank contents is analyzed and the results logged. In this way, a record is kept of all planned releases of radioactive liquid waste. The liquid waste is discharged from the monitor tank in a batch operation, and the discharge flowrate is restricted as necessary to maintain an acceptable concentration when diluted by the circulating water discharge flow. These provisions preclude uncontrolled releases of radioactivity. In addition, the discharge line contains a stop valve interlocked with a radiation monitor on the system discharge line. The valve automatically closes, and an alarm is actuated if the activity in the discharge stream reaches the monitor setpoint. The stop valve is also interlocked to isolate discharge flow if sufficient dilution flow is not available.

To minimize leakage from the LWPS, the system is of all-welded construction except where flanged connections are required to facilitate component maintenance. The use of canned rotor design pumps for most applications minimizes system leakage and the release of radioactive gas that might be entrained in the leaking fluid to the building atmosphere.

Provisions are made to preclude uncontrolled spills of radioactive liquids due to tank overflows. Tables 11.2.1-4 and 11.4.5-1 lists the provisions for tank level indication, level annunciation, and overflow disposition for tanks located outside the containment building that potentially could contain radioactive liquids.

11.2.1.2.1 Expected Releases

The LWPS design ensures that the annual average concentration limits established by Appendix B, Table II, column 2 of 10 CFR 20.1 - 20.601 for liquid releases are not exceeded during plant operation with expected levels of fuel cladding defects. Subsection 11.2.3 discusses the calculated releases of radioactive materials from the LWPS and other portions of the liquid waste management systems resulting from normal plant operation.

An evaluation is provided in subsection 11.2.3 to demonstrate that the doses to individuals, at or beyond the site boundary, resulting from the expected releases from the liquid waste management systems are within the numerical design objectives of Appendix I of 10 CFR 50.

11.2.1.2.2 Off-Normal Operation

Paragraph 11.2.1.1 discusses the capability of the LWPS to accommodate various equipment failures and anticipated operational occurrences. During these anticipated occurrences, the effectiveness of the LWPS in controlling releases of radioactivity remains essentially unaffected, so releases are limited to approximately the same as during normal operation.

The GALE code,[(1)] used to calculate the release concentration described in paragraph 11.2.1.2.1, contains an adjustment factor of 0.15 Ci/year per reactor to account for abnormal occurrences resulting in unplanned releases.

11.2.1.3 Equipment Design

The LWPS equipment design parameters are provided in table 11.2.1-2.

The seismic design classification and safety classification for the LWPS components and structures are listed in table 3.2.2-1. Safety class designations are also indicated on the LWPS piping and instrumentation diagram, drawings 1X4DB124, 1X4DB125, 1X4DB126, 1X4DB127, AX4DB124-2, AX4DB124-3, AX4DB124-4, and AX4DB124-5.

11.2.1.4 Reference

1. U.S. Nuclear Regulatory Commission, "Calculation of Releases from Pressurized Water Reactors," NUREG-0017, April 1976.

11.2.2 SYSTEM DESCRIPTIONS

The liquid waste processing system (LWPS) collects and processes potentially radioactive wastes for recycling or release to the environment. Provisions are made to sample and analyze fluids before discharge. Based on the laboratory analysis, these wastes are either retained for further processing or released under controlled conditions through the cooling water system, which dilutes the discharge flow. A permanent record of liquid releases is provided by analyses of known volumes of effluent.

The radioactive liquid discharged from the reactor coolant system (RCS) is processed by the radwaste processing facility systems and may be discharged or recycled.

The LWPS is arranged to recycle reactor grade water if desired. This is implemented by the segregation of equipment drains and waste streams to prevent intermixing of liquid wastes. The LWPS can be divided into the following subsystems:

A. Reactor Coolant Drain Tank (RCDT) Subsystem

   This portion of the LWPS collects nonaerated, reactor grade effluent from sources inside the containment.

B. Drain Channel A

   This portion of the LWPS collects aerated, reactor grade effluent that can be recycled.

C. Drain Channel B

   This portion of the LWPS processes all effluent that is not suitable for recycling.

D. Radwaste Processing Facility Demineralizers

   The radwaste processing facility demineralizer systems consist of portable demineralizers installed in subterranean enclosures inside the radwaste processing facility. The radwaste processing facility is described in paragraph 11.4.2.4. The radwaste processing facility demineralizers can be aligned to process any of the three waste drain streams.

E. The radwaste processing facility filtration system consists of a portable, vendor supplied system located within a shielded area inside the radwaste processing facility. The filtration system associated tanks, pumps, accumulator, piping, valves, and controls located within a shielded area inside the radwaste

processing facility. The peripheral equipment is located adjacent to the filter assembly. The filter system can be aligned to process any of the three waste drain streams. Details of this equipment are shown on drawing AX4DB124-1.

In addition, the LWPS provides capability for handling and storage of spent ion exchange resins.

The LWPS does not include provisions for processing secondary system wastes. Secondary system effluent is handled by the steam generator blowdown processing system (SGBPS), as described in subsection 10.4.8, and by the turbine building drain system. Estimated releases from these systems are discussed in subsection 11.2.3. The LWPS design, which segregates primary and secondary wastes, minimizes the amount of water that must be processed by discharging low activity wastes directly, where permissible, with no treatment.

Instrumentation and controls necessary for the operation of the LWPS are located on a control board in the auxiliary building. Any alarm on this control board (except for the waste processing holdup control panel) is relayed to the main control board in the control room.

The LWPS piping and instrumentation diagrams are shown in drawings 1X4DB124, 1X4DB125, 1X4DB126, 1X4DB127, AX4DB124-1, AX4DB124-2, AX4DB124-3, AX4DB124-4, and AX4DB124-5 and process flow diagram for the LWPS is shown on figure 11.2.2-1. Table 11.2.1-1 lists the assumptions regarding flows and activity levels that were used in preparation of table 11.2.1-3, which gives nuclide concentrations for key locations within the LWPS as shown on figure 11.2.2-1. The process flow data is calculated using the data in table 11.2.1-1, the flow paths indicated on figure 11.2.2-1, realistic primary coolant activity levels from section 11.1, and decontamination factors as given in reference 1 of subsection 11.2.1.

11.2.2.1 Reactor Coolant Drain Tank Subsystem

Recyclable reactor grade effluents enter this subsystem from valve leakoffs, reactor coolant pump No. 2 seal leakoffs, reactor vessel flange leakoff, and other deaerated, tritiated water sources inside the containment. Connections are provided for draining the RCS loops and the safety injection system (SIS) accumulators and for cooling the pressurizer relief tank. In addition, refueling canal drains can be routed to the refueling water storage tank using the RCDT pumps.

The RCDT contents are continuously recirculated through the RCDT heat exchanger to maintain the desired temperature. Level is prevented from varying significantly by a control valve which automatically opens a path from the recirculation line to the BRS when normal tank level is exceeded. The RCDT is also connected to the gaseous waste processing system (GWPS) vent header. Hydrogen gas bottles connected to the RCDT ensure a hydrogen blanket. Maintaining a constant level minimizes the amount of gas sent to the GWPS and minimizes the amount of hydrogen used. Provisions for sampling the gas are provided.

Details of the RCDT subsystem are shown on drawing 1X4DB127. A separate RCDT subsystem is provided for each of the two units.

11.2.2.2 Drain Channel A Subsystem

Aereated, tritiated liquid enters drain channel A through lines connected to the waste holdup tank. Sources of this aerated liquid are as follows:

A. Accumulator drainage (via RCDT pump suction).

B. Sample room sink drains (excess primary sample volume only).

C. Ion exchanger, filter, pump, and other equipment drains.

The containment sump or auxiliary building sump may be directed to the waste holdup tank or the floor drain tank for processing as necessary.

The collected aerated drainage is pumped or flows to the waste holdup tank prior to processing through the radwaste processing facility filtration system and/or the radwaste processing facility demineralizers before reuse or discharge. Details of this equipment are shown on drawings AX4DB124-2, AX4DB124-3, AX4DB124-4, and AX4DB124-5.

The basic composition of the liquid collected in the waste holdup tank is boric acid and water with some radioactivity.

A separate drain channel A subsystem is provided for each of the two units. Details are shown on drawings 1X4DB124 and 1X4DB127. Table 11.2.1-1 lists the estimated flows entering the waste holdup tank.

11.2.2.3 Drain Channel B Subsystem

Drain channel B is provided to collect and process nonreactor grade liquid wastes. These include:

- Wastes from floor drains.
- Equipment drains containing nonreactor grade water.
- Laundry and hot shower drains.
- Other nonreactor grade sources.

Drain channel B is comprised of three drain subchannels, each associated with one of the following tanks.

A. Laundry and Hot Shower Tank

The laundry and hot shower tank is provided to collect and process waste effluents from the plant laundry and personnel decontamination showers and hand sinks.

Laundry and hot shower drains normally need no treatment for removal of radioactivity. This water is transferred to a waste monitor tank through the laundry and hot shower tank filter for eventual discharge. If sample analysis indicates that decontamination is necessary, the water can be directed through the Unit 1 or Unit 2 waste monitor tank demineralizer or the radwaste processing facility for cleanup.

The laundry and hot shower tank and filter are shared by the two units. Details of this portion of the LWPS are shown on drawing 1X4DB126. Table 11.2.1-1 lists estimated flows entering the laundry and hot shower tank.

B. Floor Drain Tank

Water may enter the floor drain tank from system leaks inside the containment through the containment sump, from system leaks in the auxiliary building through auxiliary building sumps and the floor drains, and floor drains in the

radwaste facilities. Sources of water to the containment sump and auxiliary building sumps and floor drains are the following:

1. Fan cooler leaks.
2. Secondary side steam and feedwater leaks.
3. Primary side process leaks.
4. Decontamination water.

The containment sump or auxiliary building sumps may be directed to the waste holdup tank.

Another source of water to the floor drain tank is the chemical laboratory drains. Excess nonreactor grade samples that are not chemically contaminated and laboratory equipment rinse water are drained to the floor drain tank.

The contents of the floor drain tank are processed through the radwaste processing facility demineralizers and/or the radwaste processing facility filtration system and then pumped to a waste monitor tank for ultimate discharge.

If the activity in the floor drain tank liquid is such that the discharge limits cannot be met without cleanup, the liquid can be processed by the waste monitor tank demineralizer, the radwaste processing facility demineralizers, or the radwaste processing facility filtration system.

A separate floor drain tank and associated equipment are provided for each of the two units. Details of this portion of the LWPS are shown on drawing 1X4DB126. Table 11.2.1-1 lists the estimated flows entering the floor drain tank.

C. Chemical Drain Tank

Laboratory samples which contain reagent chemicals (and possibly tritiated liquid) are discarded through a sample room sink which drains to the chemical drain tank. Chemical drains requiring radwaste processing are sent to the solid waste management system or may be processed through the radwaste processing facility demineralizers and/or the radwaste processing facility filtration system.

The chemical drain tank and associated equipment are shared by Units 1 and 2. Details of this portion of the LWPS are shown on drawing 1X4DB125. Table 11.2.1-1 lists the estimated flow directed to the chemical drain tank.

Any liquids released to the environment by the LWPS are first directed to a waste monitor tank. Before releasing the contents of a waste monitor tank, a sample is taken for analysis. The findings are logged, and, if the activity level is within acceptable limits, the tank contents are released to the discharge canal. The discharge valve is interlocked with a process radiation monitor and closes automatically when the radioactivity concentration in the liquid discharge exceeds a preset limit. The radiation element is located upstream of the discharge valve at a distance sufficient to close the valve before passing the fluid that activated the detector trip signal. The isolation valve also blocks flow if sufficient dilution water is not available. The radiation monitor is described in section 11.5. A permanent record of the radioactive releases is provided by a sample analysis of the known volumes of waste effluent released. Liquid waste discharge flow and volume are also recorded.

If the monitor tank contents are not acceptable for discharge, the fluid can be held for a time to allow activity to decay to acceptable levels, or it can be further processed by the waste monitor

tank demineralizer, the radwaste processing facility filtration system, or radwaste processing facility demineralizers.

11.2.2.4 Spent Resin Handling Subsystem

This subsystem collects, handles, and processes spent resins from the primary fluid systems prior to their disposal.

Spent resin from the primary system demineralizers is transported to and stored in the spent resin storage tank prior to being drummed. The spent resin sluice portion of the LWPS consists of a spent resin sluice filter, spent resin sluice pump, and the spent resin storage tank. The resin sluice water, after being directed to an ion exchange vessel by the sluice pump, returns to the spent resin storage tank for reuse.

Thus, sluicing of spent resin from primary plant demineralizers is accomplished without generating a large volume of liquid waste.

Reactor makeup water storage tank system pressure may also be used as a motive force for sluicing demineralizers to the spent resin storage tank.

Resin from the SGBPS demineralizers is handled in a similar manner to that described for the primary system demineralizers. The SGBPS resin handling is discussed in subsection 10.4.8. The primary system resins and the SGBPS resins are segregated during all phases of handling so that no cross-contamination occurs.

Resin slurry from both spent resin storage tanks is sent to the radwaste processing facility by pressurizing the tank with nitrogen.

A separate resin handling subsystem is provided for each of the two units. Details of this portion of the LWPS are shown on drawing 1X4DB125.

11.2.2.5 Liquids From Sources Other Than the Liquid Waste Processing System

11.2.2.5.1 Steam Generator Blowdown Processing System

Blowdown from the steam generators of each unit is cooled, filtered, and demineralized. Normally it is then returned to the condensers for reuse as condensate makeup, but it may be discharged to the environment. The SGBPS is described in subsection 10.4.8.

11.2.2.5.2 Turbine Building Drain System

The function of the turbine building drain system is:

A. To collect the floor drains and sampling wastes in the turbine building and other miscellaneous drains.

B. To treat these wastes, if necessary, to meet the requirements of the State of Georgia Environmental Protection Agency (EPA) regulations and Nuclear Regulatory Commission regulations prior to discharge to the Savannah River.

All the wastes generated in the turbine building including drains, leakages, and the sampling wastes are collected in the sumps in the turbine building. The sump pump discharge, combining with the auxiliary building clean water sump discharge, auxiliary feedwater pumphouse drain, maintenance building drain, and the treated turbine building drain, normally passes through a radiation monitor before entering into an oil water separator to meet the EPA oil discharge limit. If the radioactivity level of this combined waste stream exceeds the setpoint of the radiation monitor, a signal is sent to stop the flow to the nonradioactive oil water separator. This stream is then sent to one of the turbine building drain tanks before processing. The amount of waste delivered to the turbine building drain tanks is kept minimal by promptly locating the specific contaminated (radioactive) stream(s) and pumping the remaining (nonradioactive) streams via their normal route. The turbine building drain system and associated components are discussed in detail in subsection 9.3.3.

11.2.2.6 Equipment Description

Principal design parameters for the LWPS equipment are given in table 11.2.1-2. All parts or components in contact with borated water are fabricated from or clad with austenitic stainless steel. Pumps are provided with vent and drain connections.

Since this system performs no function related to the safe shutdown of the plant, many components of the system are classified as nonnuclear safety. Component safety classes, seismic design, and principal codes are shown in table 3.2.2-1.

11.2.2.6.1 Pumps

Pumps in the LWPS have been standardized wherever possible. Where operation is not critical and surge capacity is available, a single pump has been provided. Spare pumps can be kept onsite in case any pump should fail. Quick replacement is possible because:

- The pumps are flanged, not welded.
- The system has surge capacity.
- Adequate vent, flush, and drain capabilities are provided.
- The pumps are standardized.

Three types of standard pumps are required, as described below.

Canned rotor pumps are used wherever possible to minimize fluid leakage and to minimize the release of entrained radioactive gases in the leaking fluid to the atmosphere.

Canned rotor pumps with similar head-flow characteristics are used for the following applications:

- Spent resin sluice pumps.
- RCDT pumps.

Another canned rotor pump design is used for the following applications:

- Waste evaporator feed pumps.

- Waste evaporator condensate tank pumps.
- Chemical drain tank pump.
- Waste monitor tank pumps.

Mechanical seal pumps are also used in some applications because these pumps have an open impeller, which will not be damaged by large particles in the water. Mechanical seal pumps with similar head-flow characteristics are used for the following applications:

- Floor drain tank pump.
- Laundry and hot shower tank pump.

Globe valves are installed in pump discharge lines where necessary to prevent pump runout. Pump miniflow lines have locked-in-position globe valves to ensure that the minimum pump flow requirements are met.

A. RCDT Pump

The design basis for this pump is that, in its function of RCS drain, the coolant level reaches the midplane of the reactor vessel nozzles within an 8-h period. Two pumps are furnished because of the relative inaccessibility of the containment during plant operation. Both pumps are operated to meet the draining time requirement. One pump provides sufficient flow for normal operation of the RCDT portion of the LWPS. The liquid is sent to the recycle holdup tanks.

B. Waste Evaporator Feed Pump

This pump supplies feed to the radwaste processing facility for processing from the waste holdup tank, and it can be used to transfer waste holdup tank contents to the floor drain tank, if desired.

C. Waste Evaporator Condensate Pump

The waste evaporator condensate tank pump is used to transfer the contents of the waste evaporator condensate tank.

D. Chemical Drain Tank Pump

This pump may be used to transfer the contents of the chemical drain tank.

E. Spent Resin Sluice Pump

One pump is provided to sluice resins from primary side demineralizers to the spent resin storage tank. Its delivery flow is based on the velocity required to sluice resin in a 3-in. pipe.

F. Laundry and Hot Shower Tank Pump

This pump is used to transfer the water from the laundry and hot shower tank to a waste monitor tank.

G. Floor Drain Tank Pump

This pump is used to transfer water from the floor drain tank to the radwaste processing facility. From the radwaste processing facility, processed waste is transferred to the waste monitor tank.

H. Waste Monitor Tank Pumps

Two pumps are provided for each unit. One pump is used for each monitor tank to discharge water from the LWPS or for recycling if further processing is required.

The pump may also be used for circulating the water in the waste monitor tank to obtain uniform tank contents, and therefore a representative sample, before discharge. These pumps can be throttled to achieve the desired discharge rate.

I. Auxiliary Waste Monitor Tank Pumps

Two pumps are provided. They are installed in Unit 2 but serve both units. One pump is used for each auxiliary waste monitor tank to discharge water from LWPS or for recycling if further processing is required. A mixer may be used for circulating the water in the auxiliary waste monitor tank to obtain uniform tank contents, thereby assuring a representative sample is acquired prior to discharge of the tank contents. The pumps can be throttled to achieve the desired discharge rate.

11.2.2.6.2 Tanks

A. Reactor Coolant Drain Tank

One tank is provided for each unit. The purpose of the RCDT is to collect leakoff-type drains inside the containment at a central collection point for further disposition through a single penetration via the RCDT pumps. The tank provides surge volume and net positive suction head (NPSH) to the pumps.

Only water which can be directed to the boron recycle holdup tanks enters the RCDT. The water is compatible with reactor coolant and does not contain dissolved air during normal plant operation, by engineering design.

A constant level is maintained in the tank to minimize the amount of gas sent to the GWPS and also to minimize the amount of hydrogen cover gas required. The level is maintained by one continuously running pump and by a control valve in the discharge line. This valve operates on a signal from a level controller to limit the flow out of the system. The remainder of the flow is recirculated to the tank.

Continuous flow is maintained through the heat exchanger in order to prevent loss of pump NPSH resulting from a sudden inflow of hot liquid into the RCDT.

B. Waste Holdup Tank

One atmospheric pressure tank is provided for each unit to collect:

1. Equipment drains.
2. Valve and pump seal leakoffs (outside the containment).
3. Boron recycle holdup tank overflows.
4. Other water from tritiated, aerated sources.

The tank size is adequate to accommodate 11 days of expected influent during normal operation.

C. Waste Evaporator Condensate Tank

One tank with a diaphragm to exclude air is provided for each unit to collect water from processing systems.

D. Chemical Drain Tank

One tank is provided to collect chemically contaminated, tritiated water from the laboratories. This tank is shared by the two units and has sufficient capacity to accept a month's laboratory waste during normal operation of both units.

E. Spent Resin Storage Tank

The purpose of the spent resin storage tank is to provide a collection point for spent resin and to allow for decay of short-lived radionuclides before disposal. The tank also serves as a head tank for the spent resin sluice pump.

One vertical, cylindrical tank with sufficient capacity to handle the spent resin storage needs is provided. A vertical, cylindrical tank is used because the symmetrical bottom facilitates the removal of resin. The tank is designed so that sufficient pressure can be applied in the gas space of the tank to move the resin slurry to the radwaste processing facility.

The spent resin storage tank and associated equipment, which can contain radioactive material, are shielded to limit the dose to personnel.

The level indicating system in the spent resin storage tank shows only total level and not the amount of resin and water separately. However, since the resin volumes flushed from demineralizers and the resin volumes transferred to the radwaste processing facility are known, the resin level in the tank is also known.

F. Laundry and Hot Shower Tank

One atmospheric pressure tank is provided to collect laundry and hot shower drains for the two units. The tank size is sufficient to furnish a 10-day surge capacity for the two units during normal operation of both units and a 2-day surge capacity during refueling of a single unit.

G. Floor Drain Tank

One atmospheric pressure tank is used to collect floor drains from the controlled areas of each unit's primary system. The tank provides sufficient surge capacity for the floor drains within the collection area and, in connection with the waste holdup tank, provides surge capacity for abnormal primary system leaks. The tank size is adequate to accommodate 3.5 days of expected influent during normal operation or 1.8 days of expected influent during shutdown operation.

H. Waste Monitor Tanks

Two atmospheric pressure waste monitor tanks are provided for each unit to monitor liquid discharged from the plant site. Each tank is sized to hold a volume large enough that sampling requirements are minimized, thereby minimizing laboratory effluent.

Two additional atmospheric pressure tanks have been provided to augment the plant capacity to handle large surges of water or to accommodate conditions in which release of the processed water is not feasible. These 20,000 gallon tanks are installed on level D of the Unit 2 auxiliary building. They are shared by both units, with one tank normally aligned to each unit.

11.2.2.6.3 Reactor Coolant Drain Tank Heat Exchanger

This heat exchanger is located in the discharge line of the RCDT pumps and is in constant service as part of the RCDT recirculation path. Continuous auxiliary component cooling water flow is maintained to the heat exchanger to accommodate, without operator action, sudden flow of hot liquid to the RCDT. The heat exchanger can also be used to cool the contents of the pressurizer relief tank in the RCS.

The heat exchanger is sized for several modes of operation:

A. It can maintain the RCDT liquid below 170°F, assuming a 10-gal/min input of 600°F reactor coolant.

B. It can cool the contents of the pressurizer relief tank from 200°F to 120°F in less than 8 h.

C. It can maintain the contents of the RCDT at less than 170°F, assuming a 25-gal/min input from the excess letdown heat exchanger (chemical and volume control system).

One RCDT heat exchanger is provided per unit.

11.2.2.6.4 Demineralizers

As part of a program of continuous pressurized water reactor operating plant followup, Westinghouse has obtained operational data on demineralizer decontamination factors for selected isotopes. The measured range of decontamination factors for these isotopes is given in table 11.2.2-1.

These values were observed across mixed bed demineralizers containing cation resin in the lithium-7 form and anion resin in the borated form. The minimum values in table 11.2.2-1 were generally observed just before resin flushing and recharging; during the operating life of the demineralizer, decontamination factors were consistently closer to the maximum values.

Although specific operating decontamination factors have not yet been measured for other isotopes, their behavior in a mixed bed demineralizer may be inferred from this data.

The process decontamination factor used for the demineralizers in the analysis of system performance is taken from reference 1, subsection 11.2.1. These decontamination factors are given in table 11.2.2-1.

A. Waste Monitor Tank Demineralizer

One mixed bed demineralizer is provided upstream of the waste monitor tanks to remove, if desired, trace ionic contaminants. The laundry and hot shower tank contents can also be processed through the demineralizer if such processing is necessary.

B. Radwaste Processing Facility Demineralizers

The radwaste processing facility houses a series of demineralizers which can be operated in various configurations. The type and loadings of resins or other filter media utilized in these vessels can be changed as necessary to optimize the performance of the system.

11.2.2.6.5 Filters

The following filters are provided in the LWPS:

- Laundry and hot shower tank filter.
- Floor drain tank filter.
- Spent resin sluice filter.
- Waste evaporator condensate filter.
- Waste monitor tank filter.

The laundry and hot shower tank filter element is normally removed since installation of the filtration system in the radwaste processing facility.

11.2.2.6.6 Strainers

Basket-type strainers of mesh construction are provided to prevent clogging of filters and lines downstream because of large particles being sluiced through the lines during liquid transfer operations.

The following strainers are provided in the LWPS:

- Laundry and hot shower tank strainer.
- Floor drain tank strainer. (This strainer may be removed to enhance system operability.)

11.2.2.7 Instrumentation Design

The system instrumentation is shown on the LWPS drawings 1X4DB124, 1X4DB125, 1X4DB126, 1X4DB127, AX4DB124-2, AX4DB124-3, and AX4DB124-4.

Instrumentation readout is located mainly on the waste processing system panel in the auxiliary building. Some instruments are read at the equipment location.

All alarms are shown separately on the waste processing system panel and are further relayed to one common waste processing system annunciator on the main control board in the control room. The waste processing holdup control panel does not relay signals to the control room annunciator.

All pumps are protected against loss of suction pressure by a control setpoint on the level instrumentation for the respective vessels feeding the pumps. In addition, the RCDT pumps and the spent resin sluice pump are interlocked with flowrate instrumentation to stop the pumps when the delivery flows reach minimum setpoints. The RCDT pumps have a keyswitch allowing the bypass of the flowrate instrumentation control logic. During certain intermittent outage flowpaths, flowrates less than those recommended by the pump vendor may be established. Administrative controls require continuous monitoring of pump performance during such flowpaths to ensure the pumps' availability.

Pressure indicators are provided to give local indication of pressure drops across demineralizers, filters, and strainers.

All releases to the environment are monitored for radioactivity. This instrumentation is described in section 11.5.

Each tank is provided with level indication instrumentation that actuates an alarm on high liquid level in the tank, thus warning of potential tank overflow.

11.2.2.8 Operating Procedures

The LWPS is manually operated except for some functions of the RCDT circuit. The system includes adequate control equipment to protect components and adequate instrumentation and alarm functions to provide operator information to ensure proper system operation.

Operation of the LWPS is essentially the same during all phases of normal and defined off-normal reactor plant operation; the only differences are in the load on the system. The term "normal operation," as used here, means all phases of plant operation except operation under emergency or accident conditions. The LWPS is not regarded as an engineered safety features system.

11.2.2.8.1 Reactor Coolant Drain Tank Subsystem Operation

A. Reactor Coolant Drain Tank Recirculation

Reactor coolant is continuously circulated through the RCDT heat exchanger to maintain ≤170°F in the event of a hot reactor coolant leak. Level is maintained by a control valve which automatically opens a path from the recirculation line to the recycle holdup tanks. Normal operation of this mode is automatic and requires no operator action. The system can be put into the manual mode, if desired.

Leakage into the RCDT can be estimated by putting the system in the manual mode, stopping the pump, and monitoring the level change. A venting system is provided to prevent wide pressure variations in the RCDT. A hydrogen blanket is automatically maintained between 2 and 6 psig. Hydrogen is supplied from bottles which must be replaced when the pressure drops to ~100 psig.

During all other operations, the recirculation mode is stopped and the RCDT is isolated.

B. Pressurizer Relief Tank Cooling

The pressurizer relief tank may be cooled by a feed- and-bleed method, by spraying cold makeup water and pumping the water to the recycle holdup tanks with an RCDT pump through the RCDT heat exchanger. This is a rapid cooldown and can be used even if the heat exchanger is out of service. However, to minimize the addition of water to the system when a rapid cooldown is not necessary, the pressurizer relief tank may be cooled from 200°F to 120°F in less than 8 h by using the RCDT heat exchanger and one pump in a recirculation mode without adding makeup water.

C. Loop Draining

Four RCS loops may be drained simultaneously to the midpoint of the reactor vessel nozzles in less than 8 h when both RCDT pumps are running. The loops are vented to 100 psig, then a spectacle flange is positioned to the pumps' suction. The water may be sent to the recycle holdup tanks, refueling water storage tank, waste holdup tank, or the spent fuel pool. Since the RCDT heat exchanger is not needed for cooling, it may be bypassed or used in parallel with the bypass.

D. Refueling Canal Draining and Cleanup

Refueling canal water is transferred, using the RCDT pumps, to the refueling water storage tank (draining), to the canal via the spent fuel pool cooling system (cleanup), or to the waste holdup tank (disposal).

Since the RCDT heat exchanger is not needed for cooling, it may be bypassed or used in parallel with the bypass.

E. Accumulator Draining

This mode is available for accumulator maintenance. After the accumulator is vented to 90 psig, the spool piece is connected to the RCDT pump suction for transfer of accumulator water to the reactor water storage tank or recycle holdup tank.

F. Excess Letdown Header

During normal plant heatup operations, excess letdown to the RCDT will permit faster heatup rate. Excess letdown flow is directed from the RCDT to the recycle holdup tanks.

11.2.2.8.2 Drain Channel A Subsystem Operation

Water is accumulated in the waste holdup tank until sufficient quantity exists to warrant starting the processing systems for a batch process.

If it is not desired to recycle the water in the waste holdup tank and analysis indicates that decontamination is not necessary, the water may be sent to the floor drain tank for eventual discharge.

The waste holdup tank contents may be processed by the radwaste processing facility filtration system and/or the radwaste processing facility demineralizers.

11.2.2.8.3 Drain Channel B Subsystem Operation

Drain channel B of the LWPS consists of the laundry and hot shower system, the floor drain tank system, and the chemical drain tank system.

Laundry and hot shower water enters the laundry and hot shower tank for holdup; it is sampled, filtered, and transferred to the monitor tank for discharge, or processed through the radwaste processing facility filtration system or the radwaste processing facility demineralizer. If demineralization is required, the resin must be replaced with clean resin thereafter.

The floor drain tank contents are recirculated, and then samples are taken and analyzed. If the floor drain tank is overloaded or the water is recyclable, the water can be transferred to the waste holdup tank.

Water leaving this system to the discharge canal is monitored for radiation. This radiation monitor is described in section 11.5. If the radiation monitor closes the discharge valve, it must be cleared before the valve can be reopened. The monitor element can be cleared by flushing it with demineralized water from the temporary connection back to the waste monitor tank. During refueling, the load on this portion of the LWPS is increased, but there is no change in operation.

Spent samples with high chemical concentrations are held up in the chemical drain tank, then sampled. The contents are drained to the auxiliary building sumps or the radioactive drain sumps.

11.2.2.8.4 Spent Resin Handling Subsystem

A. Resin Fluffing

The demineralizer is valved out of service, and the flow path is aligned from the spent resin sluice pump (or the reactor makeup water system header) through the process line of the demineralizer, through the Johnson screen at the top of the demineralizer, and back to the spent resin storage tank. The resin bed is backflushed for about 10 min to loosen it for sluicing. This operation may also be used to recover pressure drop caused by bed fouling by backwashing particulates from the top layer of the resin into the spent resin storage tank. Such a recovery is useful when the resin is not ionically depleted.

B. Resin Sluicing

The sluice pump is shut off after fluffing. The valves in the backflush circuit are closed, the sluice line to the bed screen is opened, and the sluice pump is started. The resin flows to the spent resin storage tank, initially at a slow rate; it continues for about 10 min until sluicing is completed. Finally, the pump is stopped, and the sluice inlet and outlet valves are closed. If the pump suction line screen in the tank becomes resin plugged at any time, it can be cleared with a blast of nitrogen.

C. Resin Fill

After sluicing is completed, fresh resin must be added. The path to the drain header from the demineralizer is opened to allow overflow. The resin fill line is opened and resin added. After fresh resin is added, the fill line valve is closed and the flow path is realigned for normal demineralizer operation.

D. Resin Disposal

The resin in the spent resin storage tank is loosened before disposal by sending pressurized nitrogen or sluice water through the six sparging nozzles in the tank. The valves in the resin transfer line are opened to direct the spent resin to the radwaste processing facility.

The tank is then pressurized with nitrogen to force the resin up through the resin transfer line to the disposal area. A single nozzle in the spent resin storage tank is provided to allow local fluidization with sluice water at the opening of the discharge pipe. During resin transfer, this nozzle is used to ease the flow of the resin slurry. After resin transfer is complete, the tank is vented to the plant vent and returned to atmospheric pressure. The resin transfer line is then backflushed to the spent resin storage tank to clear it of resin.

Since a certain amount of resin remains in the tank after a disposal operation, it may hinder the backflush operation. Therefore, the fluidizing nozzle is again used to facilitate the backwash operation.

11.2.3 RADIOACTIVE RELEASES

11.2.3.1 Criteria for Discharge, Recycle, or Further Treatment of Liquid Waste

Processed liquids are recycled for reuse within the plant whenever possible, provided that the following criteria are satisfied:

- The plant water inventory requires makeup.
- The water to be reused satisfies system water quality requirements.
- Tritium buildup is less than plant operating limits.

Processed liquids are discharged under the following conditions:

- The processed water does not satisfy plant operating requirements for water quality and tritium buildup.
- The effluent concentrations are within the limits specified by the Offsite Dose Calculation Manual.
- The discharge does not cause the limits of 10 CFR 50, Appendix I, to be exceeded.

Processed liquids are recycled within their respective treatment systems for additional processing when system water quality requirements are not satisfied and reuse within the plant is desirable, or when discharge of the processed liquid is planned but the discharge would result in exceeding limits in the Offsite Dose Calculation Manual.

11.2.3.2 Estimated Releases

The equipment utilized during liquid waste processing is at the discretion of the operator; therefore, the calculated releases conservatively do not address all possible treatment processes but only the expected process. Liquid releases from VEGP are calculated using the PWR-GALE computer code(1) and parameters listed in table 11.1-8, which are discussed in more detail below. Releases calculated assuming operation with expected levels of fuel cladding defects of 0.12 percent are presented in table 11.2.3-1. Primary and secondary coolant activity levels are given in section 11.1 for the realistic case. In agreement with reference 1, the total releases include an adjustment factor of 0.15 Ci/year, using the same isotopic distribution as the calculated release, to account for anticipated operational occurrences.

The tables list the calculated annual release from each of the process paths discussed below as well as the total annual release. A comparison of annual average effluent concentrations with Appendix B, Table II, column 2 values of 10 CFR 20.1 -20.601 is provided in table 11.2.3-2 for operation with expected fuel leakage.

The releases are calculated for one unit, assuming that both units are operating. This is done to reflect the impact of the second unit's operation on the operation of systems and components shared between the two units. To obtain the combined releases of the two units, simply double the values listed in table 11.2.3-1.

A survey has been performed of liquid discharges from different Westinghouse pressurized water reactor plants, with results presented in table 11.2.3-3. The data includes radionuclides released on an unidentified basis and are all within the permissible concentration for release of liquid containing an unidentified radionuclide mixture. The data in table 11.2.3-4 clearly indicate that actual releases are highly dependent upon the actual operation of the plant and can vary significantly from year to year for a given plant as well as from plant to plant.

11.2.3.2.1 Boron Recycle System (BRS)

Primary coolant is withdrawn from the reactor coolant system (RCS) and processed through the chemical and volume control system (CVCS). A side stream of 1700 gal/day of the letdown stream is assumed to be diverted to the BRS as shim bleed. The shim bleed is combined with a conservatively estimated 300 gal/day of other reactor grade wastes that are collected by the reactor coolant drain tank (equipment drain wastes). Since the BRS is shared by both units, the total process flow is 4000 gal/day. The equipment drains and shim bleed flows have an activity level equivalent to primary coolant activity (PCA).

The combined shim bleed and equipment drain wastes streams are routed to one of the recycle holdup tanks. The contents of the recycle holdup tank are then processed through the radwaste processing facility filtration system and the radwaste processing facility demineralizers. The water is either pumped to the reactor makeup water storage tank for reuse in the plant or to a waste monitor tank for monitoring and discharge. The BRS has sufficient capacity to allow total reuse of the combined shim bleed and equipment drain wastes.

Radioactive decay during collection in the recycle holdup tanks is calculated using a collection time of 22.4 days. This value is based upon filling one of the recycle holdup tanks to 80 percent of capacity using the combined shim bleed and reactor coolant drain tank flows. Radioactive decay during processing and discharge is calculated using a process time of 0.31 days. This value is based upon processing the combined shim bleed and reactor coolant drain tank flows at the design flowrate of the recycle evaporator.

The decontamination factors used in calculating radionuclide removal for iodine, cesium, rubidium, and other nuclides, are determined by applying the methodology and parameters of reference 1 to the processing capabilities of the BRS and CVCS as shown in figure 11.2.3-1. No credit is taken for the recycle evaporation condensate demineralizer since its main function is to remove boron carryover.

11.2.3.2.2 Liquid Waste Processing System (LWPS)

A. Clean Wastes (Drain Channel A - Miscellaneous Wastes)

Clean wastes are collected in the waste holdup tank for processing. Based on sample analysis, the water in the monitor tank would either be discharged or processed further by recirculation through the waste monitor tank demineralizer until sample analysis indicated that it was acceptable for discharge. The flow to the waste holdup tank is 713 gal/day at 0.051 times PCA. The LWPS has sufficient capacity to allow total reuse of the processed clean wastes. However,

in the release calculations a discharge fraction of 1.0 is used as specified by reference 1.

Radioactive decay during collection in the waste holdup tank is calculated using a collection time of 5.6 days. This value is based upon filling the waste holdup tank to 40 percent of capacity. Radioactive decay during processing and discharge is calculated using a process time of 0.028 days. This value is based upon processing 40 percent of the waste holdup tank capacity at the design flowrate of the waste evaporator (no longer in system) and the time required to pump this volume of water out of the monitor tank at the maximum pumping rate.

The decontamination factors used in calculating radionuclide removal are based upon demineralization as shown in figure 11.2.3-1.

B. Dirty Wastes (Drain Channel B - Miscellaneous Wastes)

Dirty wastes are collected in the floor drain tank. A sample is analyzed to determine whether the water can be discharged without processing. If cleanup is required, the liquid is processed by the radwaste processing facility filtration system and/or the radwaste processing facility demineralizer and the effluent directed to a waste monitor tank. Based on sample analysis, the monitor tank contents are either discharged or processed further by recirculation through the monitor tank demineralizer until sample analysis indicates that the water is acceptable for discharge. The flow to the floor drain tank is 2047 gal/day at 0.02 times PCA. Since all of the dirty wastes are normally discharged, a discharge fraction of 1.0 is used in the release calculations.

Radioactive decay during collection in the floor drain tank is calculated using a collection time of 1.43 days. This value is based upon filling the floor drain tank to 40 percent of capacity. Radioactive decay during processing is calculated using process time of 0.03 days. This value is based upon processing 40 percent of the floor drain tank capacity at the design flowrate of the waste evaporator (no longer in service) and the time required to pump this volume of water out of the monitor tank at the maximum pumping rate.

The decontamination factors used in calculating radionuclide removal are based upon the decontamination factors given in reference 1 for demineralizers as shown in figure 11.2.3-1.

C. Detergent Wastes (Laundry and Hot Shower Tank)

Detergent wastes are normally released without treatment. The releases through this path are assumed to be the same as listed in reference 1, table 2-20.

11.2.3.2.3 Steam Generator Blowdown Processing System (SGBPS)

Blowdown from the steam generators is normally processed by the two generator blowdown demineralizers (in series) and recycled back to the main condenser. If discharge of blowdown is desired, the demineralizers can be bypassed. A valve in the discharge line automatically stops discharge on a high radioactivity signal. In the event of a primary to secondary side leak, the demineralizers would be in use. For the release calculation, a 100 lb/day primary to secondary side leakage is assumed. Also, a design basis total blowdown rate of 180,096 lb/h is assumed with 100 percent of the flow being discharged.

No credit is taken for radioactive decay of the isotopes in the blowdown stream since the SGBPS design utilizes a through-flow process with no significant decay.

The decontamination factors used in calculating radionuclide removal are shown in figure 11.2.3-1. These values are based upon the decontamination factors given in reference 1 for two steam generator blowdown mixed bed demineralizers (in series).

11.2.3.2.4 Turbine Building Floor Drains

The processing of the low level radioactive water in the turbine building dirty drain tank is manually initiated by the operator based on the predetermined water level in the tank. This processing system, located in the auxiliary building, consists of an oil water separator, an activated charcoal filter, two demineralizers in series, and a discharge filter. The treated water is returned to the clean drain tank in the turbine building. After monitoring it is combined with other waste streams for disposal.

11.2.3.3 Release Points

The natural draft cooling towers are not radioactive release points. However, the following information regarding location is offered.

Blowdown from both cooling towers and flow from the waste water retention basin is collected in a common blowdown sump and is discharged to the river via the waste water effluent pipe. The radioactive release line discharges into the waste water effluent pipe at a point downstream of the blowdown sump and is then discharged into the Savannah River downstream of the river intake structure.

11.2.3.4 Dilution Factors

At 100-percent capacity factor and design basis conditions for cooling tower operation, blowdown from the cooling tower to the blowdown sump is approximately 5000 gal/min per unit. Furthermore, 10,000gal/min per unit river water can be provided to the sump through a dilution flow transfer line takeoff on the river water makeup line to the natural draft cooling towers. Thus, a total of 15,000 gal/min per unit dilution flow is available during the liquid radwaste discharge operation. Minimum dilution flow is as specified in the release permit which is prepared in accordance with the ODCM.

There will be some additional dilution following discharge due to the effect of the near field mixing zone in the Savannah River. For 15,000 gal/min (33.4 ft^3/s) effluent discharge into the Savannah River with 5800 ft^3/s minimum flow, the VEGP thermal plume analysis utilizes a dilution factor of 10 for summer discharge conditions and 20 for winter conditions.[2]

For conservatism, the lower dilution factor of 10 was chosen for the LADTAP II analysis.

11.2.3.5 Estimated Doses

Release of radioactive effluents to the Savannah River during normal plant operation and anticipated operational occurrences will result in a minimal radiological exposure to individuals as noted in table 11.2.3-4. The estimated annual average doses to the maximum exposed individual were calculated using the LADTAP II computer code.[3]

The location of maximum exposure is in that area of the river allowing for initial near field dilution of the discharge with essentially zero travel time. Since crop irrigation from the Savannah River has not been observed in the vicinity of the plant site, this pathway has not been considered in the evaluation of doses. The anticipated dose due to drinking water could be considered insignificant because the nearest location of potable use of river water is in Beaufort County, South Carolina, approximately 103 river miles downstream of the plant site. However, for conservatism, this pathway was evaluated assuming the maximum exposed individual obtains drinking water in the immediate area of the discharge plume. Shoreline use in the vicinity of the plant is very limited, with essentially no fishing from the bank, swimming, or sunbathing, and, consequently, is expected to be an insignificant pathway in comparison with the pathway of aquatic foods. Nevertheless, for purposes of conservatism, this pathway has been included in the evaluation of doses for the maximum exposed individual.

Furthermore, a single dilution factor (10.0) for the initial near field dilution of the discharge with a travel time of zero was used for the pathway's consideration. In lieu of site specific data, the pathway usage factors and shorewidth factor (0.2) outlined in Regulatory Guide 1.109, Revision 1, October 1977, were utilized in this evaluation. As noted in table 11.2.3-4, based on the criteria as outlined above, the maximum exposed individual annual doses from the discharge of radioactive materials in liquid effluent from each of the VEGP units meet the guideline of Appendix I, 10 CFR 50, and Docket RM-50-2, Annex to Appendix I.

11.2.3.6 References

1. U.S. Nuclear Regulatory Commission, "Calculation of Releases from Pressurized Water Reactors," PWR-GALE Computer Code, NUREG-0017, April 1976.
2. Georgia Power Company, "Waste Water Effluent Discharge Structure Thermal Plume Analysis," Vogtle Electric Generating Plant Units 1 and 2, Revised March 1981.
3. U.S. Nuclear Regulatory Commission, "Calculation of Radiation Exposure to Man from Routine Release of Nuclear Reactor Liquid Effluents," LADTAP II Computer Code, NUREG/CR-1276, March 1980.

TABLE 11.2.1-1 (SHEET 1 OF 2)

PARAMETERS USED IN THE CALCULATION OF ESTIMATED ACTIVITY IN LIQUID WASTES

Collection Tank and Sources	Expected Input Rates[a]	Activity[b]	Basis	Disposition
1. Reactor coolant drain tank	300 gal/day	1.0 x (RCS) tritium; 0.1 x (RCS) other nuclides	RC pump No. 2 seal leakoff of 0.05 gal/min per pump	Directed to BRS recycle holdup tanks for process and transfer to the waste monitor tank
2. Waste holdup tank	913 gal/day	0.051 x (RCS)		Processed and transferred to the waste monitor tank
a. Equipment drains	156 gal/day	0.1 x (RCS)	Drains from tanks, filters, demineralizers, and heat exchangers containing reactor grade water	
b. Containment sump[c]	530 gal/day	0.036 x (RCS)	ANSI/ANS-55.6-1979	
c. Sample room drains	27 gal/day	0.05 x (RCS)	(d)	
3. Floor drain tank	2560 gal/day (normal) 5520 gal/day (shutdown)			Processed and transferred to the waste monitor tank
a. Auxiliary building sump	200 gal/day	0.1 x (RCS)	NUREG-0017	
b. Miscellaneous	700 gal/day	0.01 x (RCS)	NUREG-0017	
c. Lab equipment rinse water	164 gal/day	0.05 x (RCS)	(d)	
d. Area decontamination	40 gal/day (normal) 3000 gal/day (shutdown)	0.01 x (RCS) 0.01 x (RCS)	ANSI/ANS-55.6-1979	
e. Spent fuel pool liner leakage	700 gal/day	0.001 x (RCS)	ANSI/ANS-55.6-1979	
4. Chemical drain tank	9 gal/day	0.148 x (RCS)	(d)	Drained to auxiliary building sumps

TABLE 11.2.1-1 (SHEET 2 OF 2)

Collection Tank and Sources	Expected Input Rates[a]	Activity[b]	Basis	Disposition
5. Laundry and hot shower tank	500 gal/day (normal) 3900 gal/day (shutdown)	(f) (f)		Discharged to environment after sampling
a. Laundry waste	300 gal/day (normal) 2000 gal/day (shutdown)	(f) (f)	ANSI/ANS-55.6-1979 ANSI/ANS-55.6-1979	
b. Hot shower waste	Negligible (normal) 400 gal/day (shutdown)	(f) (f)	ANSI/ANS-55.6-1979 ANSI/ANS-55.6-1979	
c. Hand washes	200 gal/day (normal) 1500 gal/day (shutdown)	(f) (f)	ANSI/ANS-55.6-1979 ANSI/ANS-55.6-1979	
6. Boron recycle system	500 gal/day	(g)	Two RCS volumes purged per year for tritium control.	Discharged to environment after sampling

a. Input rates are for one unit. These rates do not represent system capacity. During operation of both units, shared tanks (chemical drain tank and the laundry and hot shower tank) have double the indicated input rates.

b. Activity of the liquid wastes entering the LWPS is given in terms of a fraction of primary coolant activity as given in section 11.1 unless stated otherwise.

c. Each sump may be directed to either the waste holdup or floor drain tank, depending upon water quality.

d. Expected input rates are based on a total rate of 200 gal/day in accordance with ANSI/ANS-55.6-1979, Reactor Systems Sampling. The total rate is divided among sample room drains, lab rinse water, and chemically contaminated samples.

e. Shutdown is assumed to last 30 days.

f. Laundry and hot shower waste activity is taken from NUREG-0017, April 1976.

g. The 500 gal/day coming from the boron recycle system is a fraction of the liquid processed by that system. The activity of the water is primary coolant activity in accordance with section 11.1 but reduced by the processing performed by the boron recycle system.

TABLE 11.2.1-2 (SHEET 1 OF 8)

EQUIPMENT DESIGN PARAMETERS

Pumps

RCDT pumps	
Number	2 (per unit)
Type	Canned centrifugal
Design pressure (psig)	150
Design temperature (°F)	200
Design flow (gal/min)	100/140[(a)]
Design head (ft)	300/250[(a)]
Material	Stainless steel (SS)
Waste evaporator feed pump	
Number	1 (per unit)
Type	Canned centrifugal
Design pressure (psig)	150
Design temperature (°F)	200
Design flow (gal/min)	35/100[(a)]
Design head (ft)	250/200[(a)]
Material	SS
Waste evaporator condensate pump	
Number	1 (per unit)
Type	Canned centrifugal
Design pressure (psig)	150
Design temperature (°F)	200
Design flow (gal/min)	35/100[(a)]
Design head (ft)	250/200[(a)]
Material	SS
Chemical drain tank pump	
Number	1 (shared)
Type	Canned centrifugal
Design pressure (psig)	150
Design temperature (°F)	200
Design flow (gal/min)	35/100[(a)]
Design head (ft)	250/200[(a)]
Material	SS

TABLE 11.2.1-2 (SHEET 2 OF 8)

Spent resin sluice pump	
Number	1 (per unit)
Type	Canned centrifugal
Design pressure (psig)	240
Design temperature (°F)	200
Design flow (gal/min)	100/140[(a)]
Design head (ft)	300/250[(a)]
Material	SS
Laundry and hot shower tank pump	
Number	1 (shared)
Type	Mechanical seal, horizontal centrifugal
Design pressure (psig)	150
Design temperature (°F)	200
Design flow (gal/min)	35/100[(a)]
Design head (ft)	250/200[(a)]
Material	SS
Floor drain tank pump	
Number	1 (per unit)
Type	Mechanical seal, horizontal centrifugal
Design pressure (psig)	150
Design temperature (°F)	200
Design flow (gal/min)	35/100[(a)]
Design head (ft)	250/200[(a)]
Material	SS
Waste monitor tank pumps	
Number	2 (per unit)
Type	Canned centrifugal
Design pressure (psig)	150
Design temperature (°F)	200
Design flow (gal/min)	35/100[(a)]
Design head (ft)	250/200[(a)]
Material	SS

TABLE 11.2.1-2 (SHEET 3 OF 8)

Filters

Waste evaporator feed filter[h]	
Number	1 (per unit)
Waste evaporator condensate filter	
Number	1 (per unit)
Type	Disposable cartridge
Design pressure, (psig)	150[b]
Design temperature (°F)	200[b]
Design flow (gal/min)	35
Wp at design flow (psi)	5
Size of particles, 98% retention (μm) (nominal)	25
Materials	
Housing	SS
Filter element	Epoxy impregnated cellulose fiber
Spent resin sluice filter	
Type	Backflushable
Number	1 (per unit)
Design pressure (psig)	375
Design temperature (°F)	200
Design flow (gal/min)	150
Wp at design flow (psi) (unfouled)	5
Size of particles, 98% retention (μm) (nominal)	25
Materials	
Housing	SS
Filter element	Epoxy impregnated cellulose fiber or fine diameter 304L SS wires
Laundry and hot shower tank filter	
Number	1
Type	Disposable cartridge
Design pressure (psig)	150[b]
Design temperature (°F)	200[b]
Design flow (gal/min)	35
Wp at design flow (psi) (unfouled)	5
Size of particles, 98% retention (μm) (nominal)	25
Materials	
Housing	SS
Filter element	Epoxy impregnated[h] cellulose fiber

TABLE 11.2.1-2 (SHEET 4 OF 8)

Floor drain tank filter[(h)]	
Number	1 (per unit)
Type	Backflushable
Design pressure (psig)	375
Design temperature (°F)	200
Design flow (gal/min)	35
Wp at design flow (psi) (unfouled)	5
Size of particles, 98% retention (μm) (nominal)	25
Materials	
Housing	SS
Filter element	Chemically etched SS discs
Waste monitor tank filter	
Type	Backflushable
Number	1 (per unit)
Design pressure (psig)	375
Design temperature (°F)	200
Design flow (gal/min)	35
Wp at design flow (psi) (unfouled)	5
Size of particles, 98% retention (μm) (nominal)	25
Materials	
Housing	SS
Filter element	Chemically etched SS discs
<u>Strainers</u>	
Laundry and hot shower tank strainer	
Number	1 (shared)
Type	Basket
Design pressure (psig)	150
Design temperature (°F)	200
Design flow (gal/min)	35
Δp at design flow (psi)	Negligible
Strainer mesh number	40
Materials	SS
Floor drain tank strainer	
Number	1 (per unit)
Type	Basket
Design pressure (psig)	150
Design temperature (°F)	200
Design flow (gal/min)	35
Δp at design flow (psi)	Negligible
Strainer mesh number	40
Materials	SS

TABLE 11.2.1-2 (SHEET 5 OF 8)

Heat Exchanger

RCDT heat exchanger

Number	1
Type	U-tube
Estimated UA[(d)] (Btu/h/°F)	70,000

	Shell Side	Tube Side
Design pressure (psig)	150	240
Design temperature (°F)	250	200
Design flow (lb/h)	112,000	44,600
Temperature inlet (°F)	105	180
Temperature outlet (°F)	125	130
Material	Carbon steel	SS

Demineralizers

Waste evaporator condensate demineralizer

Number	1 (per unit)
Type	Flushable
Design pressure (psig)	150[(b)]
Design temperature (°F)	200[(b)]
Design flow (gal/min)	35
Resin volume (ft^3)	30[(e)]
Material	SS
Process decontamination factors©	100 for iodine; 2 for cesium; 100 for all other nuclides[(f)]

Waste monitor tank demineralizer

Number	1 (per unit)
Type	Flushable
Design pressure (psig)	150[(b)]
Design temperature (°F)	200[(b)]
Design flow (gal/min)	35
Resin volume (ft^3)	30[(e)]
Material	SS
Process decontamination factors[(c)]	100 for iodine; 2 for cesium; 100 for all other nuclides[(f)]

TABLE 11.2.1-2 (SHEET 6 OF 8)

Tanks

RCDT	
Number	1 (per unit)
Usable volume (gal)	350
Type	Horizontal
Internal design pressure (psig)	100
External design pressure (psig)	60
Design temperature (°F)	250
Material	SS
Waste holdup tank	
Number	1 (per unit)
Usable volume (gal)	10,000
Type	Vertical
Design pressure (psig)	Atmospheric
Design temperature (°F)	200
Material	SS
Waste evaporator condensate tank	
Number	1 (per unit)
Usable volume (gal)	5000
Type	Vertical with diaphragm
Design pressure (psig)	Atmospheric
Design temperature (°F)	200
Material	SS
Chemical drain tank	
Number	1 (shared)
Usable volume (gal)	600
Type	Vertical
Design pressure (psig)	Atmospheric
Design temperature (°F)	200
Material	SS
Spent resin storage tank	
Number	1 (per unit)
Usable volume (ft^3)(g)	350
Type	Vertical
Design pressure (psig)	150
Design temperature (°F)	200
Material	SS

TABLE 11.2.1-2 (SHEET 7 OF 8)

Laundry and hot shower tank	
Number	1 (shared)
Usable volume (gal)	10,000
Type	Vertical
Design pressure (psig)	Atmospheric
Design temperature (°F)	200
Material	SS
Floor drain tank	
Number	1 (per unit)
Usable volume (gal)	10,000
Type	Vertical
Design pressure (psig)	Atmospheric
Design temperature (°F)	200
Material	SS
Waste monitor tank	
Number	2 (per unit)
Usable volume (gal)	5000
Type	Vertical
Design pressure (psig)	Atmospheric
Design temperature (°F)	200
Material	SS
Auxiliary waste monitor tank	
Number	2 (shared)
Nominal volume (gal)	20,000
Type	Vertical
Design pressure (psig)	Atmospheric
Design temperature (°F)	200
Material	SS

a. Denotes two design conditions.

b. These are minimum conditions required. Design conditions supplied are 300 psig and 250°F.

c. Decontamination factors are taken from "Calculation of Releases of Radioactive Materials in Gaseous and Liquid Effluents from Pressurized Water Reactors (PWR-GALE Code)," Office of Standards Development, U.S. Nuclear Regulatory Commission, NUREG-0017, April 1976.

TABLE 11.2.1-2 (SHEET 8 OF 8)

d. U is the overall heat transfer coefficient. A is the heat transfer area.

e. This is the design maximum loading. Smaller volumes may be used based on operational needs and type of resin used. Media loaded into the demineralizer are evaluated through the VEGP Chemical Control Program.

f. The process decontamination factor is 10 for iodine, cesium, and all other nuclides when used as a second demineralizer in a series.

g. Total for resin and liquid.

h. The internals have been removed from these filters. They remain in place as part of the piping system.

TABLE 11.2.1-3 (SHEET 1 OF 10)

ESTIMATED LIQUID WASTE PROCESSING SYSTEM ISOTOPIC INVENTORIES AND CONCENTRATIONS[a]

Line Number	Liquid Process Line Description	Annual Average Flowrate (gal/day)	Isotopic Concentration (μCi/g)				
			Cr-51	Mn-54	Fe-55	Fe-59	Co-58
1	Reactor coolant pump seal leakoff to reactor coolant drain tank	300	1.90E-04	3.10E-05	1.60E-04	1.00E-04	1.60E-03
2	Reactor coolant drain tank outflow	300	1.88E-04	3.10E-05	1.60E-04	9.94E-05	1.59E-03
3	Drain header flow to waste holdup tank	222	1.64E-04	2.23E-05	2.67E-04	1.07E-04	1.66E-03
4	Sample sink flow to waste holdup tank	27	9.50E-05	1.55E-05	8.00E-05	5.00E-05	8.00E-04
5	Sum of lines 3 and 4	249	1.57E-04	2.16E-05	2.47E-04	1.01E-05	1.57E-03
6	Flow to waste holdup tank from containment sump	530	6.84E-05	1.12E-05	5.76E-05	3.60E-05	5.76E-04
12	Floor drain flow to floor drain tank	1137	1.66E-05	2.71E-06	1.40E-05	8.73E-06	1.40E-04
13	Building sump flow to floor drain tank	900	4.37E-05	7.13E-06	3.68E-05	2.30E-05	3.68E-04
14	Sum of lines 12 and 13	2037	2.86E-05	4.66E-06	2.41E-05	1.50E-05	2.41E-04
15	Lab rinse water flow to floor drain tank	164	9.50E-05	1.55E-05	8.00E-05	5.00E-05	8.00E-04
16	Flow from floor drain tank	2800	2.66E-05	4.35E-06	2.24E-05	1.40E-05	2.25E-04
19	Waste monitor tank demineralizer outlet flow	2800	2.66E-07	4.35E-08	2.24E-07	1.40E-07	2.25E-06
20	Flow from boron recycle system	500	1.57E-08	3.25E-09	1.71E-08	9.10E-09	1.55E-07
21	Sum of lines 19 and 20	3300	2.26E-07	3.90E-08	1.93E-07	1.20E-07	1.73E-06
22	Discharge flow from waste monitor tank No. 2	3300	2.26E-07	3.90E-08	1.93E-07	1.20E-07	1.93E-06
23	Flow of contaminated samples to chemical drain tank	9	2.81E-04	4.59E-05	2.37E-04	1.48E-04	2.37E-03
24	Chemical drain tank outflow	9	2.39E-04	4.52E-05	2.36E-04	1.34E-04	2.22E-03
25	Flow to laundry and hot shower tank	500	0.00E+00	1.61E-06	0.00E+00	0.00E+00	6.44E-06
26	Discharge from waste monitor tank No. 1	500	0.00E+00	1.61E-06	0.00E+00	0.00E+00	6.44E-06
27	Sum of lines 22 and 26	3800	1.96E-07	2.46E-07	1.68E-07	1.04E-07	2.53E-06

TABLE 11.2.1-3 (SHEET 2 OF 10)

Isotopic Concentration (μCi/g)

Line Number	Co-60	Br-83	Br-84	Br-85	Rb-86	Rb-88	Sr-89	Sr-90	Y-90
1	2.00E-04	5.40E-04	3.00E-04	3.40E-05	4.45E-05	1.15E-01	3.60E-05	1.00E-06	1.20E-07
2	2.00E-04	1.38E-04	2.11E-05	2.42E-07	4.38E-05	4.68E-03	3.58E-05	1.00E-06	2.07E-07
3	3.22E-04	3.80E-04	2.11E-04	2.39E-05	6.30E-06	1.62E-02	3.84E-05	1.83E-06	5.26E-07
4	1.00E-04	2.70E-04	1.50E-04	1.70E-05	4.45E-06	1.15E-02	1.80E-05	5.00E-07	6.00E-08
5	2.98E-04	3.68E-04	2.04E-04	2.32E-05	6.10E-06	1.57E-02	3.62E-05	1.69E-06	4.75E-07
6	7.20E-05	1.94E-04	1.08E-04	1.22E-05	3.20E-06	8.28E-03	1.30E-05	3.60E-07	4.32E-08
12	1.75E-05	4.67E-05	2.59E-05	2.94E-06	7.77E-07	2.00E-03	3.14E-06	8.73E-08	1.09E-08
13	4.60E-05	1.24E-04	6.90E-05	7.82E-06	2.05E-06	5.29E-03	8.28E-06	2.30E-07	2.76E-08
14	3.01E-05	8.09E-05	4.49E-05	5.10E-06	1.34E-06	3.45E-03	5.41E-06	1.50E-07	1.83E-08
15	1.00E-04	2.70E-04	1.50E-04	1.70E-05	4.45E-06	1.15E-02	1.80E-05	5.00E-07	6.00E-08
16	2.80E-05	7.47E-05	4.15E-05	4.70E-06	1.25E-06	6.61E-04	5.05E-06	1.40E-07	1.77E-08
19	2.80E-07	7.47E-07	4.15E-07	4.70E-08	6.25E-07	3.31E-04	5.05E-08	1.40E-09	1.77E-10
20	2.15E-08	3.32E-10	3.95E-11	4.18E-13	1.62E-08	4.22E-08	3.35E-09	1.08E-10	9.17E-11
21	2.41E-07	6.33E-07	7.52E-07	4.00E-08	5.33E-07	2.81E-04	4.33E-08	1.20E-09	1.64E-10
22	2.41E-07	6.33E-07	7.52E-07	4.00E-08	5.33E-07	2.81E-04	4.33E-08	1.20E-09	1.64E-10
23	2.96E-04	7.99E-04	4.44E-04	5.03E-05	1.32E-05	3.40E-02	5.33E-05	1.48E-06	1.78E-07
24	2.95E-04	8.64E-06	1.06E-06	1.13E-08	1.04E-05	4.55E-05	4.88E-05	1.48E-06	1.12E-06
25	1.45E-05	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
26	1.45E-05	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
27	2.12E-06	5.50E-07	3.06E-07	3.47E-08	4.63E-07	2.44E-04	3.76E-08	1.04E-09	1.41E-10

TABLE 11.2.1-3 (SHEET 3 OF 10)

Isotopic Concentration (μCi/g)

Line Number	Sr-91	Y-91m	Y-91	Y-93	Zr-95	Nb-95	Mo-99	Tc-99m	Ru-103
1	7.00E-05	4.10E-05	6.50E-06	3.70E-06	6.10E-06	5.10E-06	8.70E-03	5.20E-03	4.60E-06
2	4.07E-05	2.59E-05	6.68E-06	2.19E-06	6.07E-06	5.11E-06	7.88E-03	6.64E-03	4.57E-06
3	4.94E-05	2.89E-05	4.76E-06	2.60E-06	4.41E-06	3.62E-06	6.12E-03	3.65E-03	1.15E-05
4	3.50E-05	2.05E-05	3.25E-06	1.85E-06	3.05E-06	2.55E-06	4.35E-03	2.60E-03	2.30E-06
5	4.78E-05	2.80E-05	4.60E-06	2.52E-05	4.26E-06	3.50E-06	5.93E-03	3.54E-03	1.05E-05
6	2.52E-05	1.48E-05	2.34E-06	1.33E-06	2.20E-06	1.84E-06	3.13E-03	1.87E-03	1.66E-06
12	6.06E-06	3.55E-06	5.68E-07	3.20E-07	5.33E-07	4.45E-07	7.56E-04	4.93E-04	4.02E-07
13	1.61E-05	9.43E-06	1.49E-06	8.51E-07	1.40E-06	1.17E-06	2.00E-03	1.20E-03	1.06E-06
14	1.05E-05	6.15E-06	9.75E-07	5.55E-07	9.16E-07	7.65E-07	1.31E-03	8.05E-04	6.92E-07
15	3.50E-05	2.05E-05	3.25E-06	1.85E-06	3.05E-06	2.55E-06	4.35E-03	2.60E-03	2.30E-06
16	9.66E-06	5.68E-06	9.10E-07	5.12E-07	8.56E-07	7.16E-07	1.21E-03	8.20E-04	6.45E-07
19	9.66E-08	5.68E-08	9.10E-09	5.12E-09	8.56E-09	7.16E-09	1,21E-05	8.20E-06	6.45E-05
20	1.84E-10	1.18E-10	6.37E-10	1.02E-11	5.84E-10	5.60E-10	1.66E-07	1.75E-07	4.10E-10
21	8.20E-08	4.85E-08	7.82E-07	4.34E-09	7.37E-09	6.15E-09	1.03E-05	6.97E-06	5.52E-04
22	8.20E-08	4.85E-08	7.82E-07	4.34E-09	7.37E-09	6.15E-09	1.03E-05	6.97E-06	5.52E-04
23	1.04E-04	6.07E-05	9.62E-06	5.48E-06	9.03E-06	7.55E-06	1.29E-02	7.70E-03	6.81E-06
24	4.54E-06	2.91E-06	9.20E-06	2.52E-07	8.42E-06	7.68E-06	3.75E-03	3.95E-03	6.07E-06
25	0.00E+00	0.00E+00	0.00E+00	0.00E+00	2.25E-06	3.22E-06	0.00E+00	0.00E+00	2.25E-07
26	0.00E+00	0.00E+00	0.00E+00	0.00E+00	2.25E-06	3.22E-06	0.00E+00	0.00E+00	2.25E-07
27	7.12E-08	4.21E-08	6.79E-09	3.77E-09	3.03E-07	4.29E-07	8.94E-06	6.05E-06	3.44E-08

TABLE 11.2.1-3 (SHEET 4 OF 10)

Isotopic Concentration (μCi/g)

Line Number	Rh103m	Ru-106	Rh-106	Te125m	Te127m	Te-127	Te129m	Te-129	I-130
1	5.10E-06	1.00E-06	1.10E-06	2.90E-06	2.80E-05	9.20E-05	1.40E-04	1.80E-04	2.30E-04
2	4.63E-06	9.99E-07	9.99E-07	2.89E-06	2.79E-05	6.44E-05	1.39E-04	1.02E-04	1.48E-04
3	3.60E-06	7.16E-07	1.11E-06	2.05E-06	1.99E-05	6.49E-05	9.87E-05	1.29E-04	1.63E-04
4	2.55E-06	5.00E-07	5.50E-07	1.45E-06	1.40E-05	4.60E-05	7.00E-05	9.00E-05	1.15E-04
5	3.49E-06	6.93E-07	1.05E-06	1.98E-06	1.93E-05	6.29E-05	9.56E-05	1.25E-04	1.58E-04
6	1.84E-06	3.60E-07	3.96E-07	1.04E-06	1.01E-05	3.31E-05	5.04E-05	6.48E-05	8.28E-05
12	4.45E-07	8.73E-07	9.60E-08	2.53E-07	2.45E-06	7.98E-06	1.22E-05	1.56E-05	1.99E-05
13	1.17E-06	2.30E-07	2.53E-07	6.67E-07	6.44E-06	2.12E-05	3.22E-05	4.14E-05	5.29E-05
14	7.65E-07	1.50E-07	1.65E-07	4.36E-07	4.21E-06	1.38E-05	2.10E-05	2.70E-05	3.45E-05
15	2.55E-06	5.00E-07	5.50E-07	1.45E-06	1.40E-05	4.60E-05	7.00E-05	9.00E-05	1.15E-04
16	7.14E-07	1.59E-07	1.52E-07	4.07E-07	3.93E-06	1.28E-05	1.96E-05	2.51E-05	3.19E-05
19	7.14E-09	1.59E-09	1.52E-09	4.07E-09	3.93E-06	1.28E-07	1.96E-07	2.51E-05	3.19E-07
20	4.10E-10	1.05E-10	1.05E-10	2.74E-10	2.81E-09	2.96E-09	1.21E-08	7.78E-09	7.86E-09
21	6.13E-06	1.37E-09	1.30E-09	3.49E-09	3.38E-08	1.09E-07	1.68E-07	2.14E-07	2.72E-07
22	6.13E-06	1.37E-09	1.30E-09	3.49E-09	3.38E-08	1.09E-07	1.68E-07	2.14E-07	2.72E-07
23	7.55E-06	1.48E-06	1.63E-06	4.29E-06	4.14E-05	1.36E-04	2.07E-04	2.66E-04	3.40E-04
24	6.08E-06	1.46E-06	1.46E-06	3.97E-06	3.97E-05	4.35E-05	1.82E-04	1.17E-04	1.92E-05
25	0.00E+00	3.86E-06	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
26	0.00E+00	3.86E-06	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
27	5.32E-06	5.09E-07	1.13E-09	3.03E-09	2.94E-08	9.47E-08	1.46E-07	1.86E-07	2.36E-07

TABLE 11.2.1-3 (SHEET 5 OF 10)

Isotopic Concentration (μCi/g)

Line Number	Te131m	Te-131	I-131	Te-132	I-132	I-133	I-134	I-135	Cs-134
1	2.60E-04	1.20E-04	2.80E-02	2.80E-03	1.10E-02	4.00E-02	5.30E-03	2.10E-02	1.30E-02
2	2.11E-04	4.26E-05	2.70E-02	2.57E-03	4.66E-03	2.99E-02	5.92E-04	1.03E-02	1.30E-02
3	1.83E-04	8.43E-04	2.10E-02	1.97E-03	7.74E-03	2.84E-02	3.43E-03	1.48E-02	2.04E-03
4	1.30E-04	6.00E-05	1.40E-02	1.40E-03	5.50E-03	2.00E-02	2.65E-03	1.05E-02	1.30E-03
5	1.77E-04	8.17E-05	2.02E-02	1.91E-03	7.50E-03	2.75E-02	3.61E-03	1.43E-02	1.96E-03
6	9.36E-05	4.32E-05	1.01E-02	1.01E-03	3.96E-03	1.44E-02	1.91E-03	7.56E-03	9.36E-04
12	2.25E-05	1.04E-05	2.44E-03	2.44E-04	9.53E-03	3.47E-03	4.58E-04	1.82E-03	2.27E-04
13	5.98E-05	2.76E-05	6.44E-03	6.44E-04	2.53E-03	9.20E-03	1.22E-03	4.83E-03	5.98E-04
14	3.90E-05	1.80E-05	4.21E-03	4.21E-04	1.65E-03	6.00E-03	7.95E-04	3.15E-03	3.91E-04
15	1.30E-04	6.00E-05	1.40E-02	1.40E-03	5.50E-03	2.00E-02	6.65E-03	1.05E-02	1.30E-03
16	3.61E-05	1.66E-05	3.92E-03	3.91E-04	1.52E-03	1.54E-03	7.33E-04	2.90E-03	3.65E-44
19	3.61E-07	1.66E-07	3.92E-05	3.91E-06	1.52E-05	1.54E-05	7.33E-06	2.90E-05	1.83E-04
20	2.19E-09	4.07E-10	1.33E-05	6.18E-08	6.84E-07	2.31E-06	1.17E-08	3.73E-07	6.93E-06
21	3.07E-07	1.41E-07	3.53E-05	3.33E-06	1.30E-05	1.34E-05	6.22E-06	2.47E-05	1.56E-04
22	3.07E-07	1.41E-07	3.53E-05	3.33E-06	1.30E-06	1.34E-05	6.22E-06	2.47E-05	1.56E-04
23	3.85E-04	1.78E-04	4.14E-02	4.14E-03	1.63E-02	5.92E-02	7.84E-03	3.11E-02	3.85E-03
24	5.20E-05	9.69E-06	2.47E-02	1.37E-03	1.54E-03	5.55E-03	3.11E-05	9.34E-04	3.82E-03
25	0.00E+00	0.00E+00	9.66E-09	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	2.09E-05
26	0.00E+00	0.00E+00	9.66E-09	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	2.09E-05
27	2.67E-07	1.22E-07	3.05E-05	2.89E-06	1.13E-05	1.16E-05	5.40E-06	2.14E-05	1.38E-04

TABLE 11.2.1-3 (SHEET 6 OF 10)

Isotopic Concentration (μCi/g)

Line Number	Cs-136	Cs-137	Ba137m	Ba-140	La-140	Ce-141	Ce-143	Pr-143	Ce-144
1	7.00E-03	9.50E-03	1.80E-03	2.20E-05	1.60E-05	7.10E-06	4.20E-06	5.10E-06	3.30E-06
2	6.85E-03	9.50E-03	8.84E-03	2.15E-05	1.68E-05	7.04E-06	3.47E-06	5.07E-06	3.30E-06
3	9.94E-04	1.51E-03	1.43E-03	1.74E-05	2.08E-05	5.01E-06	2.95E-06	3.59E-06	2.36E-06
4	7.00E-04	9.50E-04	9.00E-04	1.10E-05	8.00E-06	3.55E-06	2.10E-06	2.55E-06	1.65E-06
5	9.62E-04	1.45E-03	1.37E-03	1.67E-05	1.94E-05	4.85E-06	2.86E-06	3.48E-06	2.28E-06
6	5.04E-04	6.84E-04	6.48E-04	7.92E-06	5.76E-06	2.56E-06	1.51E-06	1.84E-06	1.19E-06
12	1.22E-04	1.66E-04	1.57E-04	1.92E-06	1.40E-06	6.20E-07	3.64E-07	4.45E-07	2.88E-07
13	3.22E-04	4.37E-04	4.14E-04	5.06E-06	3.68E-06	1.63E-06	9.66E-07	1.17E-06	7.59E-07
14	2.10E-04	2.86E-04	2.71E-04	3.31E-06	2.41E-06	1.07E-06	6.30E-07	7.65E-07	4.96E-07
15	7.00E-04	9.50E-04	9.00E-04	1.10E-05	8.00E-06	3.55E-06	2.10E-06	2.55E-06	1.65E-06
16	1.96E-04	2.67E-04	2.53E-04	3.08E-06	2.25E-06	9.95E-07	5.84E-07	7.15E-07	4.63E-07
19	9.80E-05	1.34E-04	2.53E-06	3.08E-08	2.25E-08	9.95E-09	5.84E-09	7.15E-09	4.63E-09
20	2.19E-06	5.11E-06	4.78E-06	1.37E-09	1.47E-09	6.07E-10	3.90E-11	3.52E-10	3.46E-10
21	8.35E-05	1.14E-04	2.87E-06	2.63E-08	1.93E-08	8.53E-09	4.96E-09	6.12E-09	3.98E-09
22	8.35E-05	1.14E-04	2.87E-06	2.63E-08	1.93E-08	8.53E-09	4.96E-09	6.12E-09	3.98E-09
23	2.07E-03	2.81E-03	2.66E-03	3.26E-08	2.37E-05	1.05E-05	6.22E-05	7.55E-06	4.88E-06
24	1.48E-03	2.81E-03	2.63E-03	2.32E-05	2.42E-05	9.15E-06	9.24E-07	5.88E-06	4.81E-06
25	0.00E+00	3.86E-05	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	8.05E-06
26	0.00E+00	3.86E-05	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	8.05E-06
27	7.25E-05	1.04E-04	2.49E-06	2.28E-08	1.68E-08	7.41E-09	4.31E-09	5.31E-09	1.06E-06

TABLE 11.2.1-3 (SHEET 7 OF 10)

Isotopic Concentration (μCi/g)

Line Number	Pr-144	Np-239	TOTAL	H-3
1	3.80E-06	1.20E-04	2.73E-01	1.00E+00
2	3.32E-06	1.07E-04	1.36E-01	1.00E+00
3	2.72E-06	8.43E-05	1.13E-01	7.03E-02
4	1.90E-06	6.00E-05	7.86E-02	5.00E-02
5	2.63E-06	8.17E-05	1.10E-01	6.81E-02
6	1.37E-06	4.32E-05	5.66E-02	3.60E-02
12	3.31E-07	1.04E-05	1.37E-02	8.87E-03
13	8.74E-07	2.76E-05	3.62E-02	2.30E-02
14	5.71E-07	1.80E-05	2.36E-02	1.51E-02
15	1.90E-06	6.00E-05	7.86E-02	5.00E-02
16	5.32E-07	5.25E-07	1.54E-02	1.38E-02
19	5.32E-09	5.25E-09	8.85E-04	1.38E-04
20	3.46E-10	1.92E-09	3.64E-05	9.98E-01
21	4.57E-09	4.75E-09	7.85E-04	1.51E-01
22	4.57E-09	4.75E-09	7.85E-04	1.51E-01
23	5.62E-06	1.78E-04	2.33E-01	1.48E-01
24	4.81E-06	4.42E-05	5.65E-02	1.48E-01
25	0.00E+00	0.00E+00	0.00E+00	0.00E+00
26	0.00E+00	0.00E+00	1.01E-04	0.00E+00
27	3.97E-09	4.12E-09	6.94E-04	1.31E-01

TABLE 11.2.1-3 (SHEET 8 OF 10)

Tank Number	Tank Description	Liquid[(c)] Volume (gal)	Tank Inventories (Ci) Cr-51	Mn-54	Fe-55	Fe-59	Co-58
1	Reactor coolant drain tank	126	8.96E-05	1.48E-05	7.62E-05	4.74E-05	7.60E-04
2	Waste holdup tank	4000	3.00E-04	1.07E-02	2.82E-03	5.51E-04	6.60E-03
3	Waste evaporator condensate tank	4000	3.42E-04	5.90E-07	2.92E-06	1.82E-06	2.92E-05
4	Floor drain tank	4000	4.03E-04	6.59E-05	3.39E-04	2.12E-04	3.41E-03
5	Waste monitor tank No. 2	4000	3.42E-04	5.90E-07	2.92E-06	1.82E-06	2.92E-05
6	Chemical drain tank	510	2.17E-04	4.10E-05	2.14E-04	1.21E-04	2.02E-03
7	Laundry and hot shower tank	4000	0.00E+00	2.44E-05	0.00E+00	0.00E+00	9.74E-05
8	Waste monitor tank No. 1	4000	0.00E+00	2.44E-05	0.00E+00	0.00E+00	9.74E-05

Tank Number	Tank Inventories (Ci) Co-60	Br-83	Br-84	Br-85	Rb-86	Rb-88	Sr-89	Sr-90	Y-90
1	9.53E-05	6.58E-05	1.01E-05	1.15E-07	2.09E-05	2.23E-03	1.71E-05	4.77E-07	9.85E-08
2	2.57E-03	6.75E-04	8.40E-05	8.58E-07	1.24E-05	1.57E-06	9.64E-03	2.66E-08	7.66E-06
3	3.65E-06	9.58E-06	5.33E-06	6.06E-07	8.07E-06	4.25E-03	6.56E-07	1.82E-08	2.48E-09
4	4.24E-04	1.13E-03	6.28E-04	7.12E-05	1.89E-05	1.00E-02	7.65E-05	2.12E-06	2.68E-07
5	3.65E-06	9.58E-06	5.33E-06	6.06E-07	8.07E-06	4.25E-03	6.56E-07	1.82E-08	2.48E-09
6	2.68E-04	7.84E-06	9.61E-07	1.03E-08	9.42E-06	4.13E-05	4.43E-05	1.34E-06	1.01E-06
7	2.19E-04	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
8	2.19E-04	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

Tank Number	Tank Inventories (Ci) Sr-91	Y-91m	Y-91	Y-93	Zr-95	Nb-95	Mo-99	Tc-99m	Ru-103
1	1.94E-05	1.23E-05	3.18E-06	1.05E-06	2.89E-06	2.43E-06	3.76E-03	3.16E-03	2.18E-06
2	4.45E-07	3.07E-05	1.33E-04	4.33E-07	3.98E-05	2.38E-07	2.62E-05	4.45E-05	9.39E-05
3	1.24E-06	7.34E-07	1.18E-07	6.57E-08	1.13E-07	9.31E-08	1.56E-04	1.06E-04	8.36E-08
4	1.46E-04	8.60E-05	1.38E-05	7.75E-06	1.30E-05	1.08E-05	1.83E-02	1.24E-02	9.77E-06
5	1.24E-06	7.34E-07	1.18E-07	6.57E-08	1.13E-07	9.31E-08	1.56E-04	1.06E-04	8.36E-08
6	4.12E-06	2.64E-06	8.35E-06	2.28E-07	7.64E-06	6.97E-06	3.41E-03	3.58E-03	5.51E-06
7	0.00E+00	0.00E+00	0.00E+00	0.00E+00	3.41E-05	4.87E-05	0.00E+00	0.00E+00	3.41E-06
8	0.00E+00	0.00E+00	0.00E+00	0.00E+00	3.41E-05	4.87E-05	0.00E+00	0.00E+00	3.41E-06

TABLE 11.2.1-3 (SHEET 9 OF 10)

Tank Inventories (Ci)

Tank Number	Rh103m	Ru-106	Rh-106	Te125m	Te127m	Te-127	Te129m	Te-129	I-130
1	2.21E-06	4.76E-07	4.76E-07	1.38E-06	1.33E-05	3.07E-05	6.62E-05	4.85E-05	7.03E-05
2	2.97E-03	3.12E-05	4.16E-05	8.07E-08	7.34E-06	1.82E-05	8.66E-06	1.71E-03	9.02E-04
3	9.28E-05	2.07E-08	1.97E-08	5.28E-08	5.12E-07	1.65E-06	2.54E-06	3.24E-06	4.12E-06
4	1.08E-05	2.41E-06	2.30E-06	6.16E-06	5.95E-05	1.94E-04	2.97E-04	3.80E-04	4.82E-04
5	9.28E-05	2.07E-08	1.97E-08	5.28E-08	5.12E-07	1.65E-06	2.54E-06	3.24E-06	4.12E-06
6	5.52E-06	1.33E-06	1.33E-06	3.60E-06	3.61E-05	3.95E-05	1.65E-04	1.06E-04	1.74E-05
7	0.00E+00	5.84E-05	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
8	0.00E+00	5.84E-05	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

Tank Inventories (Ci)

Tank Number	Te131m	Te-131	I-131	Te-132	I-132	I-133	I-134	I-135	Cs-134
1	1.01E-04	2.03E-05	1.29E-02	1.22E-03	2.22E-03	1.43E-02	2.82E-04	4.89E-03	6.19E-03
2	1.45E-05	6.16E-06	2.09E-01	2.80E-05	1.32E-02	1.85E-01	4.01E-07	1.53E-02	1.39E-01
3	4.65E-06	2.13E-06	5.34E-04	5.04E-05	1.97E-04	2.02E-04	9.42E-05	3.74E-04	2.36E-03
4	5.47E-04	2.51E-04	5.93E-02	5.92E-03	2.30E-02	2.33E-02	1.11E-02	4.39E-02	5.53E-03
5	4.65E-06	2.13E-06	5.34E-04	5.04E-05	1.97E-04	2.02E-04	9.42E-05	3.74E-04	2.36E-03
6	4.72E-05	8.80E-06	2.24E-02	1.25E-03	1.40E-03	5.04E-03	2.82E-05	8.48E-04	3.47E-03
7	0.00E+00	0.00E+00	1.46E-05	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	3.17E-04
8	0.00E+00	0.00E+00	1.46E-05	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	3.17E-04

Tank Inventories (Ci)

Tank Number	Cs-136	Cs-137	Ba137m	Ba-140	La-140	Ce-141	Ce-143	Pr-143	Ce-144
1	3.26E-03	4.53E-03	4.21E-03	1.03E-05	8.02E-06	3.35E-06	1.65E-06	2.41E-06	1.57E-06
2	1.53E-02	2.10E-07	4.50E-05	2.29E-04	2.45E-04	5.45E-04	1.11E-09	1.06E-03	8.37E-04
3	1.26E-03	1.73E-03	4.35E-05	3.98E-07	2.92E-03	1.29E-07	7.51E-08	9.27E-08	6.03E-08
4	2.97E-03	4.04E-03	3.83E-03	4.66E-05	3.41E-05	1.31E-05	8.84E-06	1.08E-05	7.01E-06
5	1.26E-03	1.73E-03	4.35E-05	3.98E-07	2.92E-03	1.29E-07	7.51E-08	9.27E-08	6.03E-08
6	1.35E-03	2.55E-03	2.39E-03	2.11E-05	2.20E-05	8.30E-06	8.39E-07	5.34E-06	4.36E-06
7	0.00E+00	5.84E-04	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	1.22E-04
8	0.00E+00	5.84E-04	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	1.22E-04

TABLE 11.2.1-3 (SHEET 10 OF 10)

Tank Number	Tank Inventories (Ci) Pr-144	Np-239	TOTAL	H-3
1	1.58E-06	5.09E-05	6.48E-02	4.77E-01
2	3.95E-05	6.50E-04	7.45E-01	4.24E-05
3	6.92E-08	7.19E-08	1.19E-02	2.29E+00
4	8.05E-06	7.95E-06	2.33E-01	2.09E-01
5	6.92E-08	7.19E-08	1.19E-02	2.29E+00
6	4.36E-06	4.01E-05	5.13E-02	1.34E-01
7	0.00E+00	0.00E+00	1.52E-03	0.00E+00
8	0.00E+00	0.00E+00	1.52E-03	0.00E+00

a. Locations of lines and tanks are shown in the liquid waste processing system process flow diagram (drawings 1X4DB124, 1X4DB125, 1X4DB126, 1X4DB127, AX4DB124-2, and AX4DB124-3). Activities are based on expected reactor coolant activity levels given in section 11.1.

b. Optional flow path.

c. This volume does not represent the capacity of the tank but represents the expected liquid volume.

TABLE 11.2.1-4

SUMMARY OF TANK LEVEL INDICATION, LEVEL ANNUNCIATORS, AND OVERFLOWS

Tank	Level Indication Location	Alarm Location	Alarm	Overflow To
Volume control	MCP Local	MCP	Low, High	None
Batching	Local	MCP	Low	Waste holdup tank
Boric acid	MCP	MCP	Empty, Low-Low	Waste holdup tank
Recycle holdup	BRP Local	BRP	Low, High	Waste holdup tank
Boron injection (Unit 1 only)	None	None	None	None
Refueling water storage	MCP	MCP	Empty, Low-Low, Low, High	RWST Dike
Condensate storage	MCP AFP	MCP	High, Low, Low-Low	CST Dike
Reactor makeup water storage	MCP	MCP	High, Low, Low-Low	RMWST Dike
Waste holdup	LWPP Local	LWPP	High, High-High	Local drains
Waste evaporator condensate	LWPP Local	LWPP	Low, High	Waste holdup tank
Spent resin storage	LWPP	LWPP	Low, High	None
Chemical drain	LWPP DRP Local	LWPP	Low, High	Local drains
Floor drain	LWPP Local	LWPP	Low, High	Local drains
Waste monitor	LWPP Local	LWPP	Low, High	None
Auxiliary waste monitor	LWPP Local	LWPP	Low, High	Local drains
Laundry and hot shower	LWPP Local	LWPP	Low, High	None
Steam generator blowdown spent resin storage	SGBP	SGBP	Low, High	None

MCP = main control panel
BRP = boron recycle panel
LWPP = liquid waste processing panel
SGBP = steam generator blowdown processing panel

TABLE 11.2.2-1

RANGE OF MEASURED DECONTAMINATION
FACTORS FOR SELECTED ISOTOPES

Isotope	Minimum	Maximum
I-131	1.1×10^{1}	1.6×10^{4}
I-133	1.1×10^{1}	1.8×10^{4}
I-135	1.4×10^{1}	2.0×10^{4}
Cs-137	2.4	1.3×10^{3}
F-18	1.73×10^{1}	1.5×10^{3}
Co-58	3.2×10^{1}	8.2×10^{3}
Mn-54	$>2.5 \times 10^{1}$	$>1.3 \times 10^{2}$

These values were observed across mixed bed demineralizers containing cation resin in the lithium-7 form and anion resin in the borated form.

TABLE 11.2.3-1

CALCULATED RELEASES OF RADIOACTIVE MATERIALS IN LIQUID EFFLUENTS (PER UNIT) ASSUMING EXPECTED FUEL LEAKAGE

		Coolant Concentrations			Liquid` Effluents Annual Releases to Discharge Canal						
Nuclide	Half-Life (days)	Primary (μCi/ml)	Secondary (μCi/ml)	Boron RS (Ci)	Misc .Wastes (Ci)	Secondary (Ci)	Turb Bldg (Ci)	Total LWS (Ci)	Adjusted Total (Ci/year)	Detergent Wastes (Ci/year)	Total (Ci/year)
Corrosion and Activation Products											
Cr-51	2.78E+01	1.90E-03	4.19E-08	0.00000	0.00000	0.00000	0.00000	0.00000	0.00003	0.00000	0.00003
Mn-54	3.03E+02	3.10E-04	1.00E-08	0.00000	0.00000	0.00000	0.00000	0.00000	0.00001	0.00100	0.00100
Fe-55	9.50E+02	1.60E-03	3.50E-08	0.00000	0.00000	0.00000	0.00000	0.00000	0.00002	0.00000	0.00003
Fe-59	4.50E+01	1.00E-03	2.57E-08	0.00000	0.00000	0.00000	0.00000	0.00000	0.00002	0.00000	0.00002
Co-58	7.13E+01	1.60E-02	3.57E-07	0.00000	0.00000	0.00003	0.00000	0.00003	0.00025	0.00400	0.00420
Co-60	1.92E+03	2.00E-03	4.50E-08	0.00000	0.00000	0.00000	0.00000	0.00000	0.00003	0.00870	0.00870
Zr-95	6.50E+01	0.00E+00	0.00E+00	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00140	0.00140
Nb-95	3.50E+01	0.00E+00	0.00E+00	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00200	0.00200
Np-239	2.35E+00	1.20E-03	2.35E-08	0.00000	0.00000	0.00000	0.00000	0.00000	0.00002	0.00000	0.00002
Fission Products											
Br-83	1.00E-01	4.80E-03	6.75E-08	0.00000	0.00000	0.00000	0.00000	0.00001	0.00007	0.00000	0.00007
Rb-86	1.87E+01	8.50E-05	2.15E-09	0.00001	0.00000	0.00000	0.00000	0.00001	0.00009	0.00000	0.00009
Rb-88	1.24E-02	2.00E-01	6.97E-07	0.00000	0.00001	0.00050	0.00000	0.00051	0.00402	0.00000	0.00400
Y-91m	3.47E-02	3.60E-04	2.09E-08	0.00000	0.00000	0.00000	0.00000	0.00000	0.00001	0.00000	0.00001
Mo-99	2.79E+00	8.40E-02	2.22E-06	0.00000	0.00001	0.00016	0.00002	0.00019	0.00148	0.00000	0.00150
Tc-99m	2.50E-01	4.80E-02	7.91E-06	0.00000	0.00001	0.00057	0.00005	0.00062	0.00494	0.00000	0.00490
Ru-103	3.96E+01	4.50E-05	1.03E-09	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00014	0.00014
Ru-106	3.67E+02	1.00E-05	2.51E-10	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00240	0.00240
Ag-110m	2.53E+02	0.00E+00	0.00E+00	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00044	0.00044
Te-127	3.92E-01	8.50E-04	3.96E-08	0.00000	0.00000	0.00000	0.00000	0.00000	0.00003	0.00000	0.00002
Te-129m	3.40E+01	1.40E-03	3.12E-08	0.00000	0.00000	0.00000	0.00000	0.00000	0.00002	0.00000	0.00002
Te-129	4.79E-02	1.60E-03	3.76E-07	0.00000	0.00000	0.00003	0.00000	0.00003	0.00022	0.00000	0.00022
I-130	5.17E-01	2.10E-03	4.17E-08	0.00000	0.00001	0.00000	0.00000	0.00001	0.00009	0.00000	0.00009
Te-131m	1.25E+00	2.50E-03	5.10E-08	0.00000	0.00000	0.00000	0.00000	0.00000	0.00003	0.00000	0.00003
Te-131	1.74E-02	1.10E-03	4.12E-07	0.00000	0.00000	0.00003	0.00000	0.00003	0.00024	0.00000	0.00024
I-131	8.05E+00	2.70E-01	6.43E-06	0.00051	0.00248	0.00046	0.00063	0.00408	0.03241	0.00006	0.03200
Te-132	3.25E+00	2.70E-02	5.65E-07	0.00000	0.00000	0.00004	0.00001	0.00005	0.00038	0.00000	0.00038
I-132	9.58E-02	1.00E-01	5.07E-06	0.00000	0.00021	0.00036	0.00009	0.00066	0.00527	0.00000	0.00530
I-133	8.75E-01	3.80E-01	7.96E-06	0.00007	0.00149	0.00057	0.00065	0.00278	0.02211	0.00000	0.02200
I-134	3.67E-02	4.70E-02	3.91E-07	0.00000	0.00001	0.00003	0.00000	0.00003	0.00027	0.00000	0.00027
Cs-134	7.49E+02	2.50E-02	6.01E-07	0.00310	0.00067	0.00043	0.00001	0.00420	0.03340	0.01300	0.04600
I-135	2.79E-01	1.90E-01	3.44E-06	0.00001	0.00027	0.00025	0.00018	0.00071	0.00562	0.00000	0.00560
Cs-136	1.30E+01	1.30E-02	2.76E-07	0.00093	0.00032	0.00020	0.00000	0.00145	0.01153	0.00000	0.01200
Cs-137	1.10E+04	1.80E-02	4.00E-07	0.00225	0.00048	0.00029	0.00000	0.00302	0.02404	0.02400	0.04800
Ba-137m	1.77E-03	1.60E-02	7.92E-06	0.00211	0.00045	0.00057	0.00000	0.00313	0.02486	0.00000	0.02500
Ce-144	2.84E+02	3.30E-05	1.00E-09	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00520	0.00520
Pr-144	1.20E-02	3.30E-05	1.75E-08	0.00000	0.00000	0.00000	0.00000	0.00000	0.00001	0.00000	0.00001
All Others		5.01E-03	8.27E-08	0.00000	0.00000	0.00001	0.00000	0.00001	0.00005	0.0	0.00005
Total (Except Tritium)		1.46E+00	4.56E-05	0.00900	0.00640	0.00454	0.00165	0.02158	0.17158	0.06234	0.23000
Tritium Release			720 Curies per year								

TABLE 11.2.3-2 (SHEET 1 OF 2)

COMPARISON OF CALCULATED CONCENTRATIONS
IN EFFLUENT WATER DISCHARGE WITH 10 CFR 20 LIMITS
ASSUMING EXPECTED FUEL LEAKAGE

Isotope	Annual Release to Discharge[a] Pipe (µCi)	Concentration in Circulating Water Discharge[b] (µCi/ml)	Maximum Permissible Concentration[c] (µCi/ml)	Fraction of Maximum Permissible Concentration
Cr-51	3.00E+01	1.26E-12	2.00E-03	6.28E-10
Mn-54	1.00E+03	4.19E-11	1.00E-04	4.19E-07
Fe-55	2.00E+01	8.38E-13	8.00E-04	1.05E-09
Fe-59	2.00E+01	8.38E-13	6.00E-05	1.40E-08
Co-58	4.20E+03	1.76E-10	1.00E-04	1.76E-06
Co-60	8.70E+03	3.64E-10	5.00E-05	7.29E-06
Zr-95	1.40E+03	5.86E-11	6.00E-05	9.77E-07
Nb-95	2.00E+03	8.38E-11	1.00E-04	8.38E-07
Np-239	2.00E+01	8.38E-13	1.00E-04	8.38E-09
Br-83	7.00E+01	2.93E-12	3.00E-06	9.77E-07
Rb-86	9.00E+01	3.77E-12	7.00E-05	5.39E-08
Rb-88	4.00E+03	1.68E-10	3.00E-06	5.58E-05
Y-91m	1.00E+01	4.19E-13	3.00E-05	1.40E-08
Mo-99	1.50E+03	6.28E-11	2.00E-04	3.14E-07
Tc-99m	4.90E+03	2.05E-10	6.00E-03	3.42E-08
Ru-103	1.40E+02	5.86E-12	8.00E-05	7.33E-08
Ru-106	2.40E+03	1.01E-10	1.00E-05	1.01E-05
Ag-110m	4.40E+02	1.84E-11	3.00E-05	6.14E-07
Te-127	2.00E+01	8.38E-13	6.00E-05	1.40E-08
Te-129m	2.00E+01	8.38E-13	3.00E-05	2.79E-08
Te-129	2.20E+02	9.21E-12	8.00E-04	1.15E-08
I-130	9.00E+01	3.77E-12	3.00E-06	1.26E-06
Te-131m	3.00E+01	1.26E-12	6.00E-05	2.09E-08
Te-131	2.40E+02	1.01E-11	3.00E-06	3.35E-06
I-131	3.20E+04	1.34E-09	3.00E-07	4.47E-03
Te-132	3.80E+02	1.59E-11	3.00E-05	5.31E-07
I-132	5.30E+03	2.22E-10	8.00E-06	2.77E-05
I-133	2.20E+04	9.21E-10	1.00E-06	9.21E-04
I-134	2.70E+02	1.13E-11	2.00E-05	5.65E-07
Cs-134	4.60E+04	1.93E-09	9.00E-06	2.14E-04
I-135	5.60E+03	2.35E-10	4.00E-06	5.86E-05
Cs-136	1.20E+04	5.03E-10	9.00E-05	5.58E-06
Cs-137	4.80E+04	2.01E-09	2.00E-05	1.01E-04

TABLE 11.2.3-2 (SHEET 2 OF 2)

Isotope	Annual Release to Discharge[a] Pipe (µCi)	Concentration in Circulating Water Discharge[b] (µCi/ml)	Maximum Permissible Concentration[c] (µCi/ml)	Fraction of Maximum Permissible Concentration
Ba-137m	2.50E+04	1.05E-09	3.00E-06	3.49E-04
Ce-144	5.20E+03	2.18E-10	1.00E-05	2.18E-05
Pr-144	1.00E+01	4.19E-13	3.00E-08	1.40E-05
All Others	5.00E+01	2.09E-12	3.00E-08	6.98E-05
Total	2.33E+05			5.99E-03
H-3	7.10E+08	2.97E-05	3.00E-03	9.91E-03
Total + H-3	7.10E+08			1.59E-02[d]

a. Total $\overline{C}_i$ (µCi/year) from table 11.2.3-1.

b. $C_i = \overline{C}_i / \overline{Q}_a$ where $\overline{Q}_a$ =15,000 gpm or 2.387 x 10^{13} ml/yr.

c. From 10 CFR 20, Appendix B, Table II, Column 2.

d. $\Sigma_i C_i/MPC_i \leq 1$.

TABLE 11.2.3-3 (SHEET 1 OF 2)

RADIOACTIVE LIQUID RELEASES FROM WESTINGHOUSE-DESIGNED PRESSURIZED WATER REACTOR PLANTS (ZIRCALOY FUEL CLADDING)

Plant	Year	Primary Coolant[a] Activity, Fraction of Design Basis	Total Annual Release (Ci)	Average Discharge Concentration (μCi/ml)
R. E. Ginna	1972	0.126	0.375	5.18×10^{-10}
	1973	0.023	0.074	9.74×10^{-11}
	1974	0.025	0.138	2.15×10^{-10}
	1975	0.013	0.420	6.09×10^{-10}
	1976	0.064	0.689	1.09×10^{-9}
	1977	0.033	0.065	9.01×10^{-11}
H. B. Robinson 2	1972	0.006	0.371	3.4×10^{-9}
	1973	0.003	0.306	2.1×10^{-9}
	1974	0.003	2.9	5.2×10^{-9}
	1975	0.002	0.44	6.79×10^{-10}
	1976	<0.001	0. 375	6.07×10^{-10}
	1977	<0.001	0. 329	4.7×10^{-10}
Point Beach 1 and 2	1972	0.062	0.934	1.9×10^{-9}
	1973	0.015	0.746	7.39×10^{-10}
	1974	0.101	0.196	3.29×10^{-10}
	1975	0.049	3.35	6.03×10^{-9}
	1976	0.012	3.24	6.17×10^{-9}
	1977	0.009	1.5	2.6×10^{-9}
Surry 1 and 2	1972	<0.001	0. 083	6.04×10^{-10}
	1973	0.001	0.145	2.65×10^{-10}
	1974	<0.001	29.2	2.32×10^{-8}
	1975	0.002	27.5	1.12×10^{-8}
	1976	0.011	33.7	1.45×10^{-9}
	1977	0.005	65.5	2.58×10^{-8}
Turkey Point 3 and 4	1973	<0.001	0.037	1.97×10^{-10}
	1974	0.010	1.69	3.21×10^{-9}
	1975	0.012	3.07	2.31×10^{-9}
	1976	0.006	8.65	2.92×10^{-10}
	1977	0.005	8.9	5.06×10^{-9}
Zion 1 and 2	1973	<0.001	<0.001	2.77×10^{-13}

TABLE 11.2.3-3 (SHEET 2 OF 2)

Plant	Year	Primary Coolant[a] Activity, Fraction of Design Basis	Total Annual Release (Ci)	Average Discharge Concentration (μCi/ml)
	1974	<0.001	0.005	8.03×10^{-12}
	1975	0.003	0.0087	1.18×10^{-11}
	1976	0.013	0.16	1.58×10^{-11}
	1977	0.009	0.95	8.33×10^{-10}
Prairie Island 1 and 2	1974	0.002	<0.001	2.29×10^{-12}
	1975	0.006	0.45	1.18×10^{-9}
	1976	0.008	<0.012	$<3.59 \times 10^{-11}$
	1977	0.008	0.013	1.41×10^{-11}
Kewaunee	1974	0.012	0.422	3.49×10^{-9}
	1975	0.005	0.447	2.68×10^{-9}
	1976	0.002	2.85	2.32×10^{-8}
	1977	<0.001	1.26	8.69×10^{-9}
Donald C. Cook 1	1975	<0.001	0.26	3.19×10^{-9}
	1976	0.003	1.87	2.69×10^{-8}
	1977	0.005	1.52	2.09×10^{-8}

a. Inferred by radioiodine in reactor coolant - includes diffusion and recoil sources.

TABLE 11.2.3-4

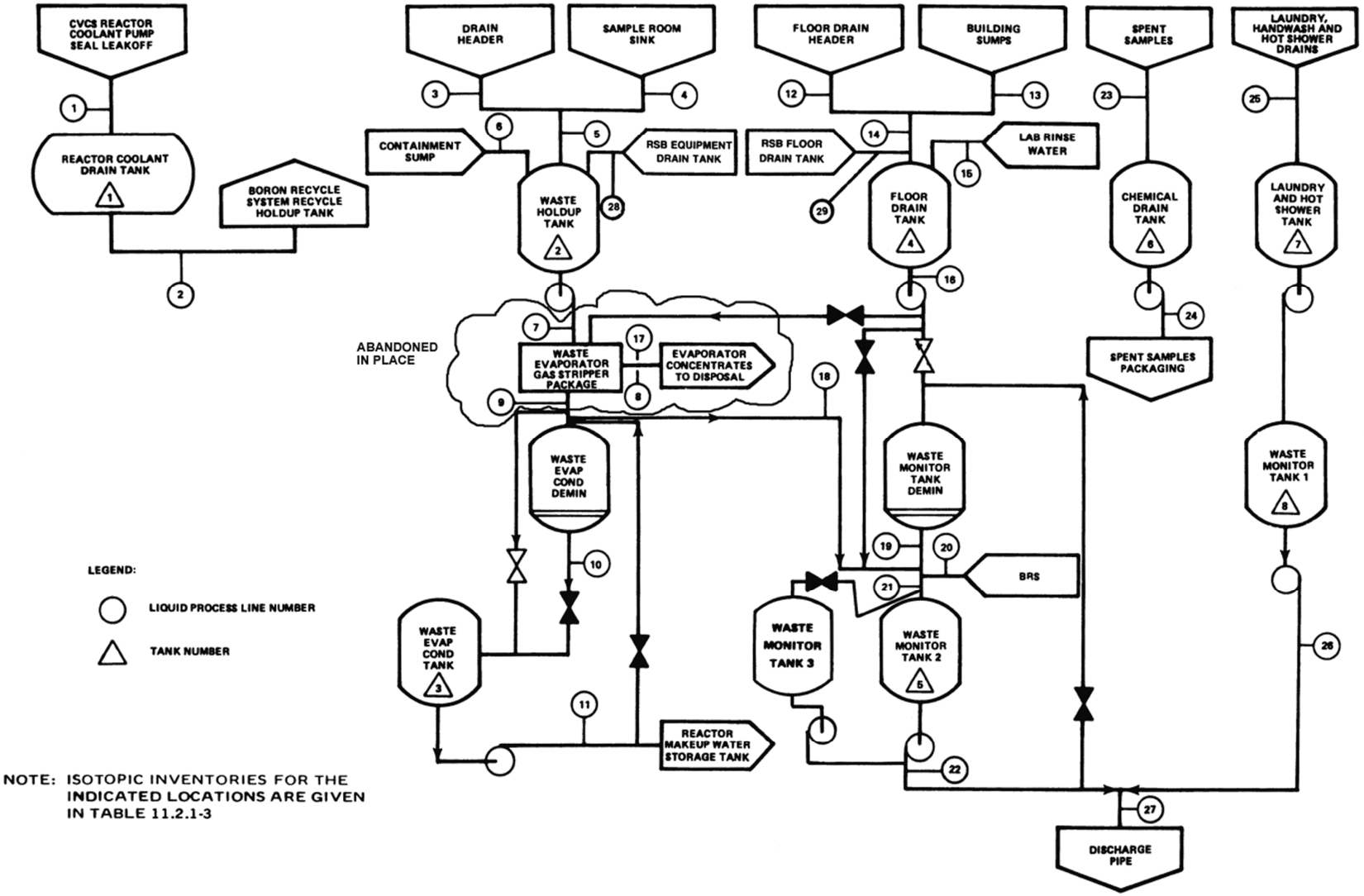
ESTIMATED INDIVIDUAL DOSES FROM LIQUID EFFLUENTS
AS LOW AS REASONABLY ACHIEVABLE[b]

Pathway	Usage (kg/yr h/yr)	Dilution	Time (h)	Shorewidth Factor = 0.2	Skin	Bone	Liver	Total Body	Thyroid	Kidney	Lung	Gi-LLi
					ADULT DOSES (mrem/year Intake)[a]							
Fish	21.0	10.0	24.01			9.51E-01	1.74E+00	1.30E+00	6.90E-02	5.88E-01	1.94E-01	1.30E-01
Drinking[c]	730.0	10.0	12.01			1.69E-02	2.13E -01	2.05E -01	3.42E-01	1.93E-01	1.86E-01	1.88E-01
Shoreline	12.0	10.0	0.01		1.21E-03	1.04E-03	1.04E -03	1.04E -03	1.04E-03	1.04E-03	1.04E-03	1.04E-03
Total					1.21E-03	9.69E-01	1.95E+00	1.51E+00	4.12E-01	7.82E-01	3.80E-01	3.19E-01
					TEENAGE DOSES (mrem/year Intake)[a]							
Pathway	Usage (kg/yr h/yr)	Dilution	Time (h)	Shorewidth Factor = 0.2	Skin	Bone	Liver	Total Body	Thyroid	Kidney	Lung	Gi-LLi
Fish	16.0	10.0	24.01			9.99E-01	1.78E+00	7.49E-01	6.39E-02	5.96E-01	2.26E-01	9.36E-02
Drinking[c]	510.0	10.0	12.01			1.63E-02	1.58E -01	1.41E-01	2.66E-01	1.39E-01	1.32E-01	1.32E-01
Shoreline	67.0	10.0	0.01		6.78E-03	5.80E-03	5.80E -03	5.80E-03	5.80E-03	5.80E-03	5.80E-03	5.80E-03
Total					6.78E-03	1.02E+00	1.95E+00	8.96E-01	3.36E-01	7.41E-01	3.64E-01	2.32E-01
					CHILD DOSES (mrem/year Intake)[a]							
Pathway	Usage (kg/yr h/yr)	Dilution	Time (h)	Shorewidth Factor = 0.2	Skin	Bone	Liver	Total Body	Thyroid	Kidney	Lung	Gi-LLi
Fish	6.9	10.0	24.01			1.24E+00	1.55E+00	3.00E-01	6.60E-02	5.01E-01	1.78E-01	3.57E-02
Drinking[c]	610.0	10.0	12.01			4.68E -02	3.05E-01	2.58E-01	5.81E-01	2.67E-01	2.53E-01	2.50E-01
Shoreline	14.0	10.0	0.01		1.42E-03	1.21E -03	1.21E-03	1.21E-03	1.21E-03	1.21E-03	1.21E-03	1.21E-03
Total					1.42E-03	1.28E+00	1.85E+00	5.59E-01	6.48E-01	7.69E-01	4.32E-01	2.87E-01
					INFANT DOSES (mrem/year Intake[a]							
Pathway	Usage (kg/yr h/yr)	Dilution	Time (h)		Skin	Bone	Liver	Total Body	Thyroid	Kidney	Lung	Gi-LLi
Fish	0.0	10.0	24.01			0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Drinking[c]	330.0	10.0	12.01			4.90E -02	3.14E -01	2.49E -01	7.68E-01	2.63E -01	2.49E-01	2.44E -01
Shoreline					0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Total					0.00E+00	4.90E -02	3.14E -01	2.49E -01	7.68E-01	2.63E -01	2.49E-01	2.44E -01

a. Individual doses calculated using the LADTAP II code. All data is on a per unit basis.

b. Appendix I Design Objectives for Liquid Effluents: total body dose = 3 mrem/year per unit from all pathways; dose to any organ = 10 mrem/year per unit from all pathways. Docket RM-50-2 Annex to Appendix I Design Objectives: total body dose = 5 mrem/year per site from all pathways; dose to any organ = 5 mrem/year per site from all pathways; nontritium releases = 5 Ci/year per unit (table 11.2.3-1).

c. Although the dose due to the drinking water pathway has been included in this evaluation; currently no river water is used for potable water within 100 river miles of the site.



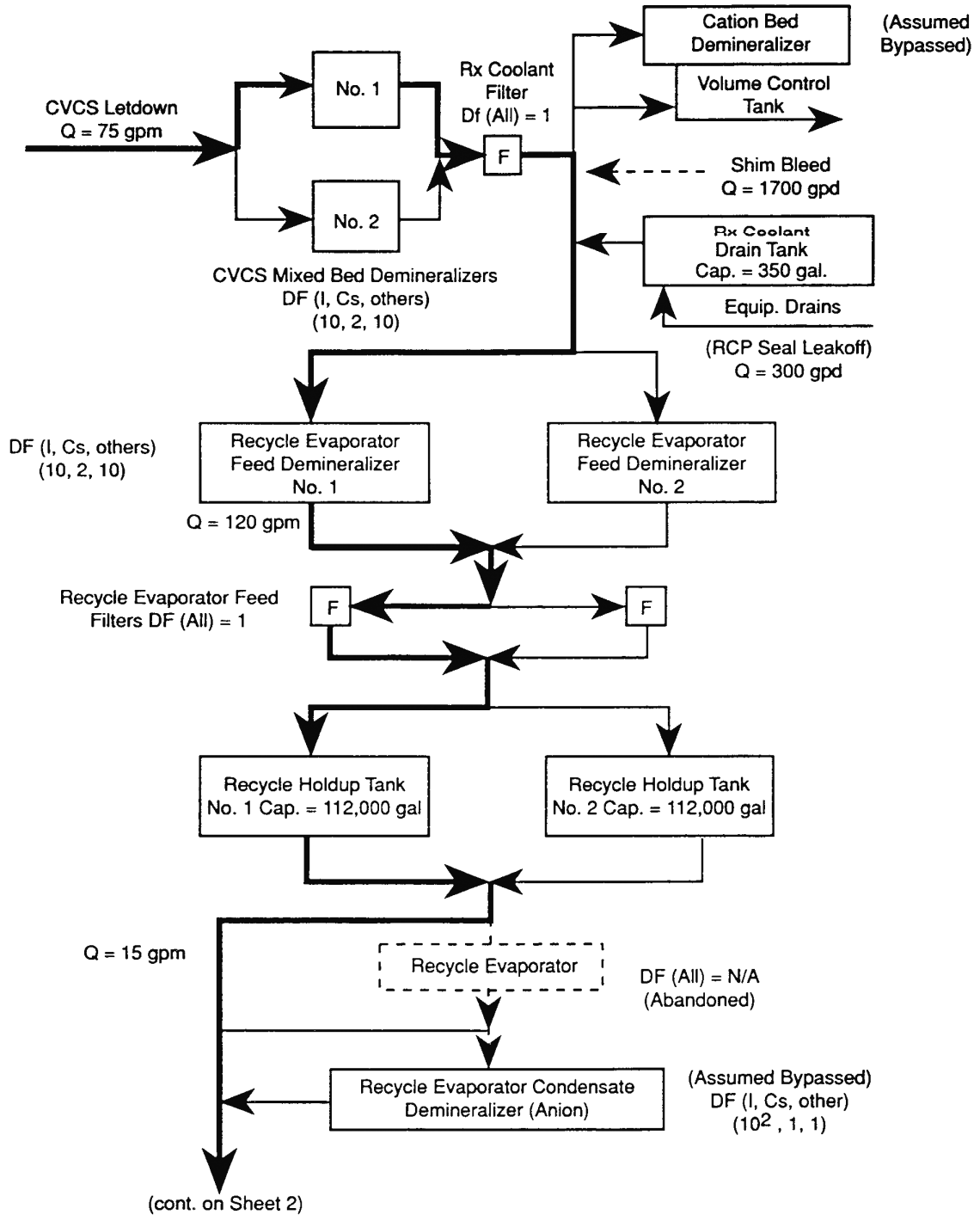
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VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

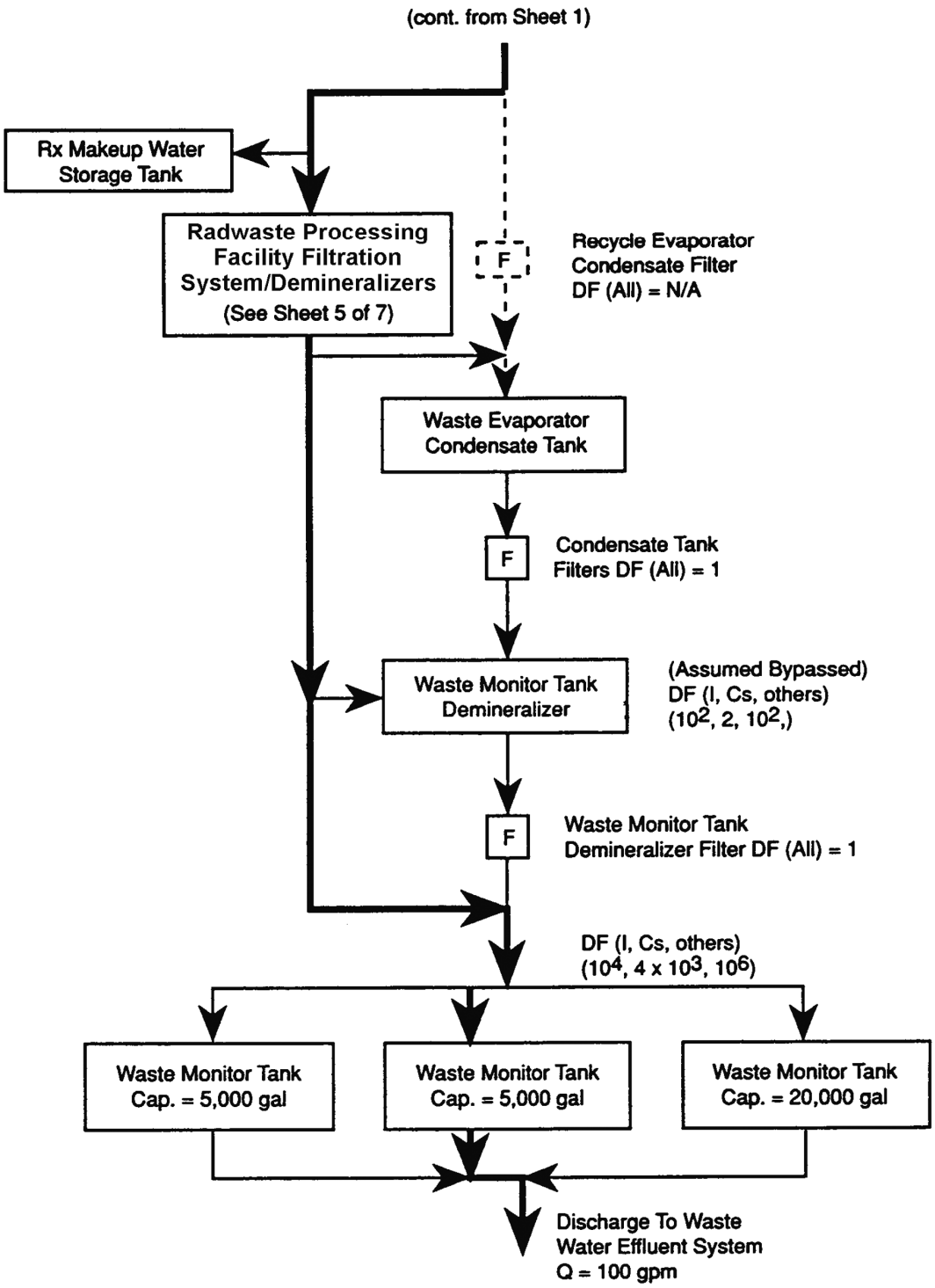
LWPS PROCESS FLOW DIAGRAM

FIGURE 11.2.2-1



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	PROCESS FLOW MODEL (FOR CVCS LETDOWN & EQUIPMENT DRAINS) FIGURE 11.2.3–1 (SHEET 1 OF 7)



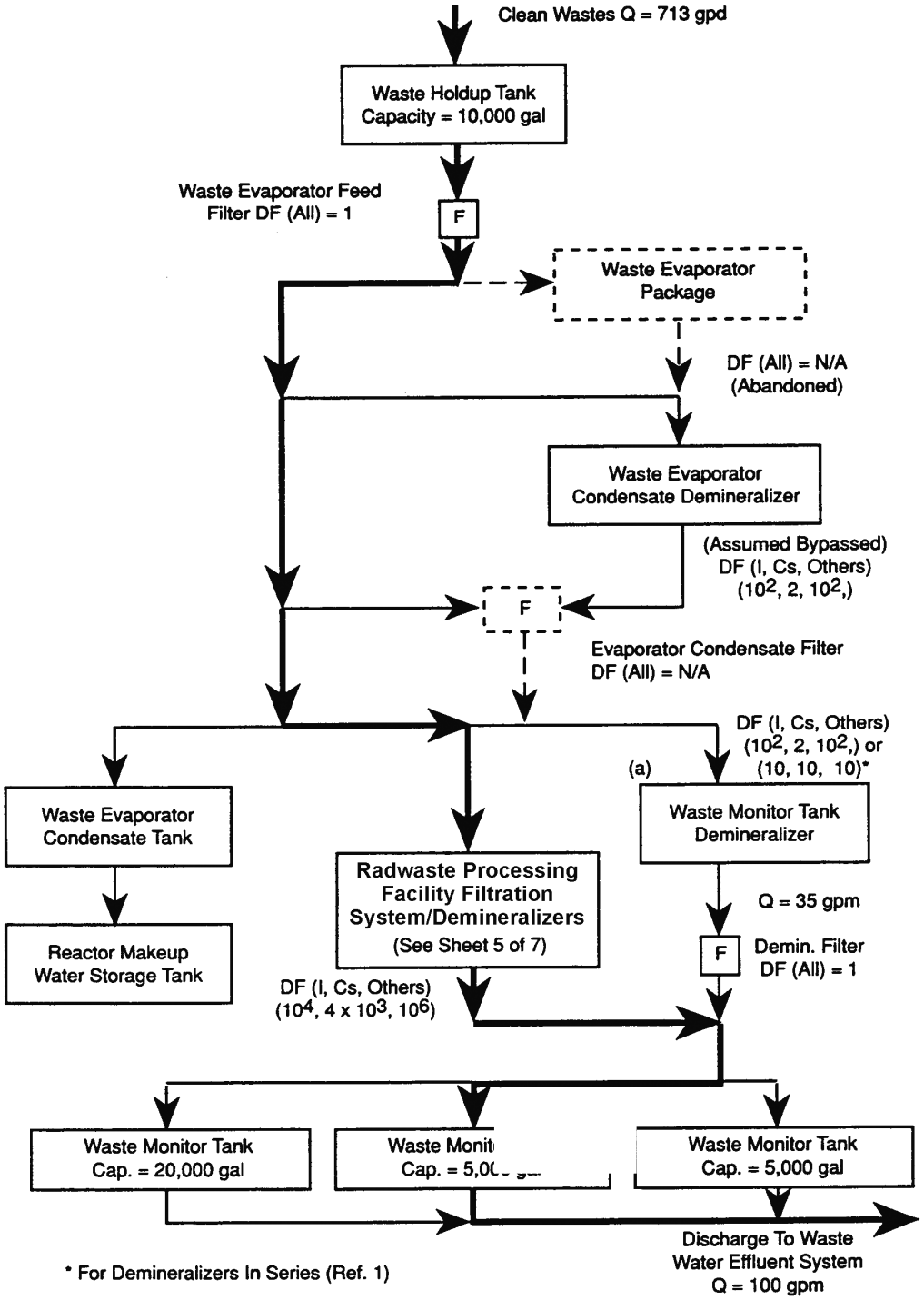
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VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

PROCESS FLOW MODEL (FOR CVCS LETDOWN & EQUIPMENT DRAINS)

FIGURE 11.2.3–1 (SHEET 2 OF 7)



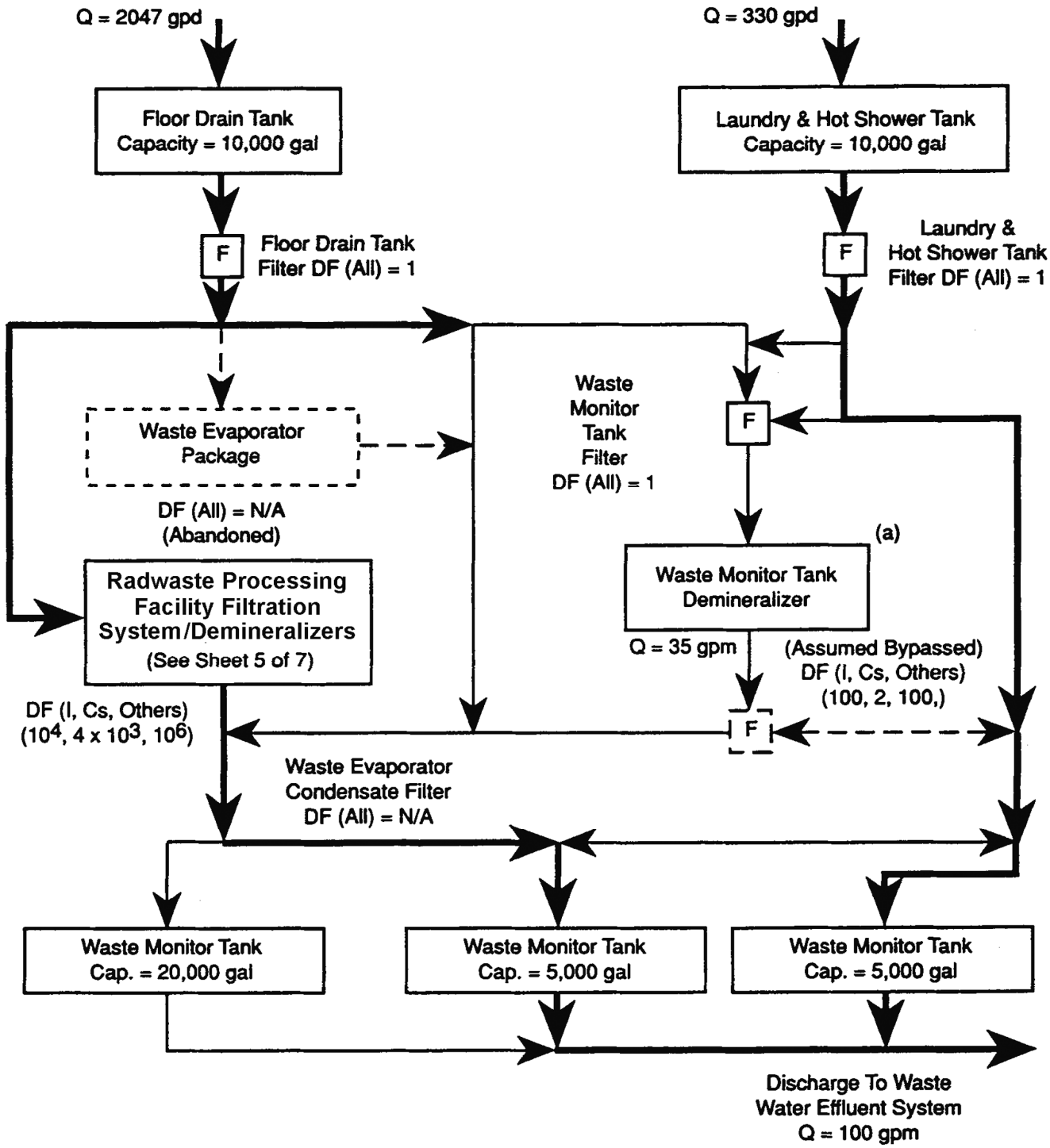
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UNIT 1 AND UNIT 2

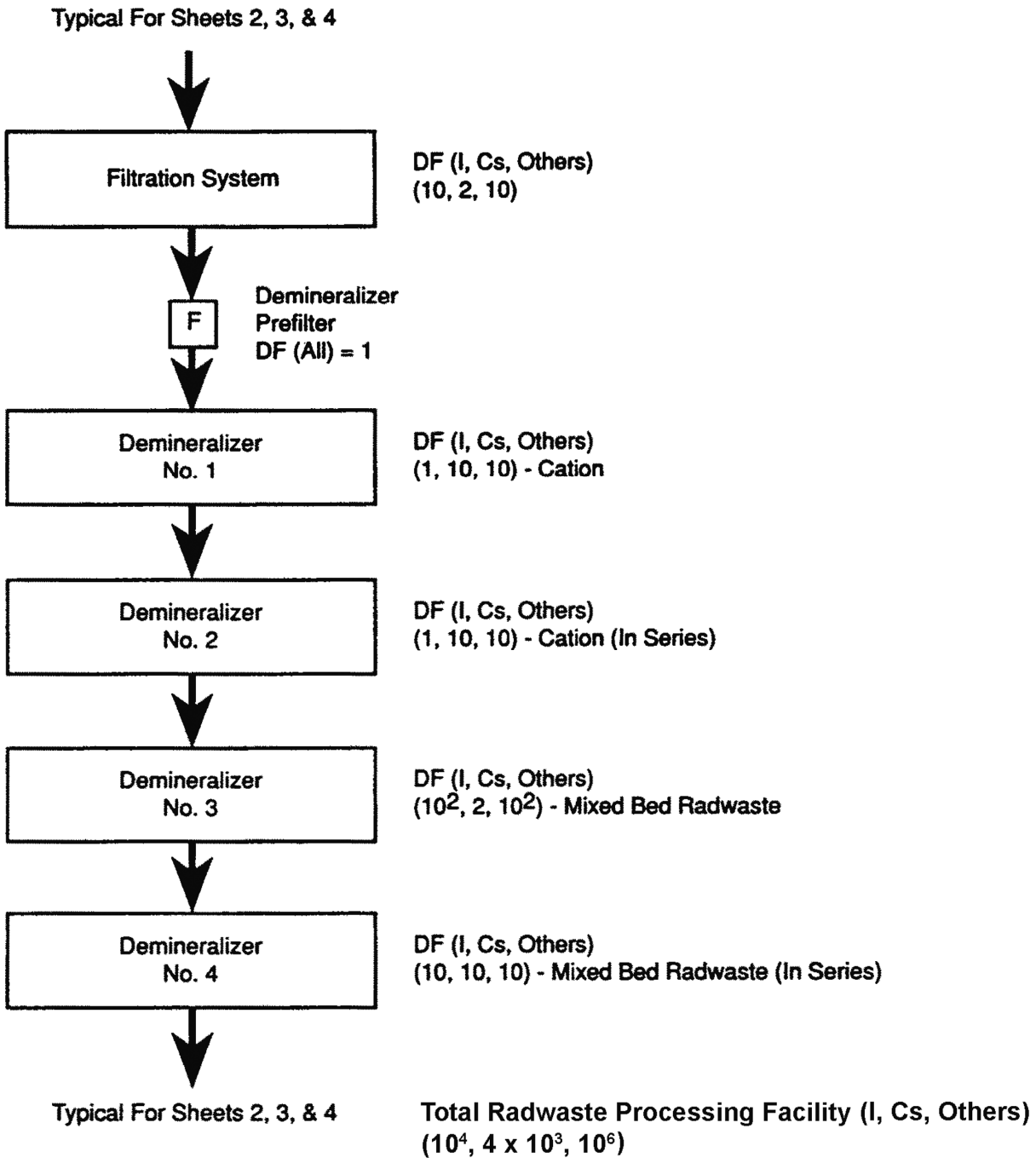
PROCESS FLOW MODEL (FOR CLEAN WASTE)

FIGURE 11.2.3–1 (SHEET 3 OF 7)



REV 13 4/06

SOUTHERN COMPANY *Energy to Serve Your World®* VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2	PROCESS FLOW MODEL (FOR DIRTY WASTE & LAUNDRY) FIGURE 11.2.3–1 (SHEET 4 OF 7)



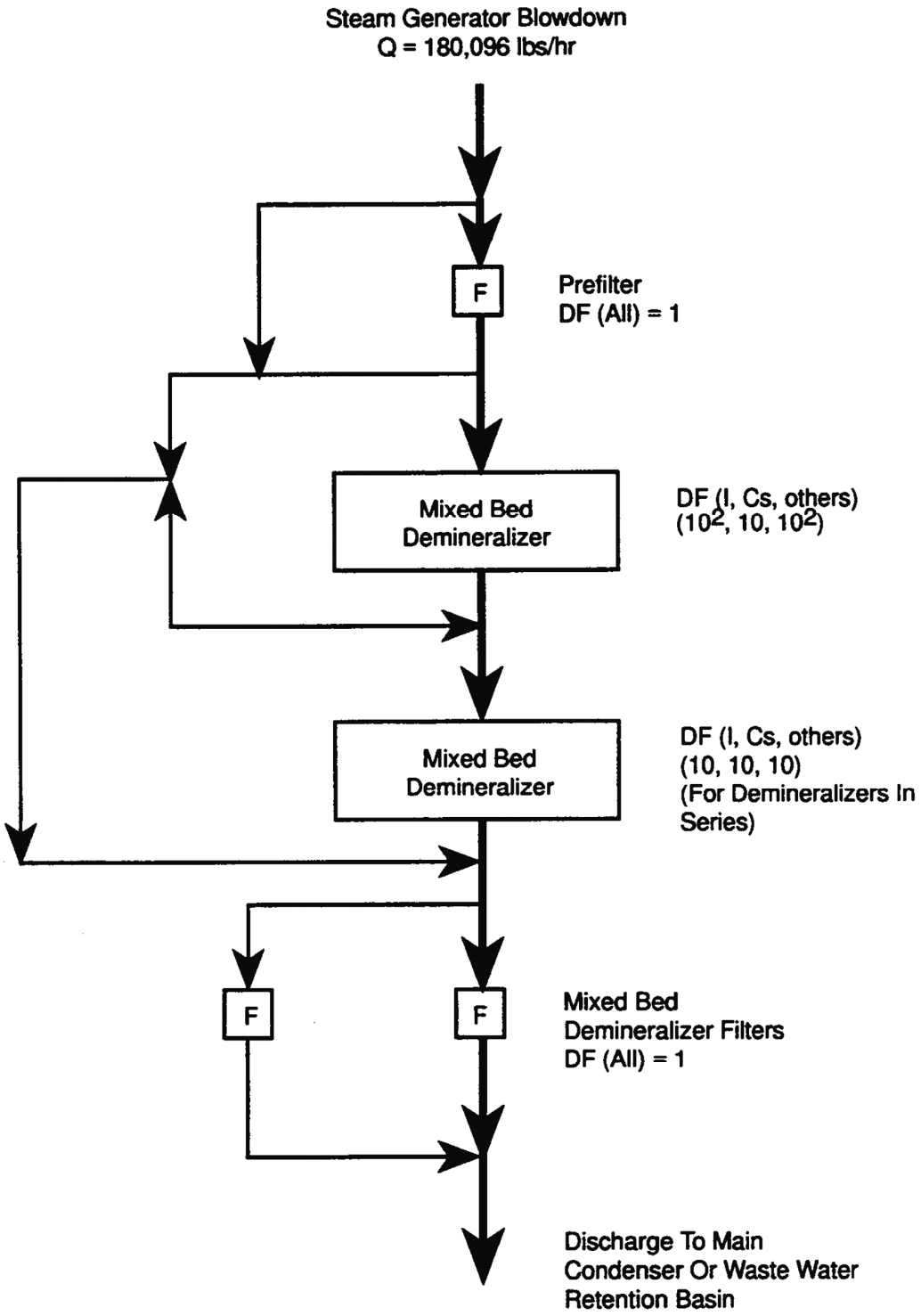
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VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

PROCESS FLOW MODEL (FOR RADWASTE PROCESSING FACILITY)

FIGURE 11.2.3–1 (SHEET 5 OF 7)



REV 13 4/06

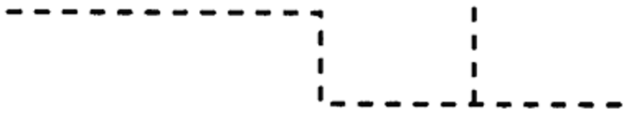
SOUTHERN COMPANY *Energy to Serve Your World®* VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2	PROCESS FLOW MODEL (FOR STEAM GENERATOR BLOWDOWN) FIGURE 11.2.3–1 (SHEET 6 OF 7)



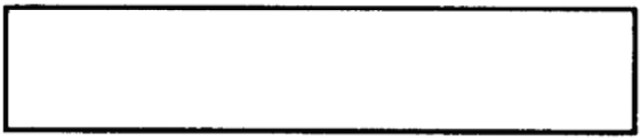
Normal Operation
Process Flow Path



Optional Process
Flow Path



Unavailable Or Deleted
Flow Path



Available Processing
Equipment



Unavailable Or Deleted
Processing Equipment

Q - Design Flow Rate

DF - Equipment Decontamination Factor For
Radioactive Iodine (I), Cesium (Cs) And Other
Miscellaneous Isotopes Defined By NUREG-0017 (Ref.1)

(a) - Note: Assumed Normally Bypassed

REV 13 4/06

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

PROCESS FLOW MODEL (LEGEND)

FIGURE 11.2.3–1 (SHEET 7 OF 7)

11.3 GASEOUS WASTE MANAGEMENT SYSTEMS

The seismic classification and safety class information for the gaseous waste processing system (GWPS) are in section 3.2.

11.3.1 DESIGN BASES

The GWPS is designed to collect, process, and store gaseous wastes generated by plant operations including anticipated operational occurrences. The system is designed to assure that the release of gaseous effluents from the plant and expected offsite doses are as low as reasonably achievable (ALARA) as defined in Appendix I of 10 CFR 50. An evaluation of plant conformance to Appendix I is given in subsection 11.3.3. The GWPS has sufficient capacity and redundancy to keep releases within the discharge concentration limits of 10 CFR 20.1 - 20.601 during periods of design basis fuel leakage, as discussed in subsection 11.3.3.

In addition, the GWPS conforms to the requirements of General Design Criterion (GDC) 60 by providing long-term holdup capacity, thus precluding the necessity of releasing radioactive effluents during unfavorable environmental conditions. All gaseous effluent discharge paths are monitored for radioactivity, in compliance with GDC 64. Monitoring of radioactive effluents is discussed in section 11.5. The system's capability for long-term storage eliminates the need for frequent routine discharge of radioactive gases to the environment; it thereby allows any discharges to be made selectively under favorable environmental conditions.

During normal operation, the annual releases by leakage and planned discharges from the GWPS will be sufficiently low to control site boundary doses from the effluent to a small fraction of regulatory requirements.

The design of the GWPS is based on continuous operation of both Unit 1 and Unit 2 of VEGP, assuming that 1 percent of the rated core power is generated by fuel rods containing cladding defects. This condition is assumed to exist over the life of the plant[a].

While the normal inflow to the GWPS is 0.7 sf^3/min per unit, either of the two main process loops of the GWPS can accommodate surges such as may result from venting a recycle holdup tank or a reactor coolant drain tank. No other anticipated operational occurrence will cause a significant surge in GWPS process flow. However, the operator may elect to increase the volume control tank purge flow above the normal 0.7 sf^3/min per unit.

If waste gases are being released to the environment, the release is automatically terminated when the radioactivity level exceeds a predetermined level. The radiological monitoring and control instrumentation are described in section 11.5.

In order to control the release of radioactive gases resulting from equipment failure or operator error, the GWPS design has all waste gas decay tanks (WGDTs) (seven tanks for Unit 1, seven tanks for Unit 2, and two shared shutdown tanks) isolated from each other with valves. The tanks are in separate compartments. Thus, the maximum uncontrolled release would be limited

[a] The renewed operating licenses authorized a 20-year period of extended operation for both VEGP units, resulting in a total plant operating life of 60 years. Since the inventory in the Waste Gas Decay Tanks (WGDTs) has been routinely released during the first 20 years of operation and is expected to continue to be routinely released during future operation, the inventory of the WGDTs accumulated during the first 20 years of operation will be released prior to entering the period of extended operation. Therefore, the stated design capacity of the GWPS remains sufficient, and the analysis of the maximum fission product inventory in the GWPS over a 40-year plant life remains bounding for a 60-year plant life.

to the contents of one WGDT. The effects of this occurrence would be the same as the postulated WGDT rupture accident which is discussed in section 15.7. To reduce GWPS downtime resulting from equipment failures, the GWPS design includes redundant waste gas compressors (two per unit) and catalytic hydrogen recombiners (one per unit and one shared).

The system is designed to preclude the possibility of an internal explosion. However, the system volume is distributed among seven isolated tanks for each unit located in separate compartments so that the dose in the unlikely event of an explosion is the same as the dose due to a WGDT rupture.

The GWPS component design parameters and a description of process instrumentation and control are provided in subsection 11.3.2.

Design features incorporated in the GWPS to reduce leakage of radioactive gases are described in subsection 11.3.2.

Conformance of the GWPS design with the criteria of Regulatory Guide 1.143 is discussed in section 1.9.

11.3.2 SYSTEM DESCRIPTION

This section describes the design, operating features, and performance of the gaseous waste processing system (GWPS) and other plant gaseous waste management systems with respect to the collection and control of radioactive gases. Detailed descriptions of the plant ventilation systems and condenser vacuum system are presented in sections 9.4 and 10.4, respectively.

The GWPS is provided with two main process loops. Each processes gaseous wastes from one of the two plant units. Interconnections are provided to allow either process loop to process waste gas from either or both units.

All equipment in the GWPS is controlled from the waste processing panel. The GWPS consists mainly of two closed loops comprised of a waste gas compressor, a catalytic hydrogen recombiner, and seven waste gas decay tanks (GDTs) to accumulate the fission product gases. All pipes containing radioactive gases are shielded as necessary, and no piping is run through normally occupied areas.

Each main process loop also includes a GDT drain pump, six gas traps, and a waste gas drain filter. All of the equipment is located in the auxiliary building.

The piping and instrumentation diagram for the system is shown in drawings 1X4DB128 and 1X4DB129. This diagram indicates safety classes for all components and piping.

The GWPS reduces the fission gas concentration in the reactor coolant system (RCS), which in turn reduces the escape of fission gases from the RCS during maintenance or through equipment leakage.

The primary source of radioactive gas to the GWPS is the volume control tank (VCT) purge. Smaller quantities of radioactive gas are received via the vent connections from the reactor coolant drain tank (RCDT), the pressurizer relief tank, and the recycle holdup tanks.

Although the GWPS functions to contain radioactive gases, at no time do the radioactive gases constitute more than a small fraction of the gases stored in the GDTs.

Since hydrogen is continuously removed in the hydrogen recombiner, this gas does not build up in the GWPS. The largest contributor to the nonradioactive gas accumulation is helium generated by a $B^{10}(n,\alpha)Li^{7}$ reaction in the reactor core. The second largest contributors are the impurities in the bulk hydrogen and oxygen supplies.

The expected accumulation rate is 575 sf^3/year per unit (1150 sf^3/year total), assuming the following:

- Two-unit operation.
- A 0.7-sf^3/min hydrogen purge for the VCT of each unit.
- An 80-percent plant load factor for each unit.
- 99.95-percent pure hydrogen.
- 99.5-percent pure oxygen.

At this rate of accumulation and assuming zero leakage from the GWPS, the 14 GDTs have sufficient combined capacity to hold all the gaseous wastes produced during 40 years[a] of plant operations without any releases to the environment. This assumes that the waste gas holdup tanks are operated with an initial charge of 5 psig of nitrogen and the pressure in the tanks is allowed to accumulate to 100 psig.

Operation of the system is such that fission gases from one unit are distributed throughout seven normal operation GDTs. Separation of the gaseous inventory into several tanks ensures that the allowable site boundary dose will not be exceeded in the event any one of the GDTs ruptures. Radiological consequences of such a postulated rupture are discussed in chapter 15.

The GWPS also provides sufficient capacity to hold indefinitely the gases generated during reactor shutdown. Nitrogen gas from previous shutdowns is contained in one of the shutdown GDTs. This is used to strip hydrogen from the RCS during subsequent shutdowns. The second shutdown tank is normally at low pressure and is used to accept relief valve discharges from the normal operation GDTs, the hydrogen recombiners, and the waste gas compressors.

Table 11.3.2-1, based on the RCS activities given in table 11.1-2, shows the maximum fission product inventory in the GWPS over the 40-year plant life[a]. Table 11.3.2-2, based on the RCS activities given in table 11.1-7, shows the expected fission product inventory in the GWPS over the 40-year life[a] assuming no releases from the system.

Figures 11.3.2-1 and 11.3.2-2 are based on the RCS activities given in tables 11.1-2 and 11.1-7, respectively; the figures show that the quantity of fission gas activity accumulated after 40 continuous years of operation[a] is about twice the activity accumulated after 30 days of operation. Most of the accumulated activity arises from short-lived isotopes reaching equilibrium in 1 month or less.

The difference between the 30-day and 40-year[a] accumulations is predominantly krypton-85. This accumulation of krypton-85 is not a hazard to the plant operator because:

A. Krypton-85 is principally a beta emitter, for which the tanks themselves provide adequate shielding.

[a] The renewed operating licenses authorized a 20-year period of extended operation for both VEGP units, resulting in a total plant operating life of 60 years. Since the inventory in the WGDTs has been routinely released during the first 20 years of operation and is expected to continue to be routinely released during future operation, the inventory of the WGDTs accumulated during the first 20 years of operation will be released prior to entering the period of extended operation. Therefore, the stated design capacity of the GWPS remains sufficient, and the analysis of the maximum fission product inventory in the GWPS over a 40-year plant life remains bounding for a 60-year plant life.

B. The activity of the system's inventory is distributed among seven normal operation GDTs, minimizing inventory in any single tank.

The removal of fission gas from the reactor coolant by the GWPS during normal operation reduces the plant activity levels caused by a leakage of reactor coolant. Operation of the GWPS allows the collection of virtually all the krypton-85 released to the reactor coolant and can reduce the fission product gas inventory in the RCS, as shown in table 11.3.2-3. Table 11.3.2-3 is based on the RCS activities given in table 11.1-2. Provisions are made to collect any gases from the RCDT and gases from the recycle holdup tanks.

Process flow diagrams are shown in figures 11.3.2-3 and 11.3.2-4. Table 11.3.2-4 gives process parameters for key locations in the GWPS for the normal operating mode with the normal operation GDTs at low pressure (less than 25 psig). This operating mode is used early in plant life and is shown in figure 11.3.2-3. Table 11.3.2-6 gives process parameters for key locations in the system for the normal operating mode at the end of plant life[a], when the waste GDTs are at high pressure. This operating mode, shown in figure 11.3.2-4, is used from the time the normal operation GDT pressure reaches 25 psig until the end of plant life[a].

Table 11.3.2-5 gives the process parameters for key locations in the GWPS based on a 90-day holdup of the waste gases.

11.3.2.1 Component Design

The gaseous waste processing equipment design parameters are given in table 11.3.2-7. Since the GWPS performs no function related to the safe shutdown of the plant, some components are classified as nonnuclear safety. Component safety classes and the corresponding code and code class are shown in section 3.2.

11.3.2.1.1 Waste Gas Compressor Packages

Two waste gas compressor packages are provided to circulate gases around each GWPS system loop. One is normally in use and the other on a standby basis.

The compressor packages are water-sealed centrifugal displacement machines which are skid mounted in a self-contained package. The waste gas compressor packages are primarily constructed of carbon steel. Mechanical seals are provided to minimize seal water leakage.

11.3.2.1.2 Catalytic Hydrogen Recombiner Packages

Three catalytic hydrogen recombiners are provided for the two units. One recombiner per unit is used in each main process loop to remove hydrogen from the hydrogen-nitrogen fission gas mixtures by oxidation to water vapor, which is removed by condensation. The third recombiner is available on a standby basis. The units are self-contained and are designed for continuous operation.

[a] The renewed operating licenses authorized a 20-year period of extended operation for both VEGP units, resulting in a total plant operating life of 60 years. Since the inventory in the WGDTs has been routinely released during the first 20 years of operation and is expected to continue to be routinely released during future operation, the inventory of the WGDTs accumulated during the first 20 years of operation will be released prior to entering the period of extended operation. Therefore, the stated design capacity of the GWPS remains sufficient, and the analysis of the maximum fission product inventory in the GWPS over a 40-year plant life remains bounding for a 60-year plant life.

11.3.2.1.3 Waste Gas Decay Tanks

The tanks are vertical cylindrical type constructed of carbon steel. There are a total of 16 waste decay tanks. Seven are used by each unit during normal operation; the remaining two are shared between the two units and are used for shutdown and startup.

11.3.2.1.4 Gas Decay Tank Drain Pump

The waste GDTs may contain water from condensation from the process gas or from maintenance operations. This pump is used to drain water from the waste GDTs when there is insufficient pressure in the GWPS to transfer the fluid. Two canned motor pumps are provided, one for each set of seven normal operation GDTs.

11.3.2.1.5 Waste Gas Drain Filter

Two waste gas drain filters are provided, one to filter the drain water from each of the two process loops.

11.3.2.1.6 Waste Gas Traps

The gas traps serve to prevent undissolved gases from leaving the GWPS via the liquid drains. For each of the two process loops there are four gas traps - one in the compressors drain line, one in the recombiner drain line, and two mounted in the waste GDT drain line. There are also gas traps on each low point of the gaseous portion of the system to avoid condensate accumulation in the piping that may block the flow path.

11.3.2.1.7 Valves and Piping

Each valve in the hydrogen recombiner packages is designed to meet the temperature, pressure, and code requirements for the specific application in which it is used. The recombiner circuits contain manual valves provided with a metal diaphragm to prevent stem leakage and control valves provided with gaseous leakoffs returned to the GWPS. Other parts of the GWPS use elastomer diaphragm valves and control valves with bellows seals. Relief valves have soft seats and operate at pressures which are normally less than two-thirds of the relief valve set pressure. The relief valves of the major components discharge to one of the shutdown GDTs. This allows the discharge to be monitored before being released. It also provides a means of containing and detecting seal leakage across the relief valves.

All piping from the waste GDTs, up to and including the isolation valves, is designed to Seismic Category 1 requirements to preclude any accidental release of gas to the environment.

11.3.2.2 Instrumentation and Control Design

The GWPS instrumentation is shown on the piping and instrumentation diagram, drawings 1X4DB128 and 1X4DB129.

The instrumentation readout is located mainly on the waste processing system (WPS) panel, while some instruments are read locally.

All alarms are shown separately on the WPS panel and further relayed to one common WPS annunciator on the main control board.

Where suitable, instrument lines are provided with diaphragm seals to prevent fission gas leakage through the instrument.

11.3.2.2.1 Waste Gas Compressor Package Instrumentation and Control

Figure 11.3.2-5 shows the location of the instruments on the waste gas compressor package.

The compressors are interlocked with the seal water inventory in the moisture separators; they trip off on either high or low moisture separator level.

During normal operation the proper seal water inventory is maintained automatically.

11.3.2.2.2 Hydrogen Recombiner Package Instrumentation and Control

The catalytic hydrogen recombiner packages are designed for automatic operation with a minimum of operator attention.

Figure 11.3.2-6 shows the location of the instruments on the hydrogen recombiner. A multipoint temperature recorder monitors temperatures at several locations in the packages. Process gas flowrate is measured by an orifice located upstream of the hydrogen recombiner preheater. Local pressure gauges indicate the hydrogen recombiner process inlet pressure and the oxygen supply pressure.

The oxygen concentration is monitored and controlled to ensure that a flammable hydrogen-oxygen mixture does not occur. The GWPS is provided with two analyzers to monitor oxygen concentrations. One is between the oxygen supply and the hydrogen recombiner package, and one is downstream of the hydrogen recombiner. When the hydrogen concentration is above the lower flammability limit of 4 percent, the minimum concentration of oxygen necessary for deflagration is 5 percent. The control function assigned to these analyzers is to automatically terminate the oxygen supply before reaching GWPS oxygen concentrations favorable for hydrogen flammability. Each hydrogen recombiner package also includes two hydrogen analyzers. One monitors the process stream entering the hydrogen recombiner, and one monitors the discharge stream.

The controls and alarms incorporated to maintain the gas composition outside the range of flammable and explosive mixtures are described below.

A. The maximum concentration of hydrogen that the hydrogen recombiner can process in a single pass is 6-percent hydrogen by volume.

B. If the hydrogen recombiner feed concentration exceeds 6-percent hydrogen by volume, a high-hydrogen alarm sounds to warn that all hydrogen entering the hydrogen recombiner is not reacted. This alarm will be followed by a second alarm indicating high hydrogen in the hydrogen recombiner discharge. These alarms warn of a possible hydrogen accumulation in the GWPS.

C. If the hydrogen concentration in the hydrogen recombiner feed reaches 9 percent by volume, a high-high hydrogen alarm sounds; the oxygen feed is terminated; and the VCT hydrogen purge flow is terminated. These controls limit the possible accumulation of hydrogen in the system to 3 percent by volume.

D. If the oxygen concentration in the hydrogen recombiner feed reaches 3 percent by volume, an alarm sounds and oxygen feed flow is limited so that no further increase in flow is possible. This control maintains the system oxygen concentration at 3 percent or less, which is below the flammable limit for hydrogen-oxygen mixtures.

E. If hydrogen in the hydrogen recombiner discharge exceeds 1.5 percent by volume, an alarm sounds. This alarm warns of high hydrogen feed, possible reactor malfunction, or loss of oxygen feed.

F. If oxygen in the hydrogen recombiner discharge exceeds 15 ppm, a high alarm sounds. If it exceeds 60 ppm, a high-high alarm sounds and oxygen feed is terminated. This control prevents any accumulation of oxygen in the system in case of reactor malfunction.

G. On low flow through the hydrogen recombiner, oxygen feed is terminated. This control prevents an accumulation of oxygen following system malfunction.

H. On high discharge temperature from the cooler-condenser (downstream from the catalytic reactor) oxygen feed is terminated. This protects against loss of cooling water flow in the cooler-condenser.

I. On high temperature indication by any one of six thermocouples in the catalyst bed, the oxygen feed is terminated, and the VCT hydrogen purge flow is terminated.

J. On high temperature indication at the recombiner catalytic reactor discharge, oxygen feed to the recombiner is terminated.

11.3.2.2.3 Volume Control Tank Purge Flow Control

Each VCT purge line into the GWPS is equipped with a trip valve, a pressure-reducing valve and a hand-operated flow control valve. This arrangement will maintain a constant purge flow as selected by the operator.

Pressure fluctuations caused by changes in the VCT water level are absorbed by the pressure-regulating valve, which maintains a constant downstream pressure. This provides a constant head loss across the hand-operated control valve. Design flow is 0.7 sf^3/min hydrogen, with a range of 0.3 sf^3/min to 1.2 sf^3/min. When mixed with the 40 sf^3/min nitrogen stream, the maximum hydrogen content (both units operating) in the recombiner feed is 6.0 volume percent.

The purge line trip valve closes as a result of:

A. A low-pressure signal from the VCT pressure instrumentation. This prevents depressurization of the VCT if the hydrogen supply is lost.

B. A low-pressure signal from the gas compressor suction line. This signal also stops the gas compressor.

C. Closure of the oxygen supply valve to the hydrogen recombiner. The purge line trip valve is closed only if the oxygen supply valves for both the normally operating recombiner and the standby recombiner are closed.

11.3.2.3 System Operation

11.3.2.3.1 Startup Operation

The GWPS is initially purged with nitrogen to remove all air. During startup operation, one waste gas compressor, one hydrogen recombiner, and one shutdown GDT are in service in the process loop serving that plant unit. The reactor is at cold shutdown and the VCT contains nitrogen in the gas space. Reactor coolant contains neither hydrogen nor fission gases, but it may be saturated with air. While one unit is being started up, the VCT purge for the other unit is unaffected since each unit is served by a separate process loop of the GWPS.

When the reactor startup procedure requires that a hydrogen blanket be established in the VCT gas space, fresh hydrogen is charged into the VCT. The hydrogen-nitrogen mixture vented from the VCT enters the GWPS circulating nitrogen stream at the waste gas compressor suction. Nitrogen added to the GWPS accumulates in the shutdown GDT, causing the tank's pressure to rise.

Initially, the VCT vent gas will be very lean in hydrogen, and almost all the gas entering the GWPS will accumulate in the shutdown GDT. As the operation continues, however, the vent gas hydrogen content will gradually increase until it is almost totally hydrogen at the point when all of the nitrogen has been removed from the reactor coolant. At that time, hydrogen gas is passing through the VCT and mixing with the circulating nitrogen stream to give a mixture of hydrogen in nitrogen at the hydrogen recombiner inlet. A sufficient amount of oxygen is added in the hydrogen recombiner to react with the hydrogen to yield a discharge stream with a low residual concentration of hydrogen in nitrogen. After the water vapor is condensed and removed, the gas flow is directed to a shutdown GDT and from there to the waste gas compressor and back to the hydrogen recombiner.

When the reactor coolant hydrogen concentration is within operating specifications, the shutdown GDT is isolated, and flow is routed to one of the normal operation GDTs provided for normal power service. Gas accumulated in the shutdown GDT will be retained for use during operations to strip hydrogen from the reactor coolant when a plant unit is shut down.

11.3.2.3.2 Normal Operations

During normal power operation, nitrogen gas with entrained fission gases is continuously circulated around each GWPS process loop by one of the two waste gas compressors in the loop. Fresh hydrogen gas is charged to the VCT, where it is mixed with fission gases which have been stripped from the reactor coolant into the VCT gas space. The contaminated hydrogen gas is continuously vented from the VCT into the circulating nitrogen stream to transport the fission gases into the GWPS. The resulting mixture of nitrogen-hydrogen fission gas is pumped by the waste gas compressor to the hydrogen recombiner, where enough oxygen is added to reduce the hydrogen to a low residual concentration by oxidation to water vapor on a catalytic surface. After the water vapor is removed, the resulting gas stream is circulated to a normal operation GDT and back to the waste gas compressor suction to complete the circuit.

Each normal operation GDT is capable of being isolated, and only one tank is valved into operation in each process loop at any time. This minimizes the amount of radioactive gases that could be released as a consequence of any single failure, such as the rupture of any single tank or connected piping. A normal operation GDT is valved into the GWPS recirculation loop

until the pressure or curie content of the inservice GDT reaches a value determined by approved plant procedure(s), after which it is isolated and another tank is placed in service. This process is illustrated in figure 11.3.2-3 for GDT pressure ≤ 20 psig and as illustrated in figure 11.3.2-4 when GDT pressure will exceed 20 psig. By alternating the use of these tanks, the accumulated activity from one unit is distributed among all seven normal operation GDTs in the process loop.

With continued plant operation, pressure in the normal operation will gradually increase as nonremovable gases accumulate in the system. The initial system equipment lineup, as described above, will be from the waste gas compressor to the hydrogen recombiner and then to the normal operation GDT. As the normal operation GDT pressure builds up, compensation must be made by periodic adjustment of the hydrogen recombiner backpressure control valve. When the normal operation GDT pressure reaches approximately 20 psig, the backpressure control valve will be fully open, so that no more adjustment can be made. At this time, the appropriate bypass lines are opened to line up the equipment for flow from the waste gas compressor to the normal operation GDTs and then to the hydrogen recombiner, as shown in figure 11.3.2-4. The hydrogen recombiner backpressure control valve is reset as required for the new arrangement, and the normal operation GDT pressure indicators are switched to read high range. This arrangement is suitable for operation up to 100 psig. Note that this high-pressure mode of operation will also normally be utilized during shutdown/startup operations.

11.3.2.3.3 Shutdown and Degassing of the Reactor Coolant System

Plant shutdown operations are essentially startup operations in reverse sequence. The VCT hydrogen blanket is maintained until after the reactor is shut down and reactor coolant fission gas concentrations have been reduced to desired level. During this operation hydrogen purge flow may be increased to speed up reactor coolant degassing. At this time, the VCT hydrogen purge is stopped for the plant unit being shut down. The normal operation GDT in service for that unit is then valved out and a nitrogen purge from a shutdown GDT to the VCT is begun for the unit being shut down. This shutdown GDT is placed in the GWPS process loop at the waste gas compressor discharge so that the gas mixture from the VCT vents to the waste gas compressor suction, passes through the shutdown GDT and to the hydrogen recombiner, where hydrogen is removed, and remaining gases are returned to the waste gas compressor suction. The nitrogen purge continues until reactor coolant hydrogen concentration reaches the required level or chemical degassing may be used by the operator to reduce RCS hydrogen concentrations during cold shutdown operations. Degassing is then complete, and the RCS may be opened for maintenance or refueling. At this point the shutdown GDT may be isolated.

During the first plant cold shutdown, fresh nitrogen is charged to the VCT to strip hydrogen from the reactor coolant. The resulting accumulation of nitrogen in the shutdown GDT is accommodated by allowing the tank pressure to increase. During subsequent shutdowns, however, there is no additional accumulation since the gas from the first shutdown will be reused.

11.3.2.3.4 System Drains

During operation, water may accumulate in the waste gas compressor moisture separator, in the hydrogen recombiner phase separator, and in the GDTs. Normally, the waste gas compressor and hydrogen recombiner drains will discharge to the VCT under motive head provided by internal component pressure. During maintenance, the drains are directed either to the recycle holdup tanks (if the drains contain dissolved fission gases) or to the waste holdup

tank through the drain header (if the drains contain no fission gases). A gas trap is provided in the drain lines from the waste gas compressors and hydrogen recombiners to prevent undissolved gases from leaving the GWPS.

All drains from the waste GDTs are manually operated. Depending on the internal tank pressure on the drain routing, the waste GDT drain pump is used or bypassed. As necessary, the tank is drained by pumping the accumulated condensate to the VCT. However, during tank maintenance, waste GDT drains are routed to the recycle holdup tank because the volume of water in a waste GDT may be large enough to cause a noticeable dilution of reactor coolant boron concentration. Gas traps are provided in the waste GDT drain line to prevent the discharge of gases into other parts of the plant. All drains from the GWPS are filtered before entering either the VCT or the recycle holdup tank.

11.3.2.3.5 Atmospheric Releases

Although the GWPS is designed to accommodate continuous operation without atmospheric releases, the system design permits controlled discharge of gas from the system. Before a tank is emptied to the atmosphere via the plant vent, a gas sample must be analyzed to determine and record the activity to be removed. While the contents of a tank are being released to the atmosphere, a trip valve in the discharge line will close automatically if a high activity level is detected in the plant vent effluent.

11.3.3 RADIOACTIVE RELEASES

11.3.3.1 Discharge Requirements

Radioactive gaseous effluents and particulates discharged from the plant may not exceed the limits specified in the Offsite Dose Calculation Manual.

11.3.3.2 Westinghouse Pressurized Water Reactor (PWR) Experience Releases

Surveys have been performed of gaseous discharges from several operating Westinghouse PWR plants. The results are presented in table 11.3.3-1.

11.3.3.3 Estimated Releases

Radioactive effluent releases from the plant for normal operation are given in table 11.3.3-2. These release rates were calculated using the PWR-GALE code (reference 1) and plant operating parameters referenced in paragraph 11.1.1.2 and table 11.1-8. The releases are calculated for one unit; to obtain the combined releases for the two units, double the values listed in table 11.3.3-2.

Potential effluent releases from the radwaste solidification building vent are provided in table 11.3.3-2. These releases were conservatively estimated using the design basis reactor coolant activities in table 11.1-2 adjusted to 0.12-percent failed fuel as the source term. These sources were used in conjunction with the waste stream flow and processing methodology outlined in the GALE code to determine the activities of the wastes being input to the volume reduction

system. These releases were determined using the decontamination factors for the system outlined in section 11.4.

11.3.3.4 Release Points for Determination of Dilution Factors

Gaseous and particulate radioactive effluents may be normally released through the following three points: the plant vent, turbine building vent, and radwaste processing facility vent. Subsection 2.3.5 outlines the methodology and information used to determine the long-term atmospheric dilution (χ/Q) and deposition (D/Q) factors for these release points. Effluent sources and associated vents are listed in table 2.3.5-2. Vent design information and input assumptions utilized for the long-term diffusion estimates are given in tables 2.3.5-3 and 2.3.5-4.

11.3.3.5 Effluent Concentration and Dilution Factors

A comparison of maximum offsite (at site boundary) airborne gaseous and particulate effluent concentrations with 10 CFR 20.1 - 20.601 limits is given in table 11.3.3-3. These concentrations are based on the sum of concentrations from the plant vent and turbine building vent using the site boundary dilution factors for the wake split model (plant vent) and the ground release model (turbine building vent) as given in subsection 2.3.5 and tables 2.3.5-10 and 2.3.5-11. For this calculation, the GALE code isotopic release rates in table 11.3.3-2 were grouped as follows:

A. The plant vent releases are composed of contributions from gas stripping, reactor building, auxiliary building, and waste gas system.

B. The turbine building vent releases are composed of contributions from turbine building and air ejector exhaust.

11.3.3.6 Estimated Doses from Atmospheric Releases

Estimated annual average doses from radionuclides released to the atmosphere from the plant during normal operation are given in table 11.3.3-4. These doses were calculated using the GASPAR code (reference 2). Various exposure pathways and nearest potential receptors within 5.0 miles (~8000 meters) of the plant were considered. As in paragraph 11.3.3.5, the isotopic release rates in table 11.3.3-2 were grouped according to discharge vent and associated atmospheric diffusion model. Thus, table 11.3.3-4 represents the total dose based on the releases from the plant vent and turbine building vent using the diffusion data referenced in paragraph 11.3.3.4. Note that all of these calculated doses are well within the guidelines of Appendix I to 10 CFR 50, and the Annex to Appendix I, Docket RM-50-2.

11.3.3.7 References

1. U.S. Nuclear Regulatory Commission, "Calculation of Releases from Pressurized Water Reactors," PWR-GALE Computer Code, NUREG-0017, April 1976.

2. U.S. Nuclear Regulatory Commission, "Calculation of Radiation Exposure to Man from Routine Air Releases of Nuclear Reactor Effluents," GASPAR Computer Code, NUREG-0597, June 1980 (based on the methodology outlined in Regulatory Guide 1.109, Revision 1, for atmospheric releases).

TABLE 11.3.2-1

DESIGN BASIS ACCUMULATED RADIOACTIVITY IN THE GWPS
AFTER 40 YEARS' OPERATION OF TWO UNITS[a]

	Activity (Ci) Following Plant Shutdown[b]		
Isotope	Zero Decay	30 Days	50 Days
Kr-85	78,000	77,600	77,300
All other noble gases			
Kr-83m	9	0	0
Kr-85m	116	0	0
Kr-87	10.8	0	0
Kr-88	110	0	0
Xe-131m	1240	230	69
Xe-133	132,000	2550	180
Xe-133m	6600	0	0
Xe-135	960	0	0
Xe-135m	3.8	0	0
Xe-138	0.26	0	0

The table is based on 40 years continuous operation[a] with 1 percent of rated core power generated by fuel rods containing cladding defects; therefore, the values are conservatively high. Power is assumed to be 3565 MWt. The data are based on a VCT purge rate of 0.7 sf^3/min for each of 2 units, the stripping fractions listed in table 11.1-1, and the reactor coolant concentrations given in table 11.1-2.

[a] The renewed operating licenses authorized a 20-year period of extended operation for both VEGP units, resulting in a total plant operating life of 60 years. Since the inventory in the Waste Gas Decay Tanks (WGDTs) has been routinely released during the first 20 years of operation and is expected to continue to be routinely released during future operation, the inventory of the WGDTs accumulated during the first 20 years of operation will be released prior to entering the period of extended operation. Therefore, the stated design capacity of the GWPS remains sufficient, and the analysis of the maximum fission product inventory in the GWPS over a 40-year plant life remains bounding for a 60-year plant life.

[b] A review of the table values for the effects of MUR power uprate determined those values need not be updated.

TABLE 11.3.2-2

EXPECTED ACCUMULATED RADIOACTIVITY IN THE GWPS
AFTER 40 YEARS' OPERATION OF TWO UNITS[a]

	Activity (Ci) Following Plant Shutdown[b]		
Isotope	Zero Decay	30 Days	50 Days
Kr-85	9400	9350	9320
All other noble gases			
Kr-83m	0.32	0	0
Kr-85m	5.6	0	0
Kr-87	0.52	0	0
Kr-88	5.6	0	0
Xe-131m	70	12	4
Xe-133	8600	170	12
Xe-135	74	0	0
Xe-135m	0.006	0	0
Xe-138	0.018	0	0

Inventories are based on reactor coolant concentrations given in table 11.1-7. The table is based on 40 years continuous operation[a]. Power is assumed to be 3565 MWt. The data are based on a VCT purge rate of 0.7 sf^3/min for each of 2 units and on the stripping fractions listed in table 11.1-7.

[a] The renewed operating licenses authorized a 20-year period of extended operation for both VEGP units, resulting in a total plant operating life of 60 years. Since the inventory in the Waste Gas Decay Tanks (WGDTs) has been routinely released during the first 20 years of operation and is expected to continue to be routinely released during future operation, the inventory of the WGDTs accumulated during the first 20 years of operation will be released prior to entering the period of extended operation. Therefore, the stated design capacity of the GWPS remains sufficient, and the analysis of the maximum fission product inventory in the GWPS over a 40-year plant life remains bounding for a 60-year plant life.

[b] A review of the table values for the effects of MUR power uprate determined those values need not be updated.

TABLE 11.3.2-3

REDUCTION IN RCS GASEOUS FISSION PRODUCTS
RESULTING FROM NORMAL OPERATION OF THE GWPS[a] [c]

	Reactor Coolant Gaseous Fission Product Activities (μCi/g)	
Isotope	GWPS Operating[b]	GWPS Not Operating
Kr-83m	0.45	0.46
Kr-85	0.059	7.3
Kr-85m	1.9	2.0
Kr-87	1.3	1.3
Kr-88	3.6	3.6
Xe-131m	0.25	2.2
Xe-133	62	270
Xe-133m	7.4	17
Xe-135	6.1	7.2
Xe-135m	0.48	0.48
Xe-138	0.64	0.64

a. Based on operating with cladding defects in fuel generating 1 percent of the rated core thermal power (3565 MWt) and a purification letdown rate of 75 gal/min.

b. The VCT purge rate is 0.7 sf^3/min.

c. A review of the table values for the effects of MUR power uprate determined those values need not be updated.

TABLE 11.3.2-4 (SHEET 1 OF 2)
PROCESS PARAMETERS FOR GWPS, INDEFINITE HOLDUP, 1-MONTH ACCUMULATION [a][b][j]
(FIRST HALF)

Item	Description Gas Streams	Temp (°F)	Press (psig)	Flow (sf^3/min)	N_2 + He (%)	H_2 (%)	Isotopic Concentration (μCi/cm^3) Kr-83m	Kr-85	Kr-85m	Kr-87
1.	VCT purge[e]	130	18	0.7	0	100	4.36(-2)	2.56(-2)	3.10(-1)	9.89(-2)
2.	GDT discharge	Amb[i]	5	40	99.9	0.1	6.74(-3)	1.69(1-)	1.15(-1)	1.10(-2)
3.	Compressor suction	Amb	0.5	40.7	98.3	1.7	7.38(-3)	1.66(1-)	1.18(-1)	1.25(-2)
4.	Compressor discharge	140	30	40.7	98.3	1.7	7.38(-3)	1.66(1-)	1.18(-1)	1.25(-2)
5.	Recombiner discharge	140	5	40	99.9	0.1	7.50(-3)	1.69(-1)	1.20(-1)	1.27(-2)
6.	Misc. vents: evap. RCDT recycle holdup tank eductor	140	0.5	Neg[g]	0	100	0	0	0	0
7.	Recombiner oxygen supply	Amb	50	0.35	0	0	0	0	0	0

Item	Description Liquid Streams	Temp (°F)	Press (psig)	Flow (gal/d)	Isotopic Concentration (mCi/cm^3)[g][n] Kr-83m	Kr-85	Kr-85m	Kr-87
1.	Waste gas compressor drain	140	40	0	1.80(-3)	4.06(-2)	2.89(-2)	3.04(-3)
2.	Recombiner drain	140	30	12	1.50(-3)	3.38(-2)	2.40(-2)	2.53(-3)
3.	GDT drain	Amb	5	36	5.98(-4)	1.50(-2)	1.02(-2)	9.71(-4)
4.	System drains to VCT	140	5-40	48	7.26(-4)	1.77(-2)	1.22(-2)	1.19(-3)

Item	Component	Temp (°F)	Press (psig)	Vol (ft^3)	N_2 + He (%)	H_2 (%)	Component Inventory (Ci/)[l] Kr-83m	Kr-85	Kr-85m	Kr-87
1.	Compressor	140	40	4	98.3	1.7	2.84(-3)	7.12(-2)	4.85(-2)	4.62(-3)
2.	Recombiners	140	30	4	99.3	0.1	2.32(-3)	5.82(-2)	3.96(-2)	3.77(-3)
3.	GDT	Amb	5	600	99.9	0.1	1.55(-1)	3.88	2.64	2.52(-1)
Total System							1.6(-1)	27.3	2.8	2.6(-1)

TABLE 11.3.2-4 (SHEET 1 OF 2)
PROCESS PARAMETERS FOR GWPS, INDEFINITE HOLDUP, 1-MONTH ACCUMULATION[(a)(b)(j)]
(SECOND HALF)

Kr-88	Kr-89	Xe-131m	Xe-133	Xe-133m	Xe-135	Xe-135m	Xe-137	Xe-138
4.94(-1)	5.39(-4)	6.29(-2)	16.6	3.46(-1)	9.0(-1)	5.83(-3)	1.17(-3)	1.93(-2)
1.18(1-)	2.40(-6)	2.50(-1)	37.1	4.70(-1)	5.99(-1)	1.31(-4)	6.32(-6)	3.83(-4)
1.24(-1)	1.16(-5)	2.46(-1)	36.7	4.68(-1)	6.04(-1)	2.29(-4)	2.63(-5)	7.09(-4)
1.24(-1)	1.16(-5)	2.46(-1)	36.7	4.68(-1)	6.04(-1)	2.29(-4)	2.63(-5)	7.09(-4)
1.26(-1)	1.18(-5)	2.50(-1)	37.4	4.76(-1)	6.15(-1)	2.33(-4)	2.68(-5)	7.22(-4)
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0

Kr-88	Kr-89	Xe-131m	Xe-133	Xe-133m	Xe-135	Xe-135m	Xe-137	Xe-138
3.03(-2)	2.84(-6)	4.96(-2)	7.40	9.44(-2)	1.22(-1)	4.62(-5)	5.31(-6)	1.43(-4)
2.52(-2)	2.36(-6)	4.12(-2)	6.16	7.85(-2)	1.01(-1)	3.84(-5)	4.41(-6)	1.19(-4)
1.04(-2)	2.13(-7)	1.82(-2)	2.71	3.44(-2)	4.39(-2)	9.57(-6)	4.63(-7)	2.81(-5)
1.25(-2)	5.20(-7)	2.15(-2)	3.21	4.07(-2)	5.21(-2)	1.37(-5)	1.03(-6)	4.11(-5)

Kr-88	Kr-89	Xe-131m	Xe-133	Xe-133m	Xe-135	Xe-135m	Xe-137	Xe-138
4.96(-2)	1.01(-6)	1.05(-1)	15.6	1.98(-1)	2.53(-1)	5.51(-5)	2.66(-6)	1.62(-4)
4.05(-2)	8.27(-7)	8.57(-2)	12.8	1.62(-1)	2.06(-1)	4.50(-5)	2.18(-6)	1.32(-4)
2.70	5.52(-5)	5.72	851	10.8	13.8	3.0(-3)	1.45(-4)	8.81(-3)
2.8	5.7(-5)	35	4300	37	17	3.1(-3)	1.5(-4)	9.1(-3)

TABLE 11.3.2-4 (SHEET 2 OF 2)

NOTES:

a. Basis: Type of operation = indefinite holdup

Power level = 3656 MWt

Number of units = 1

Normal operation GDTs in rotational use = 7

GDT operating interval = 1 day

Accumulation period = 1 month

b. Concentrations based on stripping fractions and reactor coolant activities from table 11.1-2.

c. Concentrations in $\mu Ci/cm^3$ of gas at atmospheric pressure and 140°F.

d. Isotopic concentrations are maximum values for all isotopes except krypton-85. Krypton-85 is at its maximum concentration after 14 ½ years of operation.

e. Parameters reflect the gas streams from one of the two operating reactors.

f. Amb = ambient.

g. Neg = negligible.

h. Concentrations in $\mu Ci/cm^3$ at room temperature.

i. Total system inventories are maximum values except for krypton-85. The maximum inventory of krypton-85 occurs at the end of plant life[1]. For all other isotopes, the maximum inventory is reached soon after initiating plant operation and is maintained until the end of plant life.

j. A review of the table values for the effects of MUR power uprate determined those values need not be updated.

[1] The renewed operating licenses authorized a 20-year period of extended operation for both VEGP units, resulting in a total plant operating life of 60 years. Since the inventory in the Waste Gas Decay Tanks (WGDTs) has been routinely released during the first 20 years of operation and is expected to continue to be routinely released during future operation, the inventory of the WGDTs accumulated during the first 20 years of operation will be released prior to entering the period of extended operation. Therefore, the stated design capacity of the GWPS remains sufficient, and the analysis of the maximum fission product inventory in the GWPS over a 40-year plant life remains bounding for a 60-year plant life.

TABLE 11.3.2-5 (SHEET 1 OF 2)
PROCESS PARAMETERS FOR GWPS, INDEFINITE HOLDUP, 90-DAY ACCUMULATION[a][b][g]
(FIRST HALF)

Item	Description Gas Streams	Temp (°F)	Press (psig)	Flow (sf^3/min)	N_2 + He (%)	H_2 (%)	Isotopic Concentration (mCi/cm^3)[c] Kr-83m	Kr-85	Kr-85m	Kr-87
1.	VCT purge[a]	130	18	0.7	0	100	4.36(-2)	2.56(-2)	3.10(-2)	9.89(-2)
2.	GDT discharge	Amb	20	40	99.9	0.1	1.37(-3)	5.79	2.34(-2)	2.23(-3)
3.	Compressor suction	Amb	0.5	40.7	98.3	1.7	2.10(-3)	5.69	2.36(-2)	3.89(-3)
4.	Compressor discharge	140	30	40.7	98.3	1.7	2.10(-3)	5.69	2.36(-2)	3.89(-3)
5.	Recombiner discharge	140	20	40	99.9	0.1	2.14(-3)	5.79	2.40(-2)	3.96(-3)
6.	Misc. vents: evap. RCDT recycle holdup tank eductor[d]	140	0.5	Neg	0	100	0	0	0	0
7.	Recombiner oxygen supply	Amb	50	0.35	0	0	0	0	0	0

Item	Description Liquid Streams	Temp (°F)	Press (psig)	Flow (gal/d)	Isotopic Concentration (mCi/cm^3)[e] Kr-83m	Kr-85	Kr-85m	Kr-87
1.	Waste gas compressor drain	140	40	0	5.12(-4)	1.39	5.75(-4)	9.50(-4)
2.	Recombiner drain	140	30	12	4.26(-4)	1.16	4.78(-3)	7.90(-4)
3.	GDT drain	Amb	20	36	6.14(-4)	2.59	1.05(-2)	9.97(-4)
4.	System drains to VCT	140	30-45	48	5.87(-4)	2.38	9.66(-3)	9.68(-4)

Item	Component	Temp (°F)	Press (psig)	Vol (ft^3)	N_2 + He (%)	H_2 (%)	Component Inventory (Ci/) Kr-83m	Kr-85	Kr-85m	Kr-87
1.	Compressor	140	40	4	98.3	1.7	5.78(-4)	2.44	9.88(-3)	9.40(-4)
2.	Recombiners	140	30	4	99.9	0.1	4.73(-4)	1.99	8.07(-3)	7.68(-4)
3.	GDT	Amb	20	600	99.9	0.1	1.59(-1)	671	2.71	2.58(-1)
Total System							1.6(-1)	4700	2.8	2.6(-1)

TABLE 11.3.2-5 (SHEET 1 OF 2)
PROCESS PARAMETERS FOR GWPS, INDEFINITE HOLDUP, 90-DAY ACCUMULATION[a][b][g]
(SECOND HALF)

Kr-88	Kr-89	Xe-131m	Xe-133	Xe-133m	Xe-135	Xe-135m	Xe-137	Xe-138
4.94(-1)	5.39(-4)	6.29(-2)	16.6	3.46(-1)	8.99(-1)	5.84(-3)	1.17(-3)	1.93(-2)
2.40(2-)	4.89(-7)	5.07(-2)	7.5	9.57(-2)	1.22(-1)	2.66(-5)	1.29(-6)	7.81(-5)
3.20(-2)	9.76(-6)	5.09(-2)	7.7	1.0(-1)	1.35(-1)	1.27(-4)	2.14(-5)	4.09(-4)
3.20(-2)	9.76(-6)	5.09(-2)	7.7	1.0(-1)	1.35(-1)	1.27(-4)	2.14(-5)	4.09(-4)
3.26(-2)	9.93(-6)	5.18(-2)	7.8	1.02(-1)	1.38(-1)	1.29(-4)	2.17(-5)	4.16(-4)
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0

Kr-88	Kr-89	Xe-131m	Xe-133	Xe-133m	Xe-135	Xe-135m	Xe-137	Xe-138
7.82(-3)	2.38(-6)	1.03(-2)	1.55	2.02(-2)	2.73(-2)	2.55(-5)	4.31(-6)	8.25(-5)
6.50(-3)	1.98(-6)	8.53(-3)	1.29	1.68(-2)	2.27(-2)	2.12(-5)	3.58(-6)	6.86(-5)
1.07(-2)	2.19(-7)	1.87(-2)	2.79	3.53(-2)	4.51(-2)	9.82(-6)	4.75(-7)	2.88(-5)
1.01(-2)	4.70(-7)	1.73(-2)	2.57	3.27(-2)	4.19(-2)	1.15(-5)	9.19(-7)	3.45(-5)

Kr-88	Kr-89	Xe-131m	Xe-133	Xe-133m	Xe-135	Xe-135m	Xe-137	Xe-138
1.01(-2)	2.06(-7)	2.14(-2)	3.18	4.03(-2)	5.14(-2)	1.12(-5)	5.42(-7)	3.29(-5)
8.25(-3)	1.68(-7)	1.75(-2)	2.60	3.30(-2)	4.20(-2)	9.16(-6)	4.43(-7)	2.69(-5)
2.77	5.66(-5)	5.87	874	11.1	14.1	3.08(-3)	1.49(-4)	9.04(-3)
2.8	5.7(-5)	35	4300	37	17	3.1(-3)	1.5(-4)	9.1(-3)

TABLE 11.3.2-5 (SHEET 2 OF 2)

NOTES:

a. Basis: Type of operation = periodic release of gases

Power level = 3565 MWt

Number of units = 1

Normal operation GDTs in rotational use = 7

GDT operating interval = 1 day

Accumulation period = 40 years[1]

b. Concentrations based on stripping fractions and reactor coolant activities from table 11.1-2.

c. Concentrations in $\mu Ci/cm^3$ of gas at atmospheric pressure and 140°F.

d. Parameters reflect the gas streams from one of the two operating reactors.

e. Concentrations in $\mu Ci/cm^3$ at room temperature and atmospheric pressure.

f. Total system inventories are maximum values.

g. A review of the table values for the effects of MUR power uprate determined those values need not be updated.

[1] The renewed operating licenses authorized a 20-year period of extended operation for both VEGP units, resulting in a total plant operating life of 60 years. Since the inventory in the Waste Gas Decay Tanks (WGDTs) has been routinely released during the first 20 years of operation and is expected to continue to be routinely released during future operation, the inventory of the WGDTs accumulated during the first 20 years of operation will be released prior to entering the period of extended operation. Therefore, the stated design capacity of the GWPS remains sufficient, and the analysis of the maximum fission product inventory in the GWPS over a 40-year plant life remains bounding for a 60-year plant life.

TABLE 11.3.2-6 (SHEET 1 OF 2)
PROCESS PARAMETERS FOR GWPS, INDEFINITE HOLDUP, 90-DAY HOLDUP AND RELEASE[a][b][f]
(FIRST HALF)

Item Description Gas Streams	Temp (°F)	Press (psig)	Flow (sf^3/min)	N_2 + He (%)	H_2 (%)	Isotopic Concentration (mCi/cm^3)[c] Kr-83m	Kr-85	Kr-85m	Kr-87
1. VCT purge[d]	130	18	0.7	0	100	4.36(-02)	2.56(-2)	3.10(-1)	9.89(-2)
2. GDT discharge	Amb	100	40	99.9	0.1	6.43(-3)	1.11	1.10(-1)	1.04(-2)
3. Compressor suction	Amb	0.5	40.7	98.3	1.7	7.07(-3)	1.09	1.08(-1)	1.20(-2)
4. Compressor discharge	140	100	40.7	98.3	1.7	7.07(-3)	1.09	1.08(-1)	1.20(-2)
5. Recombiner discharge	140	20	40	99.9	0.1	7.19(-3)	1.11	1.10(-1)	1.22(-2)
6. Misc. vents: evap. RCDT recycle holdup tank eductor[d]	140	0.5	Neg	0	100	0	0	0	0
7. Recombiner oxygen supply	Amb	50	0.35	0	0	0	0	0	0

Item Description Liquid Streams	Temp (°F)	Press (psig)	Flow (gal/d)	Isotopic Concentration (mCi/cm^3)[e] Kr-83m	Kr-85	Kr-85m	Kr-87
1. Waste gas compressor drain	140	40	0	1.72(-3)	2.67(-1)	2.65(-2)	2.92(-3)
2. Recombiner drain	140	30	12	1.43(-3)	2.22(-1)	2.20(-2)	2.43(-3)
3. GDT drain	Amb	100	36	5.99(-4)	1.04(-1)	1.02(-2)	9.73(-4)
4. System drains to VCT	140	30-100	48	7.18(-4)	1.21(-1)	1.19(-2)	1.18(-3)

Item Component	Temp (°F)	Press (psig)	Vol (ft^3)	N_2 + He (%)	H_2 (%)	Component Inventory (Ci/)[f] Kr-83m	Kr-85	Kr-85m	Kr-87
1. Compressor	140	40	4	98.3	1.7	2.71(-3)	4.69(-1)	4.63(-2)	4.40(-3)
2. Recombiners	140	30	4	99.9	0.1	2.21(-3)	3.84(-1)	3.78(-2)	3.60(-3)
3. GDT	Amb	100	600	99.9	0.1	1.55(-1)	26.9	2.65	2.52(-1)
Total System						1.6(-1)	81.4	2.8	2.6(-1)

TABLE 11.3.2-6 (SHEET 1 OF 2)
PROCESS PARAMETERS FOR GWPS, INDEFINITE HOLDUP, 90-DAY HOLDUP AND RELEASE[(a)(b)(f)]
(SECOND HALF)

Kr-88	Kr-89	Xe-131m	Xe-133	Xe-133m	Xe-135	Xe-135m	Xe-137	Xe-138
4.94(-1)	5.39(-4)	6.29(-2)	16.6	3.46(-1)	8.99(-1)	5.84(-3)	1.17(-3)	1.93(-2)
1.12(1-)	2.29(-6)	4.96(-1)	65.3	6.55(-1)	5.74(-1)	1.25(-4)	6.03(-6)	3.66(-4)
1.19(-1)	1.15(-5)	4.89(-1)	64.4	6.50(-1)	5.79(-1)	2.23(-4)	2.60(-5)	6.92(-4)
1.19(-1)	1.15(-5)	4.89(-1)	64.4	6.50(-1)	5.79(-1)	2.23(-4)	2.60(-5)	6.92(-4)
1.21(-1)	1.17(-5)	4.97(-1)	65.6	6.61(-1)	5.89(-1)	2.27(-4)	2.65(-5)	7.04(-4)
0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	0

Kr-88	Kr-89	Xe-131m	Xe-133	Xe-133m	Xe-135	Xe-135m	Xe-137	Xe-138
2.90(-2)	2.81(-6)	9.85(-2)	13	1.31(-1)	1.17(-1)	4.50(-5)	5.25(-6)	1.40(-4)
2.41(-2)	2.34(-6)	8.19(-2)	10.8	1.09(-1)	9.71(-2)	3.74(-5)	4.36(-6)	1.16(-4)
1.04(-2)	2.13(-7)	3.82(-2)	5	5.04(-2)	4.41(-2)	9.58(-6)	4.64(-7)	2.81(-5)
1.24(-2)	5.17(-7)	4.44(-2)	5.8	5.88(-2)	5.17(-2)	1.36(-5)	1.02(-6)	4.07(-5)

Kr-88	Kr-89	Xe-131m	Xe-133	Xe-133m	Xe-135	Xe-135m	Xe-137	Xe-138
4.73(-2)	9.65(-7)	2.09(-1)	27.5	2.76(-1)	2.42(-1)	5.25(-5)	2.54(-6)	1.54(-4)
3.86(-2)	7.89(-7)	1.71(-1)	22.5	2.26(-1)	1.98(-1)	4.29(-5)	2.08(-6)	1.26(-4)
2.71	5.52(-5)	12	1575	15.8	13.8	3.0(-3)	1.45(-4)	8.82(-3)
2.8	5.7(-5)	35	4300	37	17	3.1(-3)	1.5(-4)	9.1(-3)

TABLE 11.3.2-6 (SHEET 2 OF 2)

NOTES:

a. Basis: Type of operation = indefinite holdup

Power level = 3565 MWt

Number of units = 1

Normal operation GDTs in rotational use = 3

GDT operating interval = 1 day

Accumulation period = 90 days

b. Concentrations based on stripping fractions and reactor coolant activities from table 11.1-2.

c. Concentrations in $\mu Ci/cm^3$ of gas at atmospheric pressure and 140°F.

d. Parameters reflect the gas streams from one of the two operating reactors.

e. Concentrations in $\mu Ci/cm^3$ at room temperature and atmospheric pressure.

f. A review of the table values for the effects of MUR power uprate determined those values need not be updated.

TABLE 11.3.2-7 (SHEET 1 OF 2)

GWPS COMPONENT DESCRIPTION

Waste Gas Compressor Packages	
Number (two units)	4
Design pressure (psig)	150
Design temperature (°F)	180
Normal operating temperature (°F)	70-130
Normal operating pressure (psig)	
Suction	0.5-2.0
Discharge	25-100
Design flowrate (N_2 at 140°F, 110 psig discharge) (sf^3/min)	40
Material of construction	Carbon steel
Waste GDTs	
Number (7 per unit plus 2 shared)	16
Design pressure (psig)	150
Design temperature (°F)	180
Volume (each) (ft^3)	600
Normal operating pressure (psig)	5-100
Normal operating temperature (°F)	50-140
Material of construction	Carbon steel
Catalytic Hydrogen Recombiner Packages	
Number (2 units)	3
Design pressure (psig)	150
Design temperature (°F)	(a)
Inlet pressure (psig)	25-100
Inlet temperature (°F)	70-140
Design flowrate (sf^3/min)	50
Design hydrogen recombiner rate (sf^3/min)	3.0
Discharge pressure (psig)	0-20
Discharge temperature (°F)	70-140
Material of construction	Stainless steel

TABLE 11.3.2-7 (SHEET 2 OF 2)

GDT Drain Pump

Quantity (2 units)	2
Design temperature (°F)	180
Design pressure (psig)	150
Design flow (gal/min)	10
Design head (ft)	90
NPSH required (ft)	10
Maximum operating temperature (°F)	140
Maximum operating suction pressure (psig)	30[b]
Fluid pumped	Reactor makeup water
Material	Stainless steel

Waste Gas Drain Filter

Quantity (2 units)	2
Type	Disposable cartridge
Design temperature (°F)	200
Design pressure (psig)	150
Design flow (gal/min)	35
Retention of 25-μm particles (%)	98
Materials	Stainless steel

Gas Traps

Quantity (2 units)	2	10
Design temperature (°F)	180	180
Design pressure (psig)	150	150
Operating inlet pressure (psig)		
Maximum	40	100
Minimum	5	5
Design flow (gal/min)	25	5
Material	Carbon steel	Carbon steel

a. Varies by component but exceeds component operating temperature by 100°F.

b. Pump not required when waste GDT pressure exceeds 30 psig.

TABLE 11.3.3-1

GASEOUS RELEASES FROM WESTINGHOUSE-DESIGNED OPERATING REACTORS[a]

	Noble Gases (Ci)		
	1975	1976	1977
R. E. Ginna	1.04(4)	5.52(3)	3.20(3)
San Onofre 1	1.11(3)	4.16(2)	1.54(2)
Surry 1 and 2	8.04(3)	1.91(4)	1.90(4)
H. B. Robinson	1.17(3)	6.40(2)	4.76(2)
Point Beach 1 and 2	4.45(4)	1.91(3)	1.13(3)
Connecticut Yankee	4.80(2)	4.52(2)	3.12(3)

	Iodine (Ci) I-131 and Particulate with half-lives > 8 days		
	1975	1976	1977
R. E. Ginna	2.0(-2)	3.17(-2)	2.55(-2)
San Onofre 1	4.0(-2)	<1.0(-2)	1.86(-4)
Surry 1 and 2	5.0(-2)	3.46(-1)	1.20(-1)
H. B. Robinson	2.0(-2)	9.96(-2)	3.88(-3)
Point Beach 1 and 2	7.0(-2)	1.85(-2)	5.02(-3)
Connecticut Yankee	<1.00(-2)	<1.00(-2)	1.74(-3)

a. "Radioactive Materials Released from Nuclear Power Plants," NUREG-0521, Annual Report, 1977.

TABLE 11.3.3-2

EXPECTED ANNUAL AVERAGE RELEASE OF AIRBORNE RADIONUCLIDES[a][b]

Gaseous Release Rate (Ci/year)[c]

			Gas Stripping		Building Ventilation					
Nuclide	Primary Coolant (μCi/g)	Secondary Coolant (μCi/g)	Shutdown	Continuous	Containment	Auxiliary	Turbine	Blowdown Vent Offgas	Air Ejector Exhaust	Total
Kr-83m	2.119E-02	5.387E-09	0.0	0.0	4.0E+00	0.0	0.0	0.0	0.0	4.0E+00
Kr-85m	1.049E-01	2.721E-08	0.0	0.0	3.2E+01	2.0E+00	0.0	0.0	1.0E+00	3.5E+01
Kr-85	8.231E-03	2.121E-09	4.0E+00	2.6E+02	6.0E+00	0.0	0.0	0.0	0.0	2.7E+02
Kr-87	6.139E-02	1.507E-08	0.0	0.0	8.0E+00	1.0E+00	0.0	0.0	0.0	9.0E+00
Kr-88	1.976E-01	5.001E-08	0.0	0.0	4.5E+01	4.0E+00	0.0	0.0	3.0E+00	5.2E+01
Kr-89	5.265E-03	1.357E-09	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Xe-131m	1.941E-02	5.034E-09	0.0	3.0E+00	1.4E+01	0.0	0.0	0.0	0.0	1.7E+01
Xe-133m	1.044E-01	2.708E-08	0.0	0.0	6.9E+01	2.0E+00	0.0	0.0	1.0E+00	7.2E+01
Xe-133	5.246E+00	1.341E-06	0.0	1.0E+00	3.7E+03	1.1E+02	0.0	0.0	7.0E+01	3.9E+03
Xe-135m	1.362E-02	3.471E-09	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Xe-135	3.026E-01	7.719E-08	0.0	0.0	1.3E+02	6.0E+00	0.0	0.0	4.0E+00	1.4E+02
Xe-137	9.475E-03	2.422E-09	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Xe-138	4.608E-02	1.157E-08	0.0	0.0	1.0E+00	1.0E+00	0.0	0.0	0.0	1.0E+00
Total Noble Gases										4.5E+03
I-131	2.845E-01	6.773E-06	0.0	0.0	2.0E-02	4.5E-03	3.7E-04	0.0	2.8E-02	5.3E-02
I-133	4.005E-01	8.392E-06	0.0	0.0	2.2E-02	6.4E-03	4.5E-04	0.0	4.0E-02	6.9E-02

Tritium gaseous release = 730 Ci/year

Airborne Particulate Release Rate (Ci/year)

		Building Ventilation		
Nuclide	Waste Gas System	Containment	Auxiliary	Total
Mn-54	4.5E-05	2.2E-04	1.8E-04	4.4E-04
Fe-59	1.5E-05	7.4E-05	6.0E-05	1.5E-04
Co-58	1.5E-04	7.4E-04	6.0E-04	1.5E-03
Co-60	7.0E-05	3.4E-04	2.7E-04	6.8E-04
Sr-89	3.3E-06	1.7E-05	1.3E-05	3.3E-05
Sr-90	6.0E-07	3.0E-06	2.4E-06	6.0E-06
Cs-134	4.5E-05	2.2E-04	1.8E-04	4.4E-04
Cs-137	7.5E-05	3.8E-04	3.0E-04	7.5E-04

a. For one unit.

b. Twenty-five Ci/year of argon-41 are released from the containment, and 8 Ci/year of carbon-14 are released from the waste gas processing system (from PWR-GALE code, section 11.1, reference 1). 710 Ci/year of tritium are released via vapor pathways.

c. The appearance of 0.0 in the table indicates release is less than 1.0 Ci/year for noble gas and 0.0001 Ci/year for I.

TABLE 11.3.3-3 (SHEET 1 OF 2)

COMPARISON OF CALCULATED MAXIMUM OFFSITE AIRBORNE CONCENTRATION WITH 10 CFR 20 ASSUMING EXPECTED FUEL LEAKAGE

Isotope	Total Annual Release from One Unit[a] (Ci/year)	Maximum Site Boundary Concentration[b] (μCi/ml)	Maximum Permissable Concentration (MPC)[c] (Ci/ml)	Fraction of MPC
Cr-51	2.66E-07	3.11E-19	4.0E-07	7.78E-13
Mn-54	4.40E-04	5.15E-16	1.0E-08	5.15E-08
Fe-55	3.62E-07	4.23E-19	3.0E-08	1.41E-11
Fe-59	1.50E-04	1.76E-16	5.0E-09	3.52E-08
Co-58	1.50E-03	1.76E-15	3.0E-08	5.87E-08
Co-60	6.80E-04	7.96E-16	1.0E-08	7.96E-08
Br-83	6.89E-10	8.06E-22	1.0E-10	8.06E-12
Br-84	1.98E-11	2.32E-23	3.0E-08	7.73E-16
Br-85m	2.02E-14	2.36E-26	3.0E-08	7.87E-19
Rb-86	1.02E-08	1.19E-20	1.0E-08	1.19E-12
Sr-89	3.41E-05	3.99E-17	3.0E-10	1.33E-07
Sr-90	6.00E-06	7.02E-18	3.0E-11	2.34E-07
Sr-91	1.26E-09	1.47E-21	2.0E-08	7.35E-14
Y-90	2.50E-11	2.93E-23	4.0E-09	7.33E-15
Y-91	5.22E-08	6.11E-20	1.0E-09	6.11E-11
Y-91m	1.23E-10	1.44E-22	8.0E-07	1.80E-16
Y-93	7.42E-11	8.68E-23	6.0E-09	1.45E-14
Zr-95	1.13E-08	1.32E-20	4.0E-09	3.30E-12
Nb-95	9.52E-09	1.11E-20	2.0E-08	5.55E-13
Mo-99	1.77E-06	2.07E-18	3.0E-08	6.90E-11
Tc-99m	7.61E-07	8.90E-19	1.0E-06	8.90E-13
Ru-103	7.34E-09	8.59E-21	2.0E-08	4.30E-13
Ru-106	2.14E-08	2.50E-20	3.0E-09	8.33E-12
Rh-103m	8.56E-12	1.00E-23	3.0E-06	3.33E-18
Rh-105m	1.53E-10	1.79E-22	1.0E-10	1.79E-12
Te-129	3.55E-10	4.15E-22	2.0E-07	2.08E-15
Te-129m	2.13E-07	2.49E-19	3.0E-09	8.30E-11
Te-131	3.22E-10	3.77E-22	3.0E-06	1.26E-16
Te-131m	2.26E-08	2.64E-20	1.0E-08	2.64E-12
Te-132	7.02E-07	8.21E-19	7.0E-09	1.17E-10
I-130	6.06E-07	7.09E-19	1.0E-10	7.09E-10
I-131	5.35E-02	6.26E-14	1.0E-10	6.26E-04
I-132	3.42E-06	4.00E-18	3.0E-09	1.33E-09
I-133	7.24E-02	8.47E-14	4.0E-10	2.12E-04
I-134	9.36E-08	1.10E-19	6.0E-09	1.83E-11
I-135	2.03E-05	2.38E-17	1.0E-09	2.38E-08
Cs-134	4.46E-04	5.22E-16	1.0E-09	5.22E-07
Cs-136	1.30E-06	1.52E-18	1.0E-08	1.52E-10
Cs-137	7.57E-08	8.86E-20	2.0E-09	4.43E-11
Ba-137m	1.84E-12	2.15E-24	3.0E-06	7.17E-19
Ba-140	1.94E-08	2.27E-20	4.0E-09	5.68E-12
La-140	5.26E-06	6.15E-18	5.0E-09	1.23E-09
Ce-141	1.06E-08	1.24E-20	2.0E-08	6.20E-13
Ce-143	4.14E-10	4.84E-22	9.0E-09	5.38E-14
Ce-144	7.22E-09	8.45E-21	3.0E-10	2.82E-11

TABLE 11.3.3-3 (SHEET 2 OF 2)

Isotope	Total Annual Release from One Unit[a] (Ci/year)	Maximum Site Boundary Concentration[b] (μCi/ml)	Maximum Permissable Concentration (MPC)[c] (Ci/ml)	Fraction of MPC
Np-239	2.15E-08	2.52E-20	3.0E-08	8.40E-13
Rb-88	5.42E-10	6.34E-22	3.0E-06	2.11E-16
Pu-239	7.61E-09	8.90E-21	6.0E-14	1.48E-07
Xe-131m	1.60E+01	1.87E-11	4.0E-07	4.68E-05
Xe-133m	6.70E+01	7.84E-11	3.0E-07	2.61E-04
Xe-133	3.58E+03	4.19E-09	3.0E-07	1.40E-02
Xe-135	1.41E+02	1.65E-10	1.0E-07	1.65E-03
Xe-138	2.00E+00	2.34E-12	3.0E-06	7.80E-07
Kr-83m	3.00E+00	3.51E-12	3.0E-06	1.17E-06
Kr-85m	3.40E+01	3.98E-11	1.0E-07	3.98E-04
Kr-85	2.60E+02	3.04E-10	3.0E-07	1.01E-03
Kr-87	8.00E+00	9.36E-12	2.0E-08	4.68E-04
Kr-88	5.10E+01	5.97E-11	2.0E-08	2.99E-03
Ar-41	2.50E+01	2.93E-11	4.0E-08	7.33E-04
C-14	8.00E+00	9.36E-12	1.0E-07	9.36E-05
H-3	9.25E+02	1.08E-09	2.0E-07	5.40E-03

a. Includes total Ci/year from table 11.3.3-2.

b. Based on the sum of contributions from the plant vent, radwaste solidification vent, and turbine building vent using the total maximum dilution factors (χ/Q) at the nearest site boundary in 16 compass directions:

χ/Q (plant vent) = 1.92E-06 s/m^3 (wake split model total)
χ/Q (turbine building) = 2.75E-05 s/m^3 (ground release model total)

c. From Appendix B, Table II, column 1 of 10 CFR 20.1 - 20.601.

TABLE 11.3.3-4 (SHEET 1 OF 6)

ESTIMATED ANNUAL DOSES TO AN INDIVIDUAL FROM GASEOUS AND PARTICULATE EFFLUENTS[a]

Gaseous Dose Rate[b]

Location	Pathway	Gamma Dose Rate in Air (mrad/year)	Beta Dose Rate in Air (mrad/year)	Total Body Dose Rate (mrem/year)	Skin Dose Rate (mrem/year)
Nearest site boundary (0.68 mile NE)	Plume	2.24E-02	3.94E-02	1.41E-02	3.34E-02
Nearest residence (1.2 mile WSW)	Plume	1.20E-02	2.21E-02	7.63E-03	1.82E-02
Nearest vegetable garden (1.4 mile WSW)	Plume	1.01E-02	1.83E-02	6.35E-03	1.53E-02
Nearest meat animal (3.10 mile SW)	Plume	5.35E-03	9.91E-03	3.36E-03	8.21E-03
Nearest milk cow and goat (4.60 mile SE)	Plume	2.31E-03	4.27E-03	1.46E-03	3.54E-03

Radioiodines and Particulates Dose Rate (mrem/year)[c]

Location	Pathway	Total Body	GI Tract	Bone	Liver	Kidney	Thyroid	Lung	Skin
Nearest site boundary (0.68 mile NE)	Ground deposition	2.42E-01	2.42E-01	2.42E-01	2.42E-01	2.42E-01	2.42E-01	2.42E-01	2.83E-01
	Inhalation								
	Adult	2.70E-02	2.51E-02	2.09E-03	2.78E-02	3.22E-02	5.78E-02	2.77E-02	2.78E-02
	Teen	2.65E-02	2.54E-02	2.93E-03	2.90E-02	2.68E-02	6.62E-02	2.81E-02	2.79E-02
	Child	2.30E-02	2.24E-02	3.96E-03	2.59E-02	2.38E-02	6.97E-02	2.51E-02	2.50E-02
	Infant	1.31E-02	1.29E-02	2.42E-03	1.56E-02	1.38E-02	5.64E-02	1.54E-02	1.55E-02
	Total Dose to Receptor[d]								
	Adult	2.69E-01	2.67E-01	2.44E-01	2.70E-01	2.74E-01	3.00E-01	2.70E-01	3.11E-01
	Teen	2.69E-01	2.67E-01	2.45E-01	2.71E-01	2.69E-01	3.08E-01	2.70E-01	3.11E-01
	Child	2.65E-01	2.64E-01	2.46E-01	2.68E-01	2.66E-01	3.12E-01	2.67E-01	3.08E-01
	Infant	2.55E-01	2.55E-01	2.44E-01	2.58E-01	2.56E-01	2.98E-01	2.57E-01	2.99E-01

TABLE 11.3.3-4 (SHEET 2 OF 6)

Dose Rate (mrem/year)

Location	Pathway	Total Body	GI Tract	Bone	Liver	Kidney	Thyroid	Lung	Skin
Nearest site boundary (0.68 Mile NE)	Ground deposition	2.42E-01	2.42E-01	2.42E-01	2.42E-01	2.42E-01	2.42E-01	2.42E-01	2.83E-01
	Vegetables								
	Adult	1.43E-01	3.94E-02	1.62E-01	1.98E-01	9.18E-02	1.53E-02	1.98E-02	3.59E-02
	Teen	1.36E-01	5.04E-02	2.66E-01	3.04E-01	1.35E-01	1.45E-01	2.53E-02	4.65E-02
	Child	1.51E-01	8.80E-02	6.36E-01	5.25E-01	2.30E-01	2.37E-01	5.07E-02	8.50E-02
	Meat								
	Adult	1.72E-02	7.95E-03	2.78E-02	2.24E-02	1.27E-02	2.51E-02	9.50E-03	7.57E-03
	Teen	9.54E-03	5.59E-03	2.33E-02	1.73E-02	9.48E-03	1.81E-02	6.93E-03	5.36E-03
	Child	1.08E-02	8.54E-03	4.35E-01	2.42E-02	1.36E-02	2.76E-02	1.03E-02	8.41E-03
	Cow Milk								
	Adult	9.59E-02	1.64E-02	1.11E-01	1.40E-01	5.84E-02	5.04E-01	2.75E-02	1.35E-02
	Teen	9.75E-02	2.33E-02	2.01E-01	2.41E-01	9.89E-02	7.96E-01	4.86E-02	1.96E-02
	Child	9.59E-02	3.99E-02	4.86E-01	4.22E-01	1.68E-01	1.57E00	8.16E-02	3.70E-02
	Infant	1.23E-01	6.86E-02	8.09E-01	8.20E-01	2.79E-01	3.80E00	1.47E-01	6.58E-02
	Inhalation								
	Adult	2.70E-02	2.51E-02	2.09E-03	2.78E-02	3.22E-02	5.78E-02	2.77E-02	2.78E-02
	Teen	2.65E-02	2.54E-02	2.93E-03	2.90E-02	2.68E-02	6.62E-02	2.81E-02	2.79E-02
	Child	2.30E-02	2.24E-02	3.96E-03	2.59E-02	2.38E-02	6.97E-02	2.51E-02	2.50E-02
	Infant	1.31E-02	1.29E-02	2.42E-03	1.56E-02	1.38E-02	5.64E-02	1.54E-02	1.55E-02
	Total Dose to Receptor[e]								
	Adult	5.25E-01	3.31E-01	5.45E-01	6.30E-01	4.37E-01	9.82E-01	3.27E-01	3.68E-01
	Teen	5.12E-01	3.47E-01	7.35E-01	8.33E-01	5.12E-01	1.27E00	3.51E-01	3.82E-01
	Child	5.23E-01	4.01E-01	1.80E00	1.24E00	6.77E-01	2.15E00	4.10E-01	4.38E-01
	Infant	3.78E-01	3.24E-01	1.05E00	1.08E00	5.35E-01	4.10E00	4.04E-01	3.64E-01

TABLE 11.3.3-4 (SHEET 3 OF 6)

Dose Rate (mrem/year)

Location	Pathway	Total Body	GI Tract	Bone	Liver	Kidney	Thyroid	Lung	Skin
Nearest Residence (1.2 mile WSW)	Ground deposition	8.01E-02	8.01E-02	8.01E-02	8.01E-02	8.01E-02	8.01E-02	8.01E-02	9.35E-02
	Inhalation								
	Adult	1.12E-02	1.06E-02	7.35E-04	1.15E-02	1.10E-02	2.28E-02	1.07E-02	1.06E-02
	Teen	1.22E-02	1.07E-02	1.03E-03	1.19E-02	1.12E-02	2.59E-02	1.08E-02	1.06E-02
	Child	1.07E-02	9.41E-03	1.39E-03	1.07E-02	9.91E-03	2.71E-02	9.57E-03	9.39E-03
	Infant	6.57E-03	5.41E-03	8.53E-04	6.36E-03	5.72E-03	2.17E-02	5.52E-03	5.40E-03
	Total Dose to Receptor								
	Adult	9.13E-02	9.07E-02	8.08E-02	9.16E-02	9.11E-02	1.03E-01	9.08E-02	1.04E-01
	Teen	9.23E-02	9.08E-02	8.11E-02	9.20E-02	9.13E-02	1.06E-01	9.09E-02	1.04E-01
	Child	9.08E-02	8.95E-02	8.15E-02	9.08E-02	9.00E-02	1.07E-01	8.97E-02	1.03E-02
	Infant	8.67E-02	8.55E-02	8.10E-02	8.65E-02	8.58E-02	1.02E-01	8.56E-02	9.89E-02
Nearest Vegetable garden (1.4 mile WSW)	Ground deposition	6.08E-02	6.08E-02	6.08E-02	6.08E-02	6.08E-02	6.08E-02	6.08E-02	7.10E-02
	Vegetables								
	Adult	4.05E-02	1.48E-02	5.15E-02	5.45E-02	2.78E-02	1.53E-02	4.90E-02	1.38E-02
	Teen	4.10E-02	1.94E-02	8.46E-02	8.32E-02	4.05E-02	1.96E-02	5.27E-02	1.84E-02
	Child	5.15E-02	3.57E-02	2.04E-01	1.46E-01	7.13E-02	3.68E-02	8.77E-02	3.50E-02
	Inhalation								
	Adult	9.27E-03	8.78E-03	1.14E-03	9.50E-03	9.07E-03	9.31E-03	1.82E-02	8.75E-03
	Teen	9.19E-03	8.83E-03	1.60E-03	9.83E-03	9.25E-03	9.52E-03	2.07E-02	8.80E-03
	Child	7.95E-03	7.80E-03	2.17E-03	8.79E-03	8.20E-03	8.61E-03	2.16E-02	7.79E-03
	Infant	4.56E-03	4.48E-03	1.33E-03	5.23E-03	4.73E-03	5.23E-03	1.70E-02	4.48E-03
	Total Dose to Receptor								
	Adult	1.11E-01	8.43E-02	1.13E-01	1.25E-01	9.77E-02	8.54E-02	1.28E-01	9.36E-02
	Teen	1.11E-01	8.90E-02	1.47E-01	1.54E-01	1.11E-01	8.99E-02	1.34E-01	9.82E-02
	Child	1.20E-01	1.04E-01	2.67E-01	2.16E-01	1.40E-01	1.06E-01	1.70E-01	1.14E-01
	Infant	6.54E-02	6.53E-02	6.21E-02	6.60E-02	6.55E-02	6.61E-02	7.78E-02	7.55E-02

TABLE 11.3.3-4 (SHEET 4 OF 6)

Dose Rate (mrem/year)

Location	Pathway	Total Body	GI Tract	Bone	Liver	Kidney	Thyroid	Lung	Skin
Nearest meat animal (3.1 mile SW)	Ground deposition	1.80E-02	1.80E-02	1.80E-02	1.80E-02	1.80E-02	1.80E-02	1.80E-02	2.10E-02
	Meat								
	Adult	2.32E-03	1.63E-03	5.57E-03	2.69E-03	1.98E-03	3.05E-03	1.72E-03	1.60E-03
	Teen	1.50E-03	1.21E-03	4.70E-03	2.08E-03	1.50E-03	2.25E-03	1.31E-03	1.19E-03
	Child	2.16E-03	1.99E-03	8.80E-03	3.45E-03	2.37E-03	3.57E-03	2.12E-03	1.98E-03
	Inhalation								
	Adult	4.08E-03	3.89E-03	2.12E-04	4.16E-03	4.01E-03	7.80E-03	3.92E-03	3.88E-03
	Teen	4.06E-03	3.91E-03	2.96E-04	4.29E-03	4.08E-03	8.81E-03	3.98E-03	3.90E-03
	Child	3.52E-03	3.46E-03	4.00E-04	3.83E-03	3.61E-03	9.12E-03	3.51E-03	3.46E-03
	Infant	2.01E-03	1.99E-03	2.45E-04	2.25E-03	2.09E-03	7.18E-03	2.02E-03	1.99E-03
	Total Dose to Receptor								
	Adult	2.44E-02	2.35E-02	2.38E-02	2.48E-02	2.39E-02	2.88E-02	2.36E-02	2.65E-02
	Teen	2.36E-02	2.31E-02	2.29E-02	2.44E-02	2.36E-02	2.90E-02	2.33E-02	2.61E-02
	Child	2.37E-02	2.34E-02	2.72E-02	2.53E-02	2.40E-02	3.07E-02	2.36E-02	2.65E-02
	Infant	2.00E-02	2.00E-02	1.82E-02	2.02E-02	2.01E-02	2.52E-02	2.00E-02	2.30E-02

TABLE 11.3.3-4 (SHEET 5 OF 6)

Dose Rate (mrem/year)

Location	Pathway	Total Body	GI Tract	Bone	Liver	Kidney	Thyroid	Lung	Skin
Nearest milk cow (4.60 mile SE)	Ground deposition	7.20E-03	7.20E-03	7.20E-03	7.20E-03	7.20E-03	7.20E-03	7.20E-03	8.40E-03
	Cow Milk								
	Adult	3.57E-03	1.21E-03	4.96E-03	4.86E-03	2.47E-03	1.73E-03	1.54E-03	1.12E-03
	Teen	4.03E-03	1.82E-03	9.07E-03	8.31E-03	4.08E-03	2.74E-02	2.57E-03	1.71E-03
	Child	5.17E-03	3.51E-03	2.20E-02	1.48E-02	7.36E-03	5.43E-02	4.74E-03	3.42E-03
	Infant	8.08E-03	6.44E-03	3.89E-02	2.87E-02	1.27E-02	1.29E-01	8.74E-03	6.35E-03
	Inhalation								
	Adult	1.83E-03	1.75E-03	9.22E-05	1.86E-03	1.79E-03	3.49E-03	1.76E-03	1.74E-03
	Teen	1.81E-03	1.76E-03	1.29E-04	1.92E-03	1.82E-03	3.95E-03	1.78E-03	1.75E-03
	Child	1.58E-03	1.55E-03	1.74E-04	1.71E-03	1.62E-03	4.08E-03	1.57E-03	1.55E-03
	Infant	9.03E-04	8.91E-04	1.07E-04	1.01E-03	9.32E-04	3.22E-03	9.07E-04	8.90E-04
	Total Dose to Receptor								
	Adult	1.26E-02	1.02E-02	1.23E-02	1.39E-02	1.15E-02	2.80E-02	1.05E-02	1.13E-02
	Teen	1.31E-02	1.08E-02	1.64E-02	1.74E-02	1.31E-02	3.86E-02	1.16E-02	1.18E-02
	Child	1.39E-02	1.23E-02	2.93E-02	2.38E-02	1.62E-02	6.55E-02	1.35E-02	1.34E-02
	Infant	1.62E-02	1.45E-02	4.62E-02	3.69E-02	2.08E-02	1.40E-01	1.69E-02	1.56E-02

TABLE 11.3.3-4 (SHEET 6 OF 6)

a. All data is on a per unit basis. Doses were calculated using the GASPAR code.

b. Evaluated at a location that could be occupied during the term of plant operation.

Appendix I design objectives – gaseous effluents (noble gases only):

Gamma dose in air - 10 mrad/year per unit
Beta dose in air - 20 mrad/year per unit
Dose to total body of individual - 5 mrem/year per unit
Dose to skin of individual - 15 mrem/year per unit

Annex to Appendix I, Docket RM-50-2. Design objectives are the same as Appendix I except on a per-site basis; therefore calculated doses should be multiplied by 2.

c. Evaluated at a location where an exposure pathway and dose receptor actually exist at the time of licensing.

Appendix I design objectives - radioiodines and particulates:

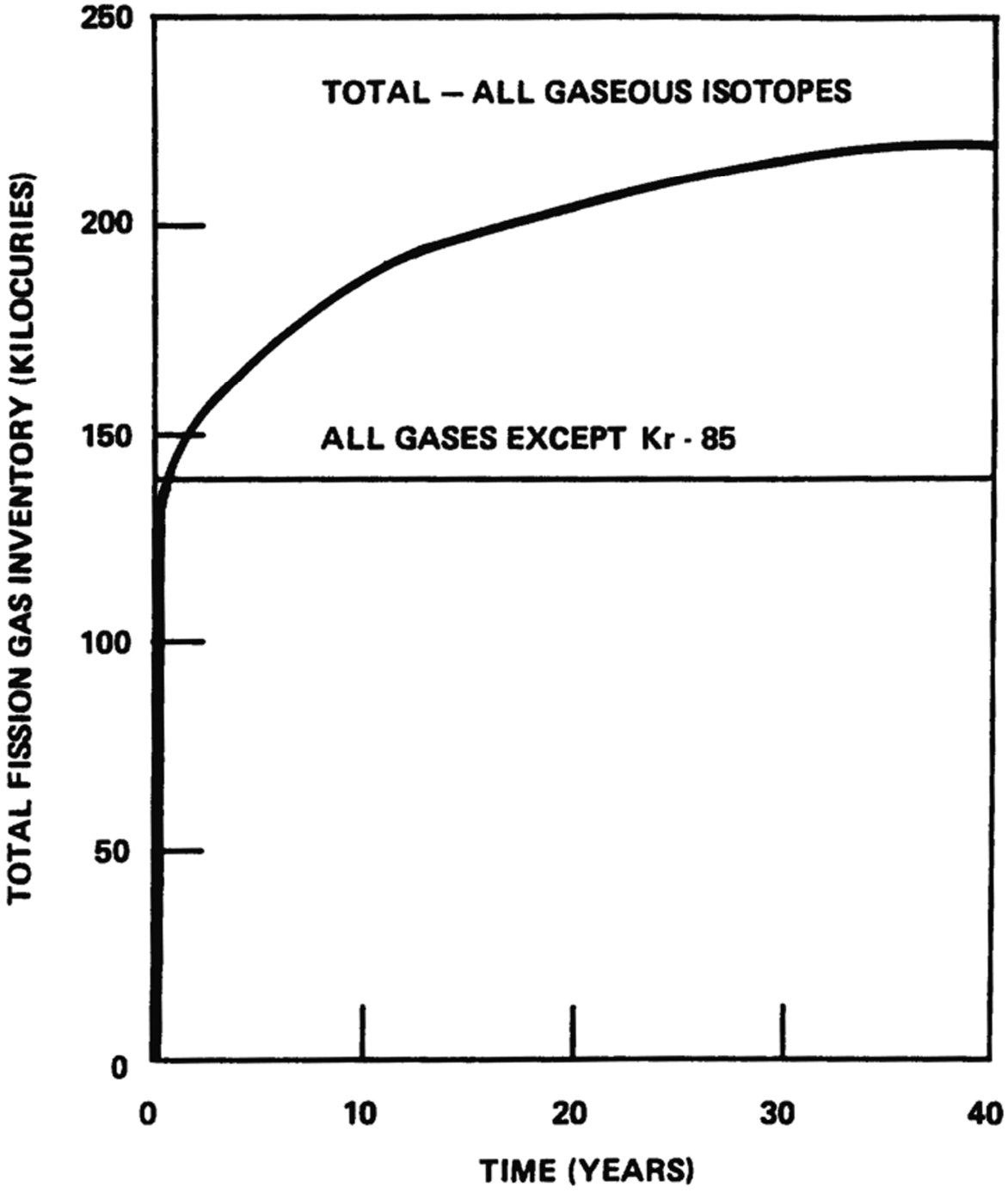
Dose to any organ from all pathways – 15 mrem/year per unit

Annex to Appendix I Docket RM-50-2 design objectives:

Dose to any organ from all pathways – 15 mrem/year per site 1-131 releases – 1 Ci/year per unit (reference table 11.3.3-3)

d. Total dose due to realistic pathways.

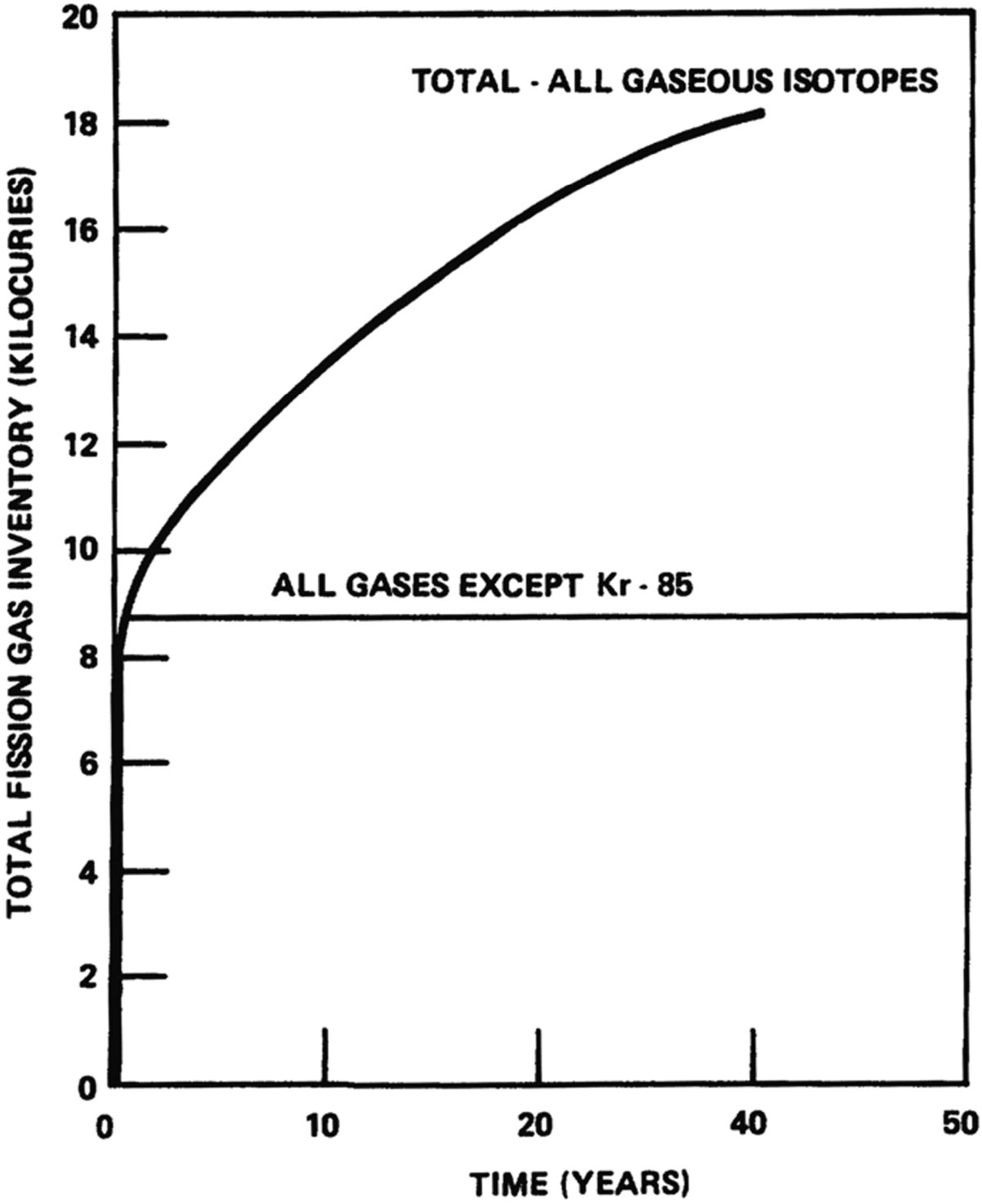
e. Provided for information only; a receptor is assumed present at the location of a potential pathway. This evaluation is based on the worst case χ /Q at the site boundary.



A review of the figure values for the effects of MUR power uprate determined the figure need not be updated.

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SOUTHERN COMPANY **VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 1 AND UNIT 2**	GWPS FISSION GAS ACCUMULATION BASED ON CONTINUOUS CORE OPERATION OF 2 UNITS AT 3565 MWt WITH 1% FUEL DEFECTS
	FIGURE 11.3.2–1



A review of the figure values for the effects of MUR power uprate determined the figure need not be updated.

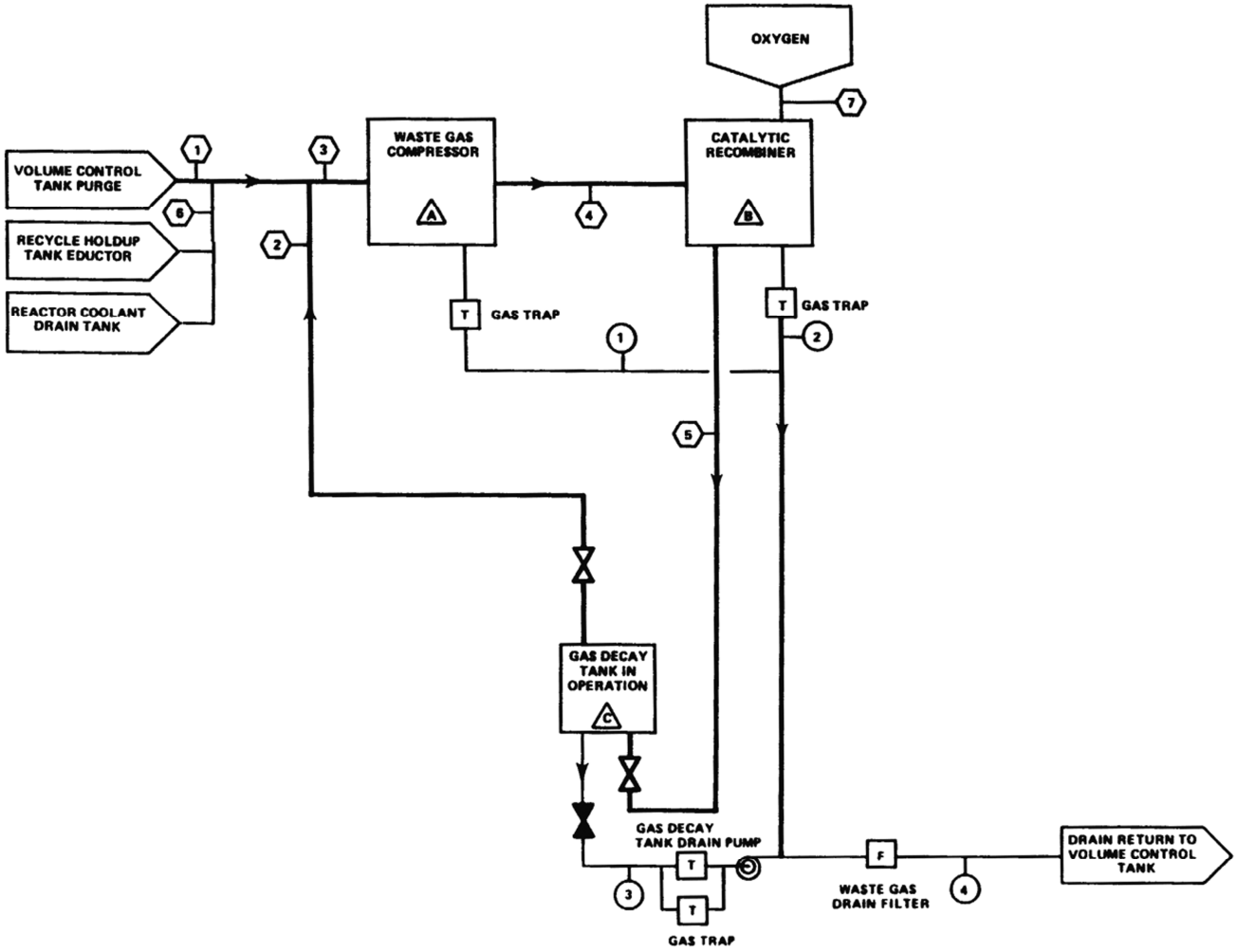
REV 16 10/10

SOUTHERN COMPANY

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

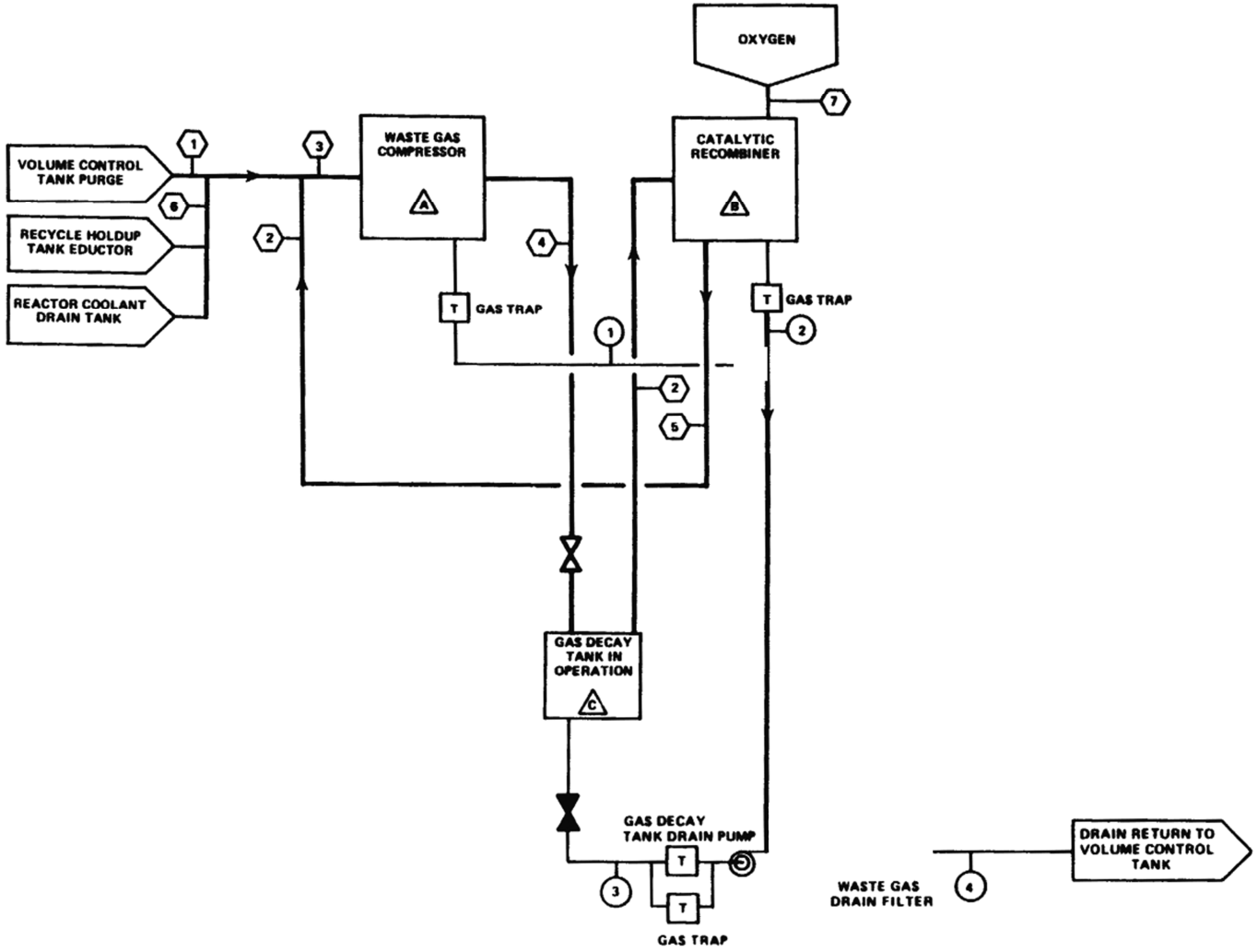
EXPECTED GWPS FISSION GAS ACCUMULATION BASED ON TABLE AND FULL–POWER OPERATION OF 2 UNITS AT 3565 MWt

FIGURE 11.3.2–2



REV 14 10/07

SOUTHERN COMPANY *Energy to Serve Your World®* **VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2**	PROCESS DIAGRAM DURING OPERATION WITH GDT PRESSURE ≤ 20 psig
	FIGURE 11.3.2–3



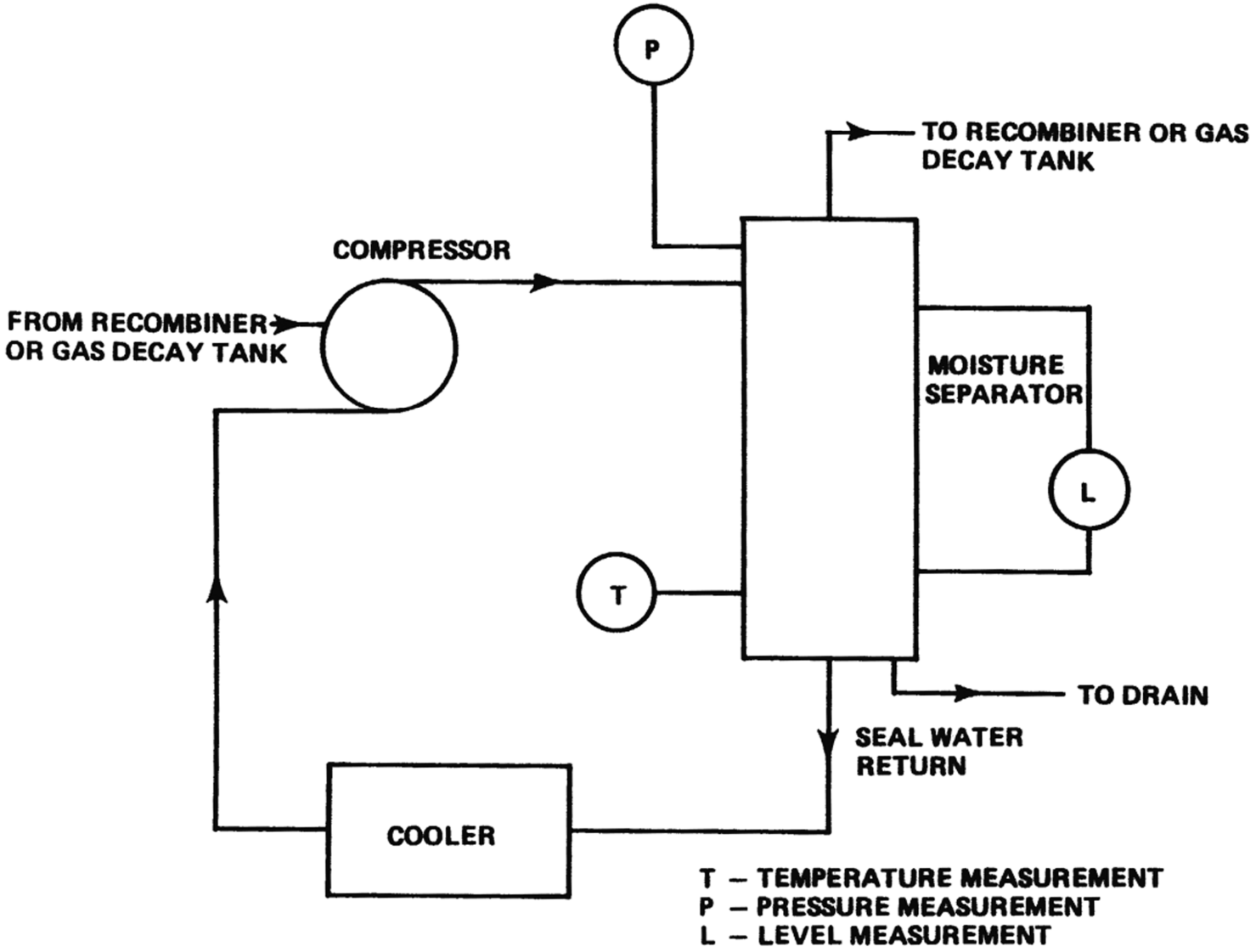
REV 14 10/07

SOUTHERN COMPANY
Energy to Serve Your World®

VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

PROCESS DIAGRAM DURING OPERATION WITH GDT PRESSURE BETWEEN 20 AND 100 psig

FIGURE 11.3.2–4



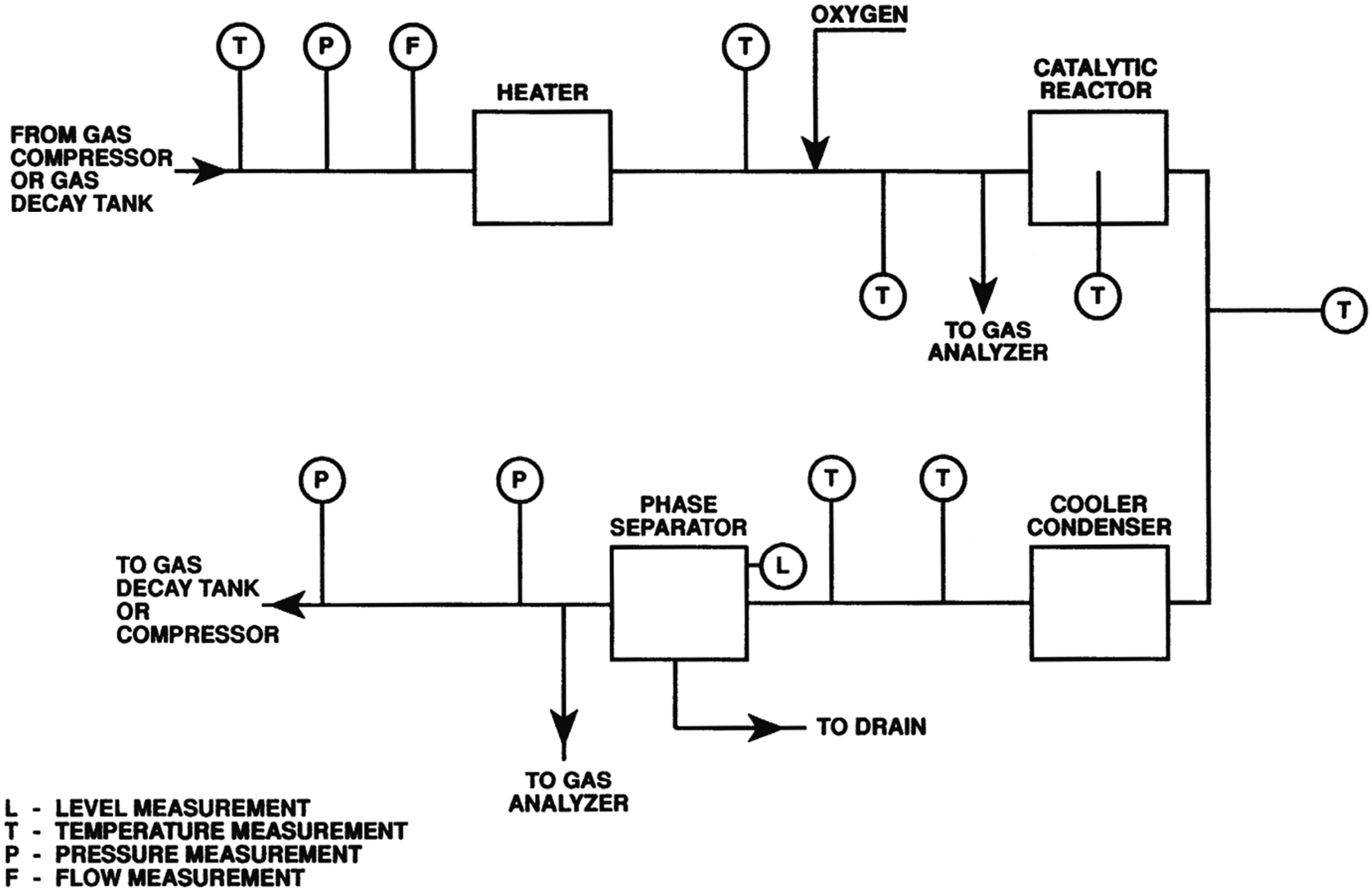
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VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

WASTE GAS COMPRESSOR PACKAGE

FIGURE 11.3.2–5



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ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

CATALYTIC HYDROGEN RECOMBINER PACKAGE

FIGURE 11.3.2–6

11.4 SOLID WASTE MANAGEMENT SYSTEM

The solid waste management system is designed to collect, process, and package all solid radioactive wastes generated as a result of normal plant operation, including anticipated operational occurrences. The packaged waste is stored until it is shipped to a waste processing facility or burial site. This system does not normally handle large waste materials such as activated core components.

11.4.1 DESIGN BASES

The solid waste management system is designed to meet the following objectives:

A. Provide packaging capability for spent radioactive resins from the liquid waste processing and steam generator blowdown systems.

B. Provide a means to package crud from backflushable filters.

C. Provide a means to package combustible dry solid wastes, such as paper, rags, and contaminated clothing.

D. Provide a means to transfer the spent filter cartridges from filter vessels to shielded disposal containers in a manner which minimizes radiation exposure to operating personnel and the spread of contamination.

E. Provide a means for compacting and packaging miscellaneous dry radioactive materials, such as paper, rags, and contaminated clothing.

F. Provide temporary onsite storage of packaged wastes in the event of delay or disruption of offsite shipping schedules.

G. Package radioactive solid wastes for offsite shipment and disposal in accordance with applicable Department of Transportation (DOT) and Nuclear Regulatory Commission (NRC) regulations, including 49 Code of Federal Regulations (CFR) 170-178 and 10 CFR 71.

The solid waste management system is designed and constructed in accordance with Regulatory Guide 1.143, as described in section 1.9, to meet the requirements of General Design Criterion (GDC) 60 of 10 CFR 50, Appendix A. The seismic design classification of the alternate radwaste, radwaste processing facility, and radwaste transfer buildings is provided in section 3.2. The radwaste processing facility is provided to allow use of portable radwaste systems (demineralizers, filtration system, dewatering/drying, solidification) supplied by contractors for handling solid wastes from plant operations during all plant conditions (paragraph 11.4.2.4). Dry active waste (DAW) locations are provided to process and store DAW (paragraph 11.4.2.5).

The solid waste management system design parameters are based on the radionuclide concentrations and volumes consistent with operating experience for similar reactor designs and with the source terms of section 11.1.

The solid waste management system airborne process effluents are released through the radwaste processing facility vent to the atmosphere as part of the 10 CFR 50, Appendix I analysis presented in subsection 11.3.3.

The solid waste management system is designed to collect, solidify, package, and store radioactive wastes so as to maintain radiation exposure to plant operation or maintenance

personnel as low as is reasonably achievable (ALARA) in accordance with GDC 60 of 10 CFR 50, Appendix A, and Regulatory Guide 8.8 in order to maintain personnel exposures below 10 CFR 20 requirements. Access to the process equipment and solid waste storage areas is controlled to minimize personnel exposure by suitable barriers such as locked doors or gates or control cards. (See paragraph 12.3.1.2 and section 12.5.) The solid waste management system has been designed to conform with the design criteria of NRC Branch Technical Position ETSB 11-3.

11.4.1.1 Safety Design Bases

The solid waste management system performs no function related to the safe shutdown of the plant, and its failure does not adversely affect any safety-related system or component; therefore, the solid waste management system has no safety design basis.

11.4.1.2 Power Generation Design Bases

The solid waste management system is designed to minimize the volume of solidified wastes and provide temporary onsite storage for the packaged wastes. The system is designed for a 40-year service life[a], maximum reliability, minimum maintenance, and minimum exposure to operating and maintenance personnel. The system is designed to have sufficient capacity based on normal waste generation rates to ensure that maintenance or repair of the solid waste management system equipment does not impact power generation.

11.4.2 SYSTEM DESCRIPTION

11.4.2.1 General Description

The solid waste management system consists of the following which are shown in drawings 1X4DB148, 1X4DB148-1, 1X4DB148-2, 1X4DB148-3, 1X4DB148-4, AX4DB148-5, AX4DB148-6, AX4DB148-7, 1X4DB148-8, 1X4DB148-9, 1X4DB148-10, 1X4DB148-11, 1X4DB148-12, and 1X4DB148-13.

- Resin transfer.
- Backflushable filter system.
- Crud transfer.

The activity of the influents to the solid waste management system is dependent on the activities of various fluid systems, such as the boron recycle system, liquid waste processing system, chemical and volume control system, spent fuel pool cooling and cleanup system, steam generator blowdown system, and condensate polishing demineralizer system. Reactor

[a] The operating licenses for both VEGP units have been renewed and the original licensed operating terms have been extended by 20 years. In accordance with 10 CFR Part 54, appropriate aging management programs and activities have been initiated to manage the detrimental effects of aging to maintain functionality during the period of extended operation (see chapter 19).

coolant system activities and the decontamination factors for the systems given above also determine the influent activities to the solid waste management system.

Table 11.4.2-1 lists the parameters used in calculation of the estimated activity fed to the solid waste management system. Table 11.4.2-2 lists the estimated expected activities of wastes to be processed on an annual basis, their physical form and source, and their isotopic makeup and curie content. Table 11.4.2-3 provides the same information based on the maximum expected activities of the input wastes. The estimated annual quantities of radwaste input to the solid waste management system and the estimated quantities of radwaste to be shipped offsite are presented in table 11.4.2-4. The estimated expected curie and isotopic content of wastes to be shipped offsite for each waste category is presented in table 11.4.2-5. Table 11.4.2-6 presents the same information based on expected maximum activities.

Section 11.1 provides the bases for determination of liquid source terms which are used to calculate the solid waste source terms. The data presented in tables 11.4.2-2, 11.4.2-3, 11.4.2-5, and 11.4.2-6 are conservatively based on section 11.1 and the following additional information:

A. As a basis for the shipped-from-site activities given in tables 11.4.2-5 and 11.4.2-6, 30 days of decay prior to shipping is assumed. The major contributors to the activity levels in the system are radionuclides having long half-lives; therefore, further onsite decay of radionuclides provides limited benefit.

B. The miscellaneous dry and compacted waste volume is based on reference 1.

C. Shipping volumes from the cartridge filter handling system are based on packaging (per 55-gal drum):

   1. One spent filter cartridge.

   2. The drum is prelined with cement with an integral cavity to accept a filter.

Section 11.1 provides the bases for the quantity of primary and secondary resins. Section 11.2 provides the basis for the quantity of chemical drain wastes, while reference 1 provides the basis for the remaining quantities presented in table 11.4.2-4.

11.4.2.2 Component Description

Codes and standards applicable to the solid waste management system are listed in table 3.2.2-1. The solid waste management system is housed within buildings designed to meet the seismic requirements of Regulatory Guide (RG) 1.143. Conformance with RG 1.143 is to the extent specified in section 1.9.

11.4.2.3 System Operation

11.4.2.3.1 Resin Transfer System

The resin transfer system provides the capability for remote transfer of spent radioactive resin from the auxiliary building to the radwaste processing facility.

11.4.2.3.2 Backflushable Filter System

The backflushable filter system provides the means to remove and deliver radioactive crud from certain process streams to the crud transfer system. The system consists of the filters summarized in table 11.4.2-7, a backflushable filter crud tank, and a centrifugal pump. Each backflushable filter is operated in a similar manner. A sufficient buildup of crud on the filter signals an annunciator and the operator then initiates the automatic filter backflush sequence (Unit 1 only). The operator can manually carry out the backflush sequence if it is desired. The backflush sequence isolates the filter, opens a line to the backflushable crud tank, and introduces approximately 8 ft^3 of high-pressure (350 psig) nitrogen from a nitrogen accumulator through the filter. The nitrogen accumulator is supplied by the auxiliary gas system. The crud removed from the filters collects in the backflushable crud tank. After each backflush operation the lines and tank are water flushed. The contents of the backflushable crud tank may be slurried to the radwaste processing facility at operator discretion or upon high tank level annunciation.

11.4.2.3.3 Filter Handling System

The filter handling system is a semiremote system which provides the capability to remove spent radioactive cartridge filters from their filter housings and place them in drums with external shielding or use of a transfer cask for transport to the radwaste processing facility. The techniques used in the filter handling system included the consideration of the following:

- Operator exposure.
- Time and manpower requirements.
- Potential for the spread of contamination.
- Potential for mechanical difficulties.
- Logistics for filter handling.

The semiremote system, as the name implies, requires the operator to be in the proximity of the filters; however, operator exposure is minimized by distance and shielding.

The filter handling system consists of a working plug, a bell-shaped shielded transfer cask, a mobile cask-to-drum transfer station, and drum capper. The working plug is designed to fit in a hatch above the filter housing and has provisions for viewing the top of the filter housing and penetrations to allow long handled tools to be used for the removal of the filter housing top. The bell-shaped shielded transfer cask is used to retract the spent filter cartridge from its housing and transport it to the filter transfer station where the filter is placed in a shielded drum and capped. The filters in the shielded drums are transported to the radwaste processing facility for packaging. The filters may also be transported using the transfer cask directly to the radwaste processing facility for packaging.

11.4.2.4 Portable Radwaste System

VEGP will utilize vendor-supplied portable radwaste equipment to provide for disposal of spent resins, radioactive cartridge filters, backflushable filter crud, and chemical wastes via dewatering or solidification. Details of this equipment are shown on drawings AX4DB124-2,

AX4DB124-3, and AX4DB124-4. In addition, a portable demineralizer system and a filtration system are available as alternate means of processing the contents of the waste holdup tank, chemical drain tank, floor drain tank, and boron recycle holdup tank. Details of this equipment are shown on drawings AX4DB124-2, AX4DB124-3, AX4DB124-4, and AX4DB124-5. These systems are housed in the radwaste processing facility which is shown on drawing AX4DE357.

Isolation valves are provided to allow processing of waste streams at the radwaste processing facility. The valves are manually and remotely operated to achieve the desired configuration. Delivery of waste streams to the radwaste processing facility is controlled from local panels near the waste stream source. Flanged connections are provided at the radwaste processing facility to interface with the vendor-supplied systems. Major components for portable radwaste systems typically include process liners, process skids, and control panels. A separate radwaste processing facility control room and dressout area is provided to facilitate system operation.

Radioactive condensate polishing demineralizer resins, backflushable filter crud, and spent resins from the liquid waste processing system and the steam generator blowdown system will be dewatered. The dewatering system supplied by vendor allows the water to be removed from the spent resins in the shipping containers. A vendor-supplied container vent is provided for the shipping containers thereby minimizing leakage into the building. A vent line to a monitored HVAC exhaust duct in the radwaste processing facility is provided. In addition, demineralizer resins from the portable demineralizers (as discussed in section 11.2) can be sluiced to the container fill skid for dewatering and disposal.

An NRC approved process control program (PCP) will be required of the vendor and appropriately referenced in the VEGP PCP prior to any actual operation. If the burial site does not accept dewatered resins or the waste form criteria cannot be met, VEGP will have the ability to solidify resins utilizing a portable solidification system. Cartridge filters will be loaded into liners for shipment offsite.

Solidification/dewatered liners will normally be shipped after filling and proper cure time, provided the proper shielding is available, without exceeding DOT radiation limits. If 49 CFR 173 dose limitations cannot be met with the available shielding, the liners are stored and allowed to decay until the appropriate shielding is available. Onsite storage for decay of short-lived radionuclides is accomplished both prior to solidification in holdup tanks and in appropriate onsite storage areas.

Short-term storage of solid radwaste for the purpose of activity reduction by radioactive decay is provided by shielded environmental containers located on a concrete pad adjacent to the southeast corner of the abandoned radwaste solidification building. High-integrity containers, containing the solid radwaste, are inserted into the shielded environmental containers which are right circular cylinders of reinforced concrete of sufficient thickness to provide appropriate shielding and normal environmental protection.

11.4.2.5 Dry Active Waste

Low level dry wastes are collected in drums, plastic bags, and other methods as approved by the health physics department, at appropriate locations throughout the plant, as dictated by the volume of these wastes generated during operation or maintenance.

Large components and equipment which have been activated during reactor operation and which are not amenable to solidification, compaction, or volume reduction are handled either by qualified plant personnel or by outside contractors specializing in radioactive materials handling and are packaged in boxes or shipping casks of appropriate size.

11.4.2.5.1 Dry Active Waste Processing

DAW, outage equipment, and other radioactively contaminated material are transported to a central location in containers or contained in some manner which will prevent any leakage of radioactive material during conditions incident to normal transportation. The radioactive material is packaged such that contamination on the outer container/containment surface is below administrative limits. The transport path remains within the owner-controlled area.

The central location determined by the Health Physics department has separate areas for incoming radwaste containers, handling of nonradioactive dry waste, and storage of empty containers. Radioactive waste is packaged such that it will comply with the criteria of 10 CFR part 71 to minimize the need for repackaging for shipment.

Outage equipment, and other radioactively contaminated material that are not shipped offsite for processing are packaged for onsite storage at several locations within the owner-controlled area.

11.4.2.5.2 Dry Active Waste Storage Building

DAW, outage equipment, and other potentially contaminated radioactive material are transported to multiple storage locations within the owner-controlled area in packaging that will comply with the criteria of 10 CFR Part 71 or in containers that will prevent leakage of any radioactive material incident to normal transportation. Lockable doors and/or a fence with lockable gates are provided. DAW may also be shipped offsite for long term processing and storage at an NRC approved facility.

11.4.2.5.3 Low-Level Waste Storage Facility

In lieu of immediate disposal of low-level waste due to the closing of the Barnwell facility to non-Atlantic Compact generators, a low-level waste storage facility is provided near the Unit 2 cooling tower for interim storage of low-level waste pending permanent disposal in an NRC-licensed facility. The low-level waste storage facility is a radiation control area and radioactive material storage area that consists of a storage pad, dense grade aggregate, and fencing with locked gates. Low-level waste stored in the facility will consist primarily of radioactive resins and filters but may also include irradiated components. Consistent with the classifications of 10 CFR Part 61, low-level waste that may be stored in the facility includes Class A, B, C, and greater-than-Class C waste. NRC-licensed disposal facilities are used when available to remove low-level waste from the facility.

Resins stored in the low-level waste storage facility for extended periods are stabilized resins that have been processed for long term storage and subsequent disposal. Dewatered resins may be stored in the low-level storage facility pending shipment of a sufficient quantity to fill a high-integrity container following offsite processing and volume reduction for long-term storage and subsequent disposal. Liquids or flammable materials are not stored in the low-level waste storage facility.

The low-level waste is stored in high-integrity containers and inserted into shielded environmental containers. The shielded environmental containers are right circular cylinders of reinforced concrete of sufficient thickness to provide appropriate shielding and normal environmental protection for the high-integrity container. Supplemental shielding is provided, as necessary, to maintain the doses outside the low-level waste facility fence to Zone I radiation area levels described in table 12.3.1-1. The shielded environmental containers incorporate a

vent and drain pipe that is routed through the container wall that provides access for performance of periodic inspection of the volume surrounding the high-integrity container to verify container integrity. The vent and drain pipe is designed to prevent rain water from entering the shielded environmental container as well as preventing any material from escaping from between the high-integrity container and the shielded environmental container. To minimize the potential for spread of contamination, high-integrity containers are not opened at the low-level waste facility.

The high-integrity containers are loaded inside the radwaste processing facility and transported to the low-level waste storage facility inside a shipping cask or transfer bell. The high-integrity containers are transferred to shielded environmental containers at the low-level storage facility using a mobile crane. Transfer of the high-integrity container to the shielded environmental container is not performed during inclement weather to minimize the potential for a tornado missile to impact the high-integrity container when missile protection is not provided by the shielded environmental container.

Other suitable containers designed for a 300-year life equipped with a special vent design that allows depressurization without the migration of radioactive material may be used as described above in lieu of high-integrity containers as appropriate based on the content of each container.

11.4.2.6 References

1. "A Waste Inventory Report for Reactor and Fuel-Fabrication Facility Wastes," ONWI-20, NUS-3314, March 1979.

11.4.3 SAFETY EVALUATION

The solid radwaste system has no safety design basis.

TABLE 11.4.2-1

PARAMETERS USED IN THE CALCULATION OF ESTIMATED ACTIVITY TO BE SOLIDIFIED

Source	Input Rate (ft^3/year)	Activity and Basis
Condensate polishing spent resin	2850	(a)
Waste processing solidification spent resin	861	(a)
Decant	861	(a)
Evaporator concentrates tanks	2810	(b)
Contaminated oil	200	(c)
Steam generator blowdown spent resin	768	(a)
Waste processing system liquid spent resin	516	(a)
Backflushable filter crud	345	(d)

a. Spent resin design bases and activities are identified in section 12.2.1, and tables 12.2.1-10 and 12.2.1-29.

b. Evaporator concentrates activities and bases are identified in sections 11.2 and 12.2.1.

c. Contaminated oil does not contain significant quantities of radioactivity.

d. For the purposes of determining activities collected by filters, it was assumed that 1 percent of the upstream demineralizer activity was deposited on the respective filter.

TABLE 11.4.2-2 (SHEET 1 OF 2)

ESTIMATED EXPECTED ANNUAL ACTIVITIES OF THE INFLUENTS TO THE SOLID RADWASTE SYSTEM (Ci/year/unit)

Isotope	Spent Resins (Primary)	Spent Resins (Secondary)	Chemical Waste	Radwaste Processing Facility Resins	Backflushable Filter	Cartridge Filters	Dry and Compacted Waste[a]
Br-83	6.07(-1)	1.34(-3)	8.59(-5)	5.57(3)	2.92(1)	-	-
Br-84	-	-	1.05(-5)	-	-	-	-
I-130	1.37(0)	8.05(-3)	1.90(-4)	1.23(-2)	1.63(-1)	3.78(-1)	-
I-131	2.47(3)	2.35(1)	2.48(-1)	8.94(0)	1.20(2)	5.78(-1)	-
I-132	1.19(1)	3.73(-1)	1.53(-2)	7.56(-1)	5.72(-1)	9.63(-3)	-
I-133	3.78(2)	2.81(0)	5.51(-2)	3.54(0)	1.85(1)	1.05(-1)	-
I-134	2.15(0)	2.90(-3)	3.09(-4)	2.61(-2)	1.06(-1)	-	-
I-135	6.38(1)	3.11(-1)	9.28(-3)	6.02(-1)	2.00(0)	1.87(-2)	-
Rb-86	-	1.72(-2)	1.03(-4)	3.22(-3)	-	1.55(-1)	-
Rb-88	3.03(0)	-	4.52(-4)	2.94(-2)	1.48(0)	-	-
Cs-134	2.61(2)	5.90(1)	3.79(-2)	1.04(0)	1.06(1)	1.66(0)	-
Cs-136	1.30(1)	1.94(0)	1.47(-2)	4.87(-1)	4.39(-1)	7.25(-2)	-
Cs-137	2.14(2)	5.03(1)	2.79(-2)	7.64(-1)	9.71(0)	1.40(0)	-
Ba-137m	2.04(2)	4.81(1)	2.61(-2)	7.14(-1)	9.31(0)	1.35(0)	-
Cr-51	4.67(1)	5.66(-1)	2.37(-3)	7.12(-2)	2.14(0)	1.29(-2)	-
Mn-54	7.87(-1)	1.30(-2)	4.50(-4)	1.24(-2)	3.80(-2)	2.81(-4)	-
Fe-55	2.32(2)	3.83(0)	2.34(-3)	6.43(-2)	1.15(1)	1.06(-1)	-
Fe-59	5.20(1)	4.81(0)	1.33(-3)	3.86(-2)	1.77(0)	1.10(-2)	-
Co-58	8.16(2)	1.19(1)	2.20(-2)	6.26(-1)	3.98(1)	2.68(-1)	-
Co-60	2.73(2)	5.11(0)	2.93(-3)	8.03(-2)	1.36(1)	1.13(-1)	-
Sr-89	1.99(1)	1.98(-1)	4.85(-4)	1.39(-2)	9.91(-1)	9.68(-3)	-
Sr-90	1.69(0)	2.72(-2)	1.47(-5)	-	8.44(-2)	6.01(-4)	-
Sr-91	2.98(-1)	1.31(-3)	4.57(-5)	2.80(-3)	1.47(-2)	-	-
Ba-140	2.98(0)	2.98(-2)	2.30(-4)	7.62(-3)	1.49(-1)	1.96(-3)	-
Y-90	6.62(-1)	1.09(-2)	1.11(-5)	-	3.31(-2)	2.19(-4)	-
Y-91m	1.86(-1)	-	2.89(-5)	1.87(-3)	-	1.56(-4)	-
Y-91	2.56(-1)	4.46(-2)	9.14(-5)	2.60(-3)	1.32(-2)	8.73(-4)	-
La-140	1.45(1)	6.88(-2)	2.40(-4)	7.32(-3)	7.25(-1)	1.37(-3)	-
Zr-95	1.46(-1)	4.21(-2)	8.37(-5)	2.38(-3)	-	8.41(-4)	-
Nb-95	-	5.43(-2)	7.63(-5)	2.06(-3)	-	1.08(-3)	-
Mo-99	-	2.34(0)	3.72(-2)	1.89(0)	2.36(-2)	4.68(-2)	-
Tc-99m	-	1.44(0)	3.92(-2)	1.96(1)	1.46(-2)	2.88(-2)	-
Ru-103	1.26(-1)	1.95(-2)	6.03(-5)	1.76(-3)	-	3.89(-4)	-
Ru-106	-	2.00(-2)	1.45(-5)	3.99(-3)	-	4.00(-4)	-
Rh-103m	-	2.96(-2)	6.04(-5)	1.76(-3)	-	4.31(-4)	-
Rh-106	-	5.13(-1)	-	-	-	1.03(-2)	-
Te-132	-	9.18(-1)	1.36(-2)	6.56(-1)	-	1.84(-2)	-
Te-125m	-	1.78(-2)	3.95(-5)	1.12(-3)	-	3.04(-3)	-
Te-127m	-	2.99(-1)	3.94(-4)	1.09(-2)	-	5.98(-3)	-
Te-127	-	3.83(-1)	4.32(-4)	1.36(-2)	-	4.80(-3)	-
Te-129m	-	4.93(-1)	1.80(-3)	5.31(-2)	-	9.88(-3)	-
Te-129	-	4.19(-1)	1.16(-3)	3.46(-2)	-	8.59(-3)	-
Te-131m	-	2.86(-2)	5.19(-4)	3.20(-2)	-	5.73(-3)	-

TABLE 11.4.2-2 (SHEET 2 OF 2)

Isotope	Spent Resins (Primary)	Spent Resins (Secondary)	Chemical Waste	Radwaste Processing Facility Resins	Backflushable Filter	Cartridge Filters	Dry and Compacted Waste[a]
Te-131	-	5.23(-3)	-	-	-	1.05(-3)	-
Ce-141	-	2.47(-2)	9.09(-3)	2.68(-3)	-	4.91(-4)	-
Ce-144	-	5.92(-2)	4.78(-5)	1.31(-3)	-	1.91(-3)	-
Pr-143	-	7.94(-3)	5.84(-5)	1.88(-3)	-	1.59(-4)	-
Pr-144	-	6.85(-2)	4.78(-5)	1.31(-3)	-	1.37(-3)	-
TOTAL	5.07(3)	2.20(2)	5.63(-1)	4.02(1)	2.44(2)	6.02(0)	7.6(-1)

a. Based on data obtained from reference 1 (a total activity of 7.6(-1) Ci/year from dry and compacted wastes). Major isotopic contributions are from Cr-51, Co-58, Co-60, Mn-54, Cs-134, and Cs-137.

TABLE 11.4.2-3 (SHEET 1 OF 2)

ESTIMATED MAXIMUM ANNUAL ACTIVITIES OF THE INFLUENTS TO THE SOLID RADWASTE SYSTEM (Ci/year/unit)

Isotope	Spent Resins (Primary)	Spent Resins (Secondary)	Chemical Waste	Radwaste Processing Facility Resins	Backflushable Filter	Cartridge Filters	Dry and Compacted Waste[a]
Br-83	1.07(1)	-	1.51(-3)	9.80(-1)	5.14(-1)	1.73(-3)	-
Br-84	1.15(0)	-	1.65(-4)	1.07(-2)	-	-	-
I-129	-	2.45(0)	-	-	-	4.90(-2)	-
I-130	1.18(1)	-	1.74(-3)	1.12(-1)	5.76(-1)	3.45(-3)	-
I-131	2.49(4)	2.33(2)	2.48(0)	1.32(2)	1.21(3)	5.78(0)	-
I-132	3.02(2)	1.04(1)	3.89(-1)	1.58(1)	1.47(1)	2.45(-1)	-
I-133	3.97(3)	2.94(1)	5.79(-1)	1.34(2)	1.93(2)	1.10(0)	-
I-134	2.38(1)	3.00(-2)	3.32(-3)	2.81(-1)	1.14(0)	-	-
I-135	6.99(2)	3.36(0)	1.02(-1)	2.76(-1)	3.34(1)	2.05(-1)	-
Rb-86	4.14(1)	4.31(0)	2.56(-2)	8.05(-1)	5.26(-1)	1.55(-1)	-
Rb-88	8.43(1)	-	9.43(-2)	6.14(-1)	3.29(0)	-	-
Rb-89	3.50(0)	-	-	-	-	-	-
Cs-134	2.60(4)	5.21(3)	1.09(1)	1.13(2)	1.02(3)	1.47(2)	-
Cs-136	2.72(3)	3.97(2)	3.06(0)	1.42(2)	1.06(2)	1.51(1)	-
Cs-137	2.13(4)	3.96(3)	2.20(0)	7.30(1)	8.27(2)	1.10(2)	-
Ba-137m	2.02(4)	3.74(3)	6.91(0)	6.91(1)	7.84(2)	1.05(2)	-
Cs-138	3.26(1)	-	-	-	1.19(0)	2.06(-3)	-
Cr-51	1.35(2)	1.64(0)	6.86(-3)	2.06(-1)	6.09(0)	3.72(-2)	-
Mn-54	5.08(1)	8.13(-1)	2.90(-2)	8.00(-1)	2.42(0)	1.81(-2)	-
Mn-56	1.39(1)	-	-	-	1.93(-1)	-	-
Fe-55	2.90(2)	4.78(0)	2.93(-3)	8.04(-2)	1.45(-1)	1.06(-1)	-
Fe-59	1.81(1)	2.51(-1)	6.91(-4)	2.01(-2)	8.37(-1)	5.70(-3)	-
Co-58	7.64(2)	1.11(1)	2.06(-2)	5.87(-1)	3.70(1)	2.51(-1)	-
Co-60	2.59(2)	4.85(0)	1.88(-2)	7.63(-2)	1.28(1)	1.07(-1)	-
Sr-89	2.38(2)	2.37(0)	5.81(-3)	1.64(-1)	1.17(1)	1.16(-1)	-
Sr-90	2.03(1)	3.26(-1)	1.76(-4)	-	1.01(0)	7.21(-3)	-
Sr-91	2.65(0)	-	4.05(-4)	2.48(-2)	2.30(-1)	-	-
Ba-140	5.71(1)	5.69(-1)	4.40(-3)	1.46(-1)	2.86(0)	1.33(-2)	-
Y-90	1.99(1)	3.28(-1)	5.14(-4)	-	9.94(-1)	6.56(-3)	-
Y-91m	-	-	2.33(-4)	1.51(-2)	-	-	-
Y-91	2.24(0)	3.83(-1)	8.02(-4)	2.28(-2)	1.16(-1)	7.66(-3)	-
La-140	5.71(1)	6.01(-1)	2.10(-3)	6.40(-2)	2.86(-2)	1.20(-2)	-
Zr-95	1.56(0)	4.48(-1)	8.93(-4)	2.54(-2)	-	8.96(-3)	-
Nb-95	-	6.92(-1)	9.72(-4)	2.63(-2)	-	1.38(-2)	-
Mo-99	-	2.02(1)	3.21(-1)	3.16(1)	2.01(-1)	4.03(-1)	-
Tc-99m	-	1.91(1)	5.20(-1)	2.96(1)	1.91(-1)	3.82(-1)	-
Ru-103	1.56(0)	2.41(-1)	7.47(-4)	2.18(-2)	-	4.82(-3)	-
Ru-106	-	2.80(-1)	1.85(-4)	-	-	5.60(-3)	-

TABLE 11.4.2-3 (SHEET 2 OF 2)

Isotope	Spent Resins (Primary)	Spent Resins (Secondary)	Chemical Waste	Radwaste Processing Facility Resins	Backflushable Filter	Cartridge Filters	Dry and Compacted Waste[a]
Rh-103m	-	2.41(-1)	6.76(-4)	6.68(-2)	-	4.83(-3)	-
Rh-106	-	6.54(0)	-	-	-	1.31(-1)	-
Te-132	-	9.50(0)	1.41(-1)	1.28(1)	-	1.90(-1)	-
Te-125m	-	1.72(-1)	3.81(-4)	1.08(-2)	-	3.44(-3)	-
Te-127m	-	3.10(0)	4.08(-3)	1.13(-1)	-	6.20(-2)	-
Te-127	-	3.13(0)	5.64(-3)	1.78(-1)	-	6.26(-2)	-
Te-129m	-	6.69(0)	2.44(-2)	7.20(-1)	-	1.34(-1)	-
Te-129	-	4.30(0)	1.16(-2)	3.46(-1)	-	8.59(-2)	-
Te-131m	-	2.87(-1)	5.19(-3)	9.40(-1)	-	5.73(-3)	-
Te-131	-	-	-	-	-	1.05(-3)	-
Te-134	-	-	-	-	-	-	-
Ce-141	-	2.18(-1)	8.60(-4)	2.38(-2)	-	4.36(-3)	-
Ce-143	-	-	1.09(-4)	-	-	-	-
Ce-144	-	6.98(-1)	5.65(-4)	1.55(-2)	-	1.40(-2)	-
Pr-143	-	-	7.22(-4)	2.32(-2)	-	1.96(-3)	-
Pr-144	-	7.03(-1)	4.89(-4)	1.35(-2)	-	1.41(-2)	-
TOTAL	1.02(5)	1.37(4)	2.79(1)	7.61(2)	5.04(3)	3.86(2)	7.6(-1)

a. Based on data obtained from reference 1 (a total activity of 7.6(-1) Ci/year from dry and compacted wastes). Major isotopic contributions are from Cr-51, Co-58, Co-60, Mn-54, Cs 134, and Cs-137.

TABLE 11.4.2-4

ESTIMATED ANNUAL QUANTITIES
OF RADWASTE

Source	Influent to Solid Radwaste (ft^3)	Quantity of Drums Produced[a]
Spent resins (primary)	516	99.0
Spent resins (secondary)	768	147.4
Radwaste processing facility resins	2810	1077
Backflushable filter crud	345	65.3
Dry waste	18,480	994
Filter cartridges	200 elements	200
Condensate demineralizer powdex resins	2,850	484.7

a. Containers other than 55-gal drums are used in conjunction with the radwaste processing facility systems.

TABLE 11.4.2-5

ESTIMATED EXPECTED ANNUAL ACTIVITIES IN SHIPPED RADWASTE (Ci/year/unit)

Isotope	Spent Resins (Primary)	Spent Resins (Secondary)	Chemical Waste	Radwaste Processing Facility Resins	Backflushable Filter	Cartridge Filters	Dry and Compacted Waste[(a)]
Sr-89	1.31(1)	1.31(-1)	3.21(-4)	9.21(-3)	6.57(-1)	6.41(-3)	-
Sr-90	1.69(0)	2.72(-2)	1.47(-5)	3.97(-4)	8.42(-2)	5.60(-4)	-
Y-90	1.69(0)	2.72(-2)	1.47(-5)	3.97(-4)	8.42(-2)	5.60(-4)	-
Y-91	1.83(-1)	3.14(-2)	6.47(-5)	1.86(-3)	9.42(-3)	6.14(-4)	-
Zr-95	1.06(-1)	3.07(-2)	6.09(-5)	1.73(-3)	2.23(-3)	6.12(-4)	-
Nb-95m	-	-	1.50(-6)	-	-	-	-
Nb-95	5.51(-2)	4.60(-2)	7.39(-5)	2.04(-3)	1.50(-3)	9.15(-4)	-
Mo-99	-	1.24(-3)	1.98(-5)	1.01(-3)	-	-	-
Tc-99m	-	3.27(-3)	5.21(-5)	2.65(-3)	-	-	-
Ru-103	7.45(-2)	1.15(-2)	3.57(-5)	1.04(-3)	1.61(-3)	2.30(-4)	-
Ru-106	-	1.89(-2)	1.37(-5)	3.77(-4)	-	2.78(-4)	-
Te-129m	-	2.65(-1)	9.68(-4)	2.86(-2)	2.65(-3)	5.31(-3)	-
Te-129	-	2.12(-1)	7.75(-4)	2.29(-2)	2.12(-3)	4.26(-3)	-
Te-132	-	1.53(-3)	2.26(-5)	1.09(-3)	-	-	-
I-131	1.87(2)	1.78(0)	1.86(-2)	6.78(-1)	9.09(0)	4.38(-2)	-
I-132	-	1.57(-3)	2.33(-5)	1.31(-3)	-	-	-
Cs-134	2.54(2)	5.74(1)	3.69(-2)	1.01(0)	1.03(1)	1.62(0)	-
Cs-136	2.63(0)	3.92(-1)	2.97(-3)	9.84(-2)	8.87(-2)	1.46(-2)	-
Cs-137	2.14(2)	5.02(1)	2.79(-2)	7.63(-1)	9.69(0)	1.40(0)	-
Ba-137m	2.02(2)	4.75(1)	2.64(-2)	7.22(-1)	9.17(0)	1.32(0)	-
Ba-140	5.87(-1)	5.87(-3)	4.53(-5)	1.50(-3)	2.94(-2)	3.86(-4)	-
La-140	6.76(-1)	6.76(-3)	5.22(-5)	1.73(-3)	3.38(-2)	4.44(-4)	-
Ce-141	-	1.30(-2)	4.78(-5)	1.41(-3)	-	2.58(-4)	-
Ce-143	-	-	-	-	-	-	-
Ce-144	-	5.50(-2)	4.44(-5)	1.22(-3)	-	1.11(-3)	-
Pr-143	-	1.73(-3)	1.29(-5)	4.21(-4)	-	-	-
Pr-144	-	5.50(-2)	4.44(-5)	1.22(-3)	-	1.11(-3)	-
Cr-51	2.21(1)	2.67(-1)	1.12(-3)	3.36(-2)	1.01(0)	6.09(-3)	-
Mn-54	7.36(-1)	1.22(-2)	4.21(-4)	1.16(-2)	3.56(-2)	2.63(-4)	-
Fe-59	3.26(1)	3.02(0)	8.34(-4)	2.42(-2)	1.11(0)	6.90(-3)	-
Co-58	6.10(2)	8.89(0)	1.64(-2)	4.68(-1)	2.97(1)	2.00(-1)	-
Co-60	2.70(2)	5.06(0)	2.90(-3)	7.94(-2)	1.35(1)	1.12(-1)	-
Rh-106	-	1.89(-2)	1.37(-5)	3.77(-4)	-	3.78(-4)	-
Te-127m	-	2.47(-1)	3.26(-4)	9.01(-3)	2.47(-3)	4.94(-3)	-
Te-127	-	2.48(-1)	3.27(-4)	9.04(-3)	2.48(-3)	4.96(-3)	-
Rh-103m	7.45(-2)	1.15(-2)	3.57(-5)	1.04(-3)	1.61(-3)	2.30(-4)	-
TOTAL	1.81(3)	1.76(2)	1.38(-1)	3.99(0)	8.46(1)	4.76(0)	7.6(-1)

a. Based on data obtained from reference 1 (a total activity of 7.6(-1) Ci/year from dry and compacted wastes). Major isotopic contributions are from Cr-51, Co-58, Co-60, Mn-54, Cs-134, and Cs-137.

TABLE 11.4.2-6

ESTIMATED MAXIMUM ANNUAL ACTIVITIES IN SHIPPED RADWASTE (Ci/year/unit)

Isotope	Spent Resins (Primary)	Spent Resins (Secondary)	Chemical Waste	Radwaste Processing Facility Resins	Backflushable Filter	Cartridge Filters	Dry and Compacted Waste[a]
Sr-89	1.58(2)	1.57(0)	3.85(-3)	1.09(-1)	7.75(0)	7.68(-2)	-
Sr-90	2.03(1)	3.25(-1)	1.76(-4)	-	1.01(0)	7.20(-3)	-
Y-90	2.03(1)	3.25(-1)	1.76(-4)	-	1.01(0)	7.20(-3)	-
Y-91	1.60(0)	2.69(-1)	5.67(-4)	1.63(-2)	8.38(-2)	5.39(-3)	-
Zr-95	1.14(0)	3.26(-1)	6.50(-4)	1.85(-2)	2.60(-2)	6.52(-3)	-
Nb-95	-	5.52(-1)	8.75(-4)	2.42(-2)	1.73(-2)	1.10(-2)	-
Mo-99	-	-	1.71(-4)	1.68(-2)	-	-	-
Tc-99m	-	-	4.49(-4)	4.42(-2)	-	-	-
Ru-103	-	1.43(-1)	4.42(-1)	1.29(-2)	1.99(-2)	2.85(-3)	-
Ru-106	-	2.65(-1)	1.75(-4)	-	-	5.29(-3)	-
Te-129m	-	3.60(0)	1.31(-2)	3.87(-1)	3.59(-2)	7.21(-2)	-
Te-129	-	2.88(0)	1.05(-2)	3.10(-1)	2.88(-2)	5.77(-2)	-
Te-132	-	-	2.35(-4)	2.13(-2)	-	-	-
I-129	-	2.45(0)	-	-	2.45(-2)	4.90(-2)	-
I-131	1.89(3)	2.52(1)	1.88(-1)	1.00(1)	9.17(1)	4.38(-1)	-
Cs-134	2.53(4)	5.07(3)	1.06(1)	1.10(2)	9.92(2)	1.43(2)	-
Cs-136	5.49(2)	8.02(1)	6.18(-1)	2.87(1)	2.14(1)	3.05(0)	-
Cs-137	2.13(4)	3.95(3)	2.20(0)	7.29(1)	8.25(2)	1.10(2)	-
Ba-137m	2.01(4)	3.74(3)	2.08(0)	6.89(1)	7.81(2)	1.04(2)	-
Ba-140	1.13(1)	1.12(-1)	8.67(-4)	2.88(-2)	5.63(-1)	2.62(-3)	-
La-140	1.30(1)	1.29(-1)	9.97(-4)	3.31(-2)	6.48(-1)	3.01(-3)	-
Ce-141	-	1.15(-1)	4.53(-4)	1.25(-2)	-	2.29(-3)	-
Ce-144	-	6.49(-1)	5.25(-4)	1.44(-2)	-	1.30(-2)	-
Pr-143	-	-	1.59(-4)	-	-	-	-
Pr-144	-	6.49(-1)	5.25(-5)	1.44(-2)	-	1.30(-2)	-
Cr-51	6.38(1)	7.75(-1)	3.24(-3)	9.73(-2)	2.88(0)	1.76(-2)	-
Mn-54	4.75(1)	7.61(-1)	2.71(-2)	7.49(-1)	2.26(0)	1.69(-2)	-
Fe-59	1.14(1)	1.58(-1)	4.34(-4)	1.26(-2)	5.25(-1)	3.58(-3)	-
Co-58	5.71(2)	8.29(0)	1.54(-2)	4.39(-1)	2.76(1)	1.88(-1)	-
Co-60	2.56(2)	4.80(0)	1.86(-2)	7.55(-2)	1.27(1)	1.06(-1)	-
Rh-106	-	2.65(-1)	1.75(-4)	-	-	5.29(-3)	-
Te-127m	-	2.56(0)	3.37(-3)	9.34(-2)	2.56(-2)	5.12(-2)	-
Te-127	-	2.57(0)	3.38(-3)	9.37(-2)	2.57(-2)	5.14(-2)	-
Rh-103m	-	1.43(-1)	4.42(-4)	1.29(-2)	1.99(-2)	2.85(-3)	-
TOTAL	7.03(4)	1.29(4)	1.58(1)	2.93(2)	2.77(3)	3.61(2)	7.6(-1)

a. Based on data obtained from reference 1 (a total activity of 7.6(-1) Ci/year from dry and compacted wastes). Major isotopic contributions are from Cr-51, Co-58, Co-60, Mn-54, Cs-134, and Cs-137.

TABLE 11.4.2-7

SUMMARY OF BACKFLUSHABLE FILTERS

Filter	System Process Filtered / Stream	Number of Filters Per Unit	Location	FSAR Section in Which Filtered System is Described
Seal water return	CVCS / Reactor coolant pump seals	1	Auxiliary building level B	Paragraph 9.3.4.1
Recycle evaporator feed	BRS / Recycle evaporator feed demineralizer	2 (common)	Auxiliary building level B	Paragraph 9.3.4.2
Waste evaporator feed	WPSL / Waste holdup tank (downstream of pumps)	1	Auxiliary building level B	Subsection 11.2.2
Resin sluice	WPSL / Spent resin storage tank (downstream of pumps)	1	Auxiliary building level B	Subsection 11.2.2
Floor drain tank	WPSL / Floor drain tank (downstream of pumps)	1	Auxiliary building level B	Subsection 11.2.2
Waste monitor tank	WPSL / Waste monitor tank demineralizer	1	Auxiliary building level D	Subsection 11.2.2
Steam generator blowdown	SGBS / Steam generator blow down trim heat exchanger	1	Auxiliary building level B	Paragraph 10.4.8.2

11.5 PROCESS AND EFFLUENT RADIOLOGICAL MONITORING AND SAMPLING SYSTEMS

The process and effluent radiation monitoring systems (PERMS) monitor radiation levels in selected plant process systems and in plant effluents. A description of the area radiation monitoring system is found in subsection 12.3.4.

The PERMS operates in conjunction with regular and special radiation surveys and with a chemical and radiological sampling program performed by the plant staff to provide timely, sufficient information for continued safe plant operation.

11.5.1 DESIGN BASES

The PERMS is designed to:

A. Provide equipment and criteria to ensure plant performance which meets applicable regulatory requirements for normal operation and for transient situations that might be expected to occur.

B. Provide support data to health physics personnel to aid them in limiting release of activity to the environment and limiting exposure to operation and maintenance personnel to as low as reasonably achievable.

C. Provide early warning of a system or equipment malfunction that could lead to a health hazard or plant damage.

D. Provide a reliable and usable tool to monitor plant-related radioactivity.

E. Provide support to the plant for compliance with applicable Nuclear Regulatory Commission guidelines including:

1. 10 CFR 50, Appendix I.
2. 10 CFR 20.
3. Regulatory Guide 1.21.
4. Regulatory Guide 8.8.
5. Regulatory Guide 1.45.
6. General Design Criteria 60, 63, and 64.
7. Regulatory Guide 1.97.
8. ANSI N13.1-1969.

The scope of PERMS regarding radiological monitoring following accidents as required by General Design Criterion 64 and Regulatory Guide 1.97 is limited to releases associated with the postulated design basis accidents considered in chapter 15. Post accident radiation monitoring is discussed in subsection 11.5.5. Although PERMS is generally a surveillance system, there are certain monitors that perform a safety function. Such monitors meet the normal requirements for safety grade equipment. For safety-related monitors, redundancy is relied upon to maintain surveillance in case of a single failure.

11.5.2 SYSTEM DESCRIPTION

11.5.2.1 Digital Radiation Monitoring System

The process and effluent radiation monitoring system (PERMS) is based on a distributed digital microprocessor approach, where each radiation monitor is self-contained and consists of a detector and a data processing module (DPM) that contains the microprocessor. The DPM is the basic control center for each PERMS channel. Each complete channel consists of a monitor with power supply and preamplifier; a local dedicated data processing module containing a digital buffer and microprocessor with alarm outputs; and access to readout modules (cathode ray tube (CRT) and printer).

The PERMS collects and displays all the information available from the field-mounted detectors on a CRT and hardcopy printer on demand. This is accomplished by either the communications display computer (CDC) or the safety-related display console (SRDC) located in the control room. The CDC receives data from the DPMs (both standard and safety-related) and interfaces with the PERMS display computer (PDC) and, via a boundary isolation device (BID), the integrated plant computer (IPC) (figure 11.5.2-1). The SRDC displays data received from the safety-related channels which are composed of safety-related Class 1E components (figure 11.5.2-2). The safety-related DPMs also interface with the CDC with one-way data transmission. The CDC is shown in figure 11.5.2-3.

11.5.2.2 Monitor Description

The radiation monitors fall into five different functional classifications:

A. Process monitors, which determine concentrations of radioactive material in plant fluid systems. The primary-to-secondary leak detection monitors (N16 and noble gas leak rate detectors) are also included in this category.

B. Effluent monitors, which measure radioactivity discharged to the environs.

C. Airborne monitors, which provide operator information on airborne concentrations of radioactive gases and particulate radioactivity at various points in the ventilation system ducts. (See figure 11.5.2-4.)

D. Area monitors, which provide operator information on external gamma radiation levels at fixed points throughout the plant.

E. Post-accident (or high range) monitors, designed to assess and follow potential pathways for release of radioactive materials during accident conditions.

Table 7.5.4-2 provides cross-references of radiation monitors' conformance with NUREG-0737 guidance. Table 11.5.2-1 gives the design parameters for the PERMS. Table 11.5.2-2 lists the detectors used in the PERMS. Table 11.5.2-3 lists the conditions of service for PERMS. Figures 11.5.2-5 and 11.5.2-6 give the monitoring locations for liquid and gaseous release pathways, and table 11.5.2-5 gives the design parameters for flowrate monitoring. Drawings AX6DD200, AX6DD204, AX6DD205, AX6DD206, AX6DD207, AX6DD209, AX6DD211, AX6DD215, 1X6DD201, 1X6DD202, 2X6DD201, 2X6DD202, AX6DD218, AX6DD221, AX6DD223, AX6DD225 and AX6DD226 show the detector locations.

11.5.2.3 Monitor Functions

Most of the PERMS radiation monitors are centrally (by unit) monitored as part of the digital radiation monitoring system (DRMS). The 112 PERMS channels installed at the VEGP site consist of:

A. Twenty-four area monitors, described in subsection 12.3.4, 17 DRMS (8 per unit and 1 shared), and 7 shared non-DRMS monitors;

B. Sixty process and effluent channels, described in subsection 11.5.2 and table 11.5.2-1 are allocated as follows:

   a. Active Channels: Unit 1-25, Unit 2-26, and 6 shared.

   b. Passive Channels: Unit 1-1, Unit 2-1, and 0 shared.

   c. Inactive Channels: 1 shared non-DRMS.

C. Twenty-eight post-accident monitor DRMS channels, described in subsection 11.5.5, (10 active and 4 passive activity collection per unit).

Some of these monitors are located in areas of the site and serve functions that are common to both units. These shared monitors are identified by an "A" in the front of the monitor's tag number (i.e., "ARE - --------").

The following is a description of each monitor:

A. Airborne Process and Effluent Monitors

1. RE-12442A - Plant Vent Effluent Air Particulate Monitor (Low Range)

   The purpose of this monitor is to continuously monitor air particulate activity as it is discharged to the environment through the main plant vent and provide data necessary to ensure that particulate activity releases from the plant vent are below specified limits.

   The monitor also serves as a backup to the containment vent particulate monitor RE-2565A. Since the main plant vent discharges directly to the atmosphere, the data from this monitor are most representative of particulate activity releases to the plant environs. Should the activity exceed a specified level, a high radiation level alarm will be annunciated to indicate that an increase in particulate release has occurred. The specific source of release within the plant may be determined through analysis of the monitors upstream of the vent monitor and/or with portable monitoring devices.

2. RE-12442B - Plant Vent Effluent Iodine Monitor (Low Range)

   The purpose of this monitor is to continuously monitor iodine activity as it is discharged to the environment through the main plant vent and provide data necessary to ensure that gaseous iodine activity releases from the plant vent are below specified limits.

   The monitor also serves as a backup to the containment vent radioiodine monitor RE-2565B. Since the main plant vent discharges directly to the atmosphere, the data from this monitor are most representative of the iodine activity releases to the plant environs. Should the activity exceed a

specified level, a high radiation level alarm will be annunciated to indicate that an increase in gaseous iodine release has occurred. The specific source of release within the plant may be determined through analysis of the monitors upstream of the vent monitor and/or with portable monitoring devices.

3. RE-12442C - Plant Vent Effluent Radiogas Monitor (Low Range)

   The purpose of this monitor is to continuously monitor the gaseous activity as it is discharged to the environment through the main plant vent and provide data necessary to ensure that the gaseous activity releases from the plant vent are below specified limits.

   The monitor also serves as a backup to other monitored streams that are routed to the main plant vent, including the waste gas processing system effluent ARE-0014, fuel building ARE-2532A and B and ARE-2533A and B, and containment RE-2565C gas monitors. Since the main plant vent discharges directly to the atmosphere, the data from this monitor are most representative of gaseous activity releases to the plant environs. Should the activity exceed a specified level, a high radiation level alarm will be annunciated to indicate that an unexpected increase in gaseous release has occurred. The specific source of release within the plant may be determined through analysis of the monitors upstream of the vent monitor and/or with portable monitoring devices.

4. RE-2565A - Containment Vent Effluent Air Particulate Monitor

   The purpose of this monitor is to measure air particulate radioactivity in the containment purge vent and provide data necessary to ensure that the release rate through the containment vent during purging is below specified limits.

   The containment purge flow is filtered and routed through the main plant vent, where activity is again monitored by the plant vent air particulate monitor RE-12442A and then discharged to the atmosphere.

   This monitor initiates automatic closure of the containment purge supply and exhaust duct valves for high radiation levels.

5. RE-2565B - Containment Vent Effluent Iodine Monitor

   The purpose of this monitor is to measure gaseous iodine activity in the containment purge vent and provide data necessary to ensure that the release rate through the containment vent during purging is below specified limits.

   The containment purge flow is filtered and routed through the main plant vent, where activity is again monitored by the plant vent radioiodine monitor RE-12442B and then discharged to the atmosphere.

   This monitor initiates automatic closure of the containment purge supply and exhaust duct valves for high radiation levels.

6. RE-2565C - Containment Vent Effluent Radiogas Monitor

   The purpose of this monitor is to measure gaseous activity in the containment purge vent and provide data necessary to ensure that the release rate through the containment vent during purging is below specified limits.

   The containment purge flow is routed through the main plant vent, where the activity is again monitored by the plant vent gas monitor RE-12442C and then discharged to the atmosphere.

   This monitor initiates automatic closure of the containment purge supply and exhaust duct valves for high radiation levels.

7. RE-2562A - Containment Atmosphere Process Air Particulate Monitor

   The purpose of this monitor is to measure air particulate radioactivity in the containment atmosphere and provide information for determining allowable personnel access to the containment.

   The monitor can also be utilized to provide an indication of reactor coolant system leakage as part of the reactor coolant leak detection system. An intermediate radiation alarm alerts the operator that a predetermined activity level in the containment has been exceeded.

8. RE-2562B - Containment Atmosphere Iodine Cartridge

   The purpose of this monitor is to measure airborne iodine radioactivity in the containment atmosphere and provide information for determining allowable personnel access to the containment.

   The cartridge may also be used to provide an indication of reactor coolant system leakage.

9. RE-2562C - Containment Atmosphere Process Radiogas Monitor

   The purpose of this monitor is to measure gaseous radioactivity in the containment atmosphere and provide information for determining allowable personnel access to the containment.

   The monitor can also be utilized to provide an indication of reactor coolant system leakage as part of the reactor coolant leak detection system. An intermediate radiation alarm alerts the operator that a predetermined activity level in the containment has been exceeded.

10. Deleted

11. Deleted

12. Deleted

13. Deleted

14. Deleted

15. RE-12116 and RE-12117 - Control Room Air Intake Process Radiogas Monitor

    The purpose of this monitor is to continuously monitor the control room supply air for gaseous activity that could be present in the atmosphere following an accident.

    A high alarm signal will generate a control room isolation (CRI) signal which will automatically switch the control room ventilation system from the normal operating mode to the accident mode using safety grade filtration trains. In addition, once the CRI has been initiated, a signal to the onsite technical support center ventilation is provided to switch to its filtration mode.

16. RE-0013 - Waste Gas Processing System Process Radiogas Monitor

    The purpose of this monitor is to continuously monitor the gaseous activity of the waste gas processing system recirculation stream and provide data used to determine the radioactive inventory of the waste gas decay tanks and/or waste gas shutdown tanks.

17. Deleted

18. Deleted

19. Deleted

20. ARE-0014 - Waste Gas Processing System Effluent Radiogas Monitor

    The purpose of this monitor is to continuously monitor the gaseous activity discharged from the waste gas processing system and provide data necessary to ensure that gaseous activity releases are below specified levels.

    The activity is routed to the Unit 1 auxiliary building exhaust (and again monitored by RE-0039A) and, in turn, to the Unit 1 plant vent (and again monitored by RE-12442C).

    A high radiation alarm signal isolates the waste gas processing system discharge line.

21. RE-0039A, RE-0039B, and RE-0039C - Waste Gas Decay Tank and Compressor Area Ventilation Effluent Radiogas Monitor

    The purpose of this monitor is to continuously monitor the ventilation exhaust from the waste gas decay tank area (RE-0039A), the waste gas compressor areas along with the catalytic recombiner room in Unit 1 (RE-0039B), and catalytic recombiner room in Unit 2 (RE-0039C).

    A high radiation alarm indicates leakage in the waste gas processing system or catalytic recombiner areas being monitored.

22. ARE-2532A and B and ARE-2533A and B - Fuel Handling Building Effluent Radiogas Monitors

The purpose of this monitor is to continuously monitor the gaseous activity discharged from the fuel handling building and provide data necessary to ensure that gaseous activity releases are below specified limits.

The activity is filtered and routed to the plant vent, where it is again monitored by the plant vent monitors RE-12442A, B, and C prior to discharge to the atmosphere. In the event of a fuel handling accident inside the fuel handling building, ARE-2532 and ARE-2533 are capable of detecting the high activity in order to switch the fuel handling building ventilation system from the normal operating mode to the accident mode, which includes safety-related filtration units. (See subsection 9.4.2.)

23. Deleted.

24. Deleted.

25. ARE-16980 – Radwaste Processing Facility Vent Air Particulate Monitor

    The purpose of this monitor is to continuously monitor air particulate activity as it is discharged through the radwaste processing facility ventilation exhaust system to the atmosphere and to provide data to ensure that activity releases from the facility are below specified limits. This monitor is part of the PERMS system.

    Should activity exceed a specified level, a high radiation level alarm will be annunciated in the radwaste processing facility local control room and in the Unit 1 main control room to indicate that an increase in particulate release has occurred.

B. Liquid Process and Effluent Monitors

1. RE-810 - Steam Generator Primary-to-Secondary Noble Gas Leak Detection Monitor

   The purpose of this monitor is to monitor the condenser steam jet air ejector header to detect noble gases and determine combined leak rate (contributed by all four steam generators) and rate of change in primary-to-secondary leaks. Remote readout is provided at the integrated plant computer (IPC) and the Communications Console PERMS viewer application in the control room.

   The noble gas monitor, RE-810, is a stand-alone monitor that sends an output signal to the Communications Console which sends the information to the IPC via the BID. The IPC converts this signal to provide readout of primary-to-secondary leak rate in gallons per day (GPD), as well as rate of change (GPD/h). RE-810 monitor parameters and alarm setpoints are maintained locally at the monitor and are also changeable via the PERMS Communications Console or the PERMS display console (PDC). Parameters for GPD leak rate, GPD/h calculations, and alarm setpoints for both calculated values are maintained on the IPC.

2. RE-0017A and B - Component Cooling Water Process Monitor

The purpose of this monitor is to continuously monitor the component cooling loop for activity indicative of a leak of radioactive water into the component cooling system.

3. RE-0018 - Waste Liquid Effluent Monitor

   The purpose of this monitor is to monitor waste disposal system liquid releases from the plant and provide data necessary to ensure that liquid waste activity discharges are below specified limits.

   A high alarm signal initiates automatic valve closure on the liquid waste discharge line to prevent further activity release. Remote readout is provided at the liquid section of the waste processing system control panel.

4. RE-0019 - Steam Generator Sample Liquid Process Monitor

   The purpose of this monitor is to monitor the liquid phase of the secondary side of the steam generators for activity (which would be indicative of primary to secondary system leaks) and to provide backup information to the air ejector gas monitor RE-12839C.

   Remote readout and A/V alarm are provided at the isolation valve controls in the secondary side of sampling area.

5. RE-724 - Steam Generator Primary-to-Secondary N16 Leak Detector Monitor

   The purpose of this monitor is to monitor the main steam lines to detect N16 isotope and determine combined leak rate (contributed by all four steam generators) and rate of change in primary-to-secondary leaks. Remote readout is provided at the integrated plant computer (IPC).

   The N-16 monitor, RE-724, is a stand-alone monitor which sends an output signal to the IPC. The IPC converts this signal to provide readout of primary-to-secondary leak rate in gallons per day (GPD), as well as rate of change (GPD/h). RE-724 monitor parameters and alarm setpoints are maintained locally at the monitor. Parameters for GPD leakrate, GPD/h calculations and alarm setpoints for both calculated values are maintained on the IPC.

6. RE-0020A and B - Nuclear Service Water Process Monitor

   The purpose of this monitor is to continuously monitor the nuclear service water loop for activity and to prevent otherwise undetected activity release from the service water system, which represents a potential release path to the environment.

   During normal operation, these monitors will indicate one of the following conditions:

   a. Leak from equipment processing radioactive water into nuclear service cooling water.

b. Leak from equipment processing radioactive liquid into auxiliary component cooling water system coincident with leak to nuclear service cooling water.

c. Leak from equipment processing radioactive liquid into component cooling water system coincident with leaks to nuclear service cooling water.

7. RE-0021 - Steam Generator Blowdown Liquid Process Monitor

The purpose of this monitor is to monitor the steam generator blowdown liquid for activity in the event of primary to secondary system leakage and provide data necessary to ensure that the steam generator blowdown processing system discharges to the environment are below desired limits.

This monitor also provides indication of a steam generator blowdown processing system malfunction.

A high alarm signal automatically closes the steam generator blowdown processing system isolation valves and discharge lines. Remote readout is provided at the steam generator blowdown system panel.

8. RE-0848 - Turbine Building Drain Liquid Effluent Monitor

The purpose of this monitor is to continuously monitor the secondary side liquid drains for activity released in the event of primary to secondary system leakage.

A high alarm signal stops the flow from the turbine building drain system to the waste water retention basin. Turbine building sump pump discharges are diverted to the "dirty" turbine building drain tank. Discharges from the "clean" turbine building drain tank are recirculated back to the clean tank.

9. RE-1950 - Auxiliary Component Cooling Water Process Monitor

The purpose of this monitor is to continuously monitor the auxiliary component cooling loop for activity indicative of a leak of contaminated water into the auxiliary component cooling system.

10. RE-48000 - Chemical and Volume Control System Letdown Monitor

The purpose of this monitor is to monitor the chemical and volume control system letdown liquid process and provide indication of abnormal activity levels in the reactor coolant system.

11.5.2.4 Alarm Setpoints

Each of the PERMS airborne and liquid monitors (excluding passive cartridge monitors RE-2562B, RE-12839A and B, and RE-12444A and B) has two alarm setpoints; i.e., intermediate and high. When the radiation level being monitored reaches the intermediate setpoint, a visual

indication alerts plant personnel of the monitor reading. If the high setpoint is reached, an alarm will be annunciated. The high alarm setpoint also triggers the control function for those monitors so equipped: gas monitors RE-2565A, B, and C, RE-12839C, RE-12116, RE-12117, ARE-0014, ARE-2532A and B, and ARE-2533A and B; and liquid monitors RE-0018, RE-0021, and RE-0848.

The primary-to-secondary leak detection (N16) monitor, RE-724, setpoints and the primary-to-secondary noble gas leak detection monitor, RE-810, leak rate and leak rate change setpoints are operator controlled through the IPC interface. The RE-810 radiation setpoint is operator controlled through the PERMS Comm Console. Variable operator-selected leak rate gallons per day (GPD) and leak rate changes (GPD/h) are provided in the main control room.

Setpoints are controlled by plant procedures and the Offsite Dose Calculational Manual (ODCM) where appropriate.

11.5.2.5 Inservice Inspection, Calibration, and Maintenance

The operability of each digital radiation monitoring system (DRMS) safety-related radiation detector/channel, except for RE-0005 and RE-0006, of the PERMS, PAMS, and ARMS is checked frequently with a source positioned remotely from the SRDC in the main control room or the DPM, or manually at the monitor. The operability of each DRMS nonsafety-related radiation detector/channel, except for RE-48000, of the PERMS, PAMS, and ARMS is checked frequently with a source positioned remotely from the communications console in the main control room, the DPM, or the chemistry PERMS station, or manually at the monitor.

Channel checks are routinely done to verify parameters and alarm conditions. Electronic testing means are used to calibrate instrument subcomponents for accurate range, setpoint, and indication periodically. The PERMS, PAMS, and ARMS radiation monitors are functionally tested periodically. This periodic test is used to verify the operability of alarms, automatic valves, and dampers, if applicable. This is performed by lowering the setpoint below background; introducing an external electronic input signal or other testing means as necessary to create an alarm/actuation condition; and verifying the appropriate response of alarms, valves, and dampers.

Calibration of all PERMS, PAMS, and ARMS digital radiation monitoring system (DRMS) detectors is normally performed every 18 months. Generally this procedure is performed on all but containment area monitors during an administratively selected period of the fuel cycle other than refueling to minimize the effect on the plant and to limit occupational exposure. Calibration of non-DRMS detectors/monitors (radwaste processing facility ARMS, and N16 and noble gas primary-to-secondary leak detection) is normally performed every 18 months with periodic testing under separate procedural controls from the DRMS.

Calibration of all PERMS and PAMS detectors is performed using sources that allow indication of low and mid range. Calibration of low range ARMS monitors is performed using a source which will allow indication in a low, mid, and high range. Calibration of high range ARMS monitors is performed in accordance with manufacturer's recommendations to prevent undesirable exposure to plant personnel.

An initial calibration was performed on PERMS, PAMS, and ARMS monitors with the use of NIST traceable standards of known isotopic concentration. Secondary sealed sources with fixed geometrical configurations were tested to determine relationship to the initial calibration standards. These secondary calibration sources or National Institute of Science and Technology (NIST) traceable sources are used for periodic onsite recalibration to ensure that

the response of each detector is in agreement with the response as determined in the original initial "type" calibration.

Provisions for minimizing sample deposition in liquid effluent radiation monitors through electropolishing, decontaminating, and flushing sample chambers have been made.

11.5.3 EFFLUENT MONITORING AND SAMPLING

11.5.3.1 Plant Sampling System

The plant sampling system is described below.

11.5.3.2 Sampling Requirements for Process and Effluent Radiation Monitoring System (PERMS)

The primary means of quantitatively evaluating the isotopic activity levels in process and effluent paths is a program of sampling and laboratory measurements. Gross activity measurements, as provided by the process and effluent monitors, are generally not acceptable for showing compliance with effluent release limits. However, the continuous gross monitors (both process and effluent) can be calibrated by normalizing against the results of specific radionuclide analysis, and in this way the gross measurement from the continuous monitors may be used to determine the isotopic activity concentration in process streams and the total isotopic activity released in effluent paths. Tables 11.5.3-1 and 11.5.3-2 present information for the sampling system associated with the PERMS monitors, for both gaseous and liquid monitors. These tables include the following information for each PERMS monitor:

- Sample identification (including fluid type, sample point location, and sample type).
- Purpose.

Sample points are located at the PERMS monitor location unless a sample point that will serve to calibrate the monitor already exists in the system.

The requirements of General Design Criterion 64, Monitoring Radioactivity Releases, are satisfied by this sampling program and the associated radiological monitors described in subsection 11.5.2. VEGP will conform to Regulatory Guide 4.15, Quality Assurance for Radiological Monitoring Programs (Normal Operations) - Effluent Streams and Environment.

11.5.4 PROCESS MONITORING AND SAMPLING

The requirements of General Design Criterion 60, Control of Releases of Radioactive Materials to the Environment, have been addressed. Monitors used to initiate automatic closure of isolation valves in liquid and gaseous systems are described in subsection 11.5.2.

The requirements of General Design Criteria 63, Monitoring Fuel and Waste Storage, have been addressed. Radiation monitors used in the radioactive waste processing systems are discussed in subsection 11.5.2.

11.5.5 POST-ACCIDENT RADIATION MONITORING

In addition to the process and effluent radiation monitoring system (PERMS) described in subsection 11.5.2, radiological monitors are also included in the post-accident monitoring system (PAMS) as required by General Design Criterion 64 and Regulatory Guide 1.97. The function of the radiation monitors in this system is to detect, compute, and indicate the radiation levels at selected plant locations and to actuate alarms should these levels exceed predetermined values. Thus, the control room operating staff is provided with the information on radiation levels needed to help assess and maintain the safety of the plant and its environs following the occurrence of any design basis accident.

The PAMS radiation monitors supplement the PERMS and area radiation monitoring system (ARMS) (subsections 11.5.2 and 12.3.4, respectively) and consist of 28 DRMS monitors (10 active and 4 passive monitors per unit). The following is a description of each monitor:

A. RE-12444A - Plant Vent Air Particulate Monitor

   The purpose of this monitor is to continuously monitor air particulate activity as it is discharged to the environment through the main plant vent and to provide data necessary to estimate that particulate activity releases from the plant vent are below specified limits.

B. RE-12444B - Plant Vent Effluent Iodine Monitor

   The purpose of this monitor is to measure airborne iodine radioactivity in the plant vent.

C. RE-12444C,D,E - Plant Vent Radiogas Monitor

   The purpose of this monitor is to continuously monitor the gaseous activity as it is discharged to the environment through the main plant vent and provide data necessary to ensure that the gaseous activity releases from the plant vent are below specified limits.

D. RE-13119, RE-13120, RE-13121, and RE-13122 - Main Steam Line Monitors

   The purpose of this monitor is to detect steam generator tube leakage and to monitor this potential effluent release path. One detector is used on the main steam line of each steam generator upstream of the main steam line isolation and atmospheric dump valves.

E. RE-12839A - Condenser Air Ejector and Steam Packing Exhauster Passive Air Particulate Monitor

   The purpose of this monitor is to continuously measure particulate activity in the discharge from the air ejector exhaust header of the condensers (which is indicative of a primary to secondary system leak) and steam packing exhauster monitor effluent path and to estimate magnitude of release of radioactivity.

F. RE-12839B - Condenser Air Ejector and Steam Packing Exhauster Passive Iodine Cartridge

   The purpose of this monitor is to continuously measure iodine activity in the discharge from the air ejector exhaust header of the condensers (which is indicative of a primary to secondary system leak) and steam packing exhauster monitor effluent path and to estimate magnitude of release of radioactivity.

G. RE-12839C,D,E - Condenser Air Ejector and Steam Packing Exhauster Radiogas Monitor

The purpose of this monitor is to continuously measure gaseous activity in the discharge from the air ejector exhaust header of the condensers (which is indicative of a primary to secondary system leak) and steam packing exhauster effluent path and to estimate magnitude of release of radioactivity.

A high radiation alarm signal from this monitor will automatically divert the air ejector discharge flow through charcoal and high efficiency particulate air filtration banks.

Table 11.5.5-1 gives the design parameters for the PAMS radiation monitors. The plant vent and the condenser air ejector and steam packing exhauster radiogas monitors consist of one beta scintillation and 2 G-M detectors each to cover the required range. Table 11.5.5-2 presents the detector requirements for the PAMS radiation monitors. Table 11.5.5-3 lists the conditions of service for the PAMS radiation monitors.

Sampling requirements for the PAMS radiation monitors (where applicable) are included with the PERMS on table 11.5.3-1.

Alarm setpoints are controlled by plant procedures and the Offsite Dose Calculation Manual (ODCM) where appropriate.

Inservice inspection, calibration, and maintenance of PAMS radiation monitors are described in paragraph 11.5.2.5.

Some monitors described in this section are utilized for normal operations as well as for accident conditions. Likewise, some of the area radiation monitors described in subsection 12.3.4 are also used for post-accident monitoring. The grouping presented in this report, recognizing that some of these monitors are multifunctional, was so chosen to maintain clarity while avoiding duplication.

11.5.5.1 Potential Gaseous Accident Release Pathways

The potential gaseous accident release pathways are from:

A. The plant vent which includes discharges from the containment purge system, the auxiliary building HVAC system (which includes discharges from the gaseous radwaste system and the containment piping penetration area filter and exhaust system) and the fuel handling building HVAC.

B. The condenser air ejector and steam packing exhauster system.

C. The steam generator safety relief valves and atmospheric dump valves.

D. The auxiliary feedwater steam turbine exhaust vent.

E. The steam generator blowdown line break blowout panel.

The plant vent, condenser air ejector and steam packing exhauster, and main steam line monitor's capability and flowrate monitoring equipment are summarized in tables 11.5.5-1 through 11.5.5-3. The steam generator liquid monitor capability and flowrate monitoring equipment is summarized in tables 11.5.2-1 through 11.5.2-3 and 11.5.2-5.

The main steam line monitors are used to estimate the releases from the actuation of steam generator secondary safety relief valves, atmospheric steam dump valves, and the auxiliary feedwater steam turbine exhaust. The main steam line monitors are externally mounted Geiger-

Mueller tubes viewing the main steam line upstream of the safety valves, atmospheric dump valves, and steam supply line to the auxiliary feedwater steam turbine. The main steam line monitors are calibrated to account for the pipe thickness of the main steam line.

The activity release rate or integrated release from the steam generator safety valves and/or atmospheric dump valves during an accident may be estimated by using the appropriate readings from the main steam line monitor and the main steam line flowrate and pressure. (Main steam line flowrate and pressure instrumentation is discussed in section 7.5.) The activity release rate from the auxiliary feedwater turbine exhaust may be estimated by using the appropriate reading from the main steam line monitor and the flowrate determined from a heat balance on the auxiliary feedwater turbine.

To prevent overpressurization of the auxiliary building from a steam generator blowdown line break outside containment, a vent pathway from the auxiliary building to the atmosphere is provided with a blowout panel that is designed to open at 1 psid. Steam generator blowdown is automatically terminated if the temperature environment in the affected area or the flowrate through the line exceeds a predetermined setpoint, thereby limiting the actual release. An estimate of the activity released to the environment can be determined from the recorded readings from the steam generator liquid monitor and the appropriate steam generator blowdown line flow instrumentation. The flow instrumentation for each steam generator connection has a flowrate range of 0 to 250 gal/min.

11.5.5.2 Standard Review Plan Evaluation

Standard Review Plan (SRP) 11.5 states that the gaseous and liquid process streams or effluent release points should be monitored and sampled according to tables 1 and 2.

Although monitoring and sampling are not provided for all of the process systems specifically listed in tables 1 and 2 of SRP 11.5, all effluent streams are monitored and sampled prior to discharge from the plant. A footnote to tables 1 and 2 of SRP 11.5 states that provisions indicated within parentheses are required only for systems not monitored, sampled, or analyzed (as indicated) prior to release by downstream provisions. The process and effluent radiological monitoring instrumentation and sampling systems provided are adequate for compliance with applicable regulations and therefore meet the intent of the SRP. Also, the provisions made for post-accident monitoring and sampling meet the intent of SRP 11.5 requirements.

TABLE 11.5.2-1 (SHEET 1 OF 4)

DESIGN PARAMETERS FOR PROCESS AND EFFLUENT RADIATION MONITORS

Monitor	Monitor Type	Safety Classification	Mechanical Code	Seismic Qualification	Number of Channels per Unit	Pump	Process Flow (ft^3/min)	Automatic Control Function	Duct Size/Pipe Diameter Material, Schedule	
Gas										
Plant Vent Effluent (Low Range)										
RE-12442A (particulate)	Effluent, offline	NNS[a]	B 31.1	Sample probe and flowmeter only	1	X	150,000 (Unit 1) 90,000 (Unit 2)	None	120 in. x 60 in. duct (Unit 1) 60 in. x 78in. (Unit 2) ASTM[b] 526, 12 gauge	Non-Class 1E diesel backed
RE-12442B (iodine)	Effluent, offline	NNS	B 31.1		1	X				
RE-12442C (radiogas)	Effluent, offline	NNS	B 31.1		1	X				
Containment Vent Effluent										
RE-2565A (particulate)	Effluent, offline	NNS	B 31.1	No	1	X	5000/15,000	Yes; closes containment purge supply and exhaust ducts	32-in.-dia. ASTM A527, 14 gauge	Non-Class 1E diesel backed
RE-2565B (iodine)	Effluent, offline	NNS	B 31.1	No	1	X				
RE-2565C (radiogas)	Effluent, offline	NNS	B 31.1	No	1	X				
Containment Atmosphere										
RE-2562A (particulate)	Airborne, offline	NNS	B 31.1	Yes	1	X	NA	None	1-in.-dia. ASTM SA-312, -GR, TP 304L, 40S	Non-Class 1E diesel backed
RE-2562B (iodine)	Airborne, offline passive iodine cartridge	NNS	B 31.1	Yes	1	X	NA	None		
RE-2562C (radiogas)	Airborne, offline	NNS	B 31.1	Yes	1	X				

TABLE 11.5.2-1 (SHEET 2 OF 4)

Monitor	Monitor Type	Safety Classification	Mechanical Code	Seismic Qualification	Number of Channels per Unit	Pump	Process Flow (ft^3/min)	Automatic Control Function	Duct Size/Pipe Diameter Material, Schedule	Power Supply
Control Room Intake RE-12116 RE-12117	Airborne, inline	SC-3/1E	NA	Yes	2		3000	Yes; switches control room ventilation to safety grade filtration train and initiates TSC filtration train	60 in. x 48 in., duct ASTM A527, 14 gauge	Class 1E
Waste Gas Processing System RE-0013	Process, inline	NNS	ASME[(c)] III-3	Yes	1		40-50 nitrogen	None	2-in. dia. ASME SA-106, GR. B, 80	Non-Class1E
Waste Gas Processing System Effluent ARE-0014	Process, inline	NNS	ASME III-3	Yes	1 Common		2 nitrogen	Yes; isolates WGPS discharge	2-in. dia. SA-106 GR. B 80	Non-Class 1E
Waste Gas Decay Tank RE-0039A and B (C for Unit 2 only)	Process, inline	NNS	NA	No	2 (Unit 1) 3 (Unit 2)		500/12000 (Unit 1) 500/1100/800 (Unit 2)	None	16 in. x 16 in. ASTM A527, 16 gauge	Non-Class 1E
ARE-2532A and B ARE-2533A and B	Effluent, inline	SC-3/1E	NA	Yes	4 Common		12,700	Yes; switches building to accident mode of operation including safety-related filtration units	36 in. x 36 in. ASTM A527, 18 gauge	Class 1E
Primary-to-Secondary N16 Leak Detection RE-724	Process, offline	NNS	NA	No	1		Variable	None		Non-Class 1E

TABLE 11.5.2-1 (SHEET 3 OF 4)

Monitor	Monitor Type	Safety Classification	Mechanical Code	Seismic Qualification	Number of Channels per Unit	Pump	Process Flow (ft^3/min)	Automatic Control Function	Duct Size/Pipe Diameter Material, Schedule	Power Supply
Primary-to-Secondary Noble Gas Leak Detection RE-810	Process, offline	NNS	B 31.1	No	1		Variable	None	18-in. dia. ASTM A-106-GRB std wt	Non-Class 1E
Radwaste Processing Facility Vent ARE-16980 (particulate)	Effluent, offline	NNS	B 31.1	No	1 Common	X	16000	None	42 in., ASTM A-653, 14 gauge	Non-Class 1E
Liquid										
Component Cooling Water RE-0017A and B	Process, offline	NNS	ASME III-3	Yes	2		9000	None	20-in. dia., ASME SA-106 GR. B std wt	Non-Class 1E
Waste Liquid Effluent RE-0018	Effluent, offline	NNS	B 31.1	No	1		100		Initiates automatic valve closure of liquid waste discharge line	2-in. dia.,Non-Class 1E ASTM A-312 GR. TP 394L, 40S
Steam Generator Liquid RE-0019	Process, offline	NNS	B 31.1	No	1		360	None	4-in. dia ASTM A-106 GR. B, 40	Non-Class 1E
Nuclear Service Water RE-0020A and B	Process, offline	NNS	ASME III-3	Yes	2		14,000-16,000(A) 16,000 18,000(B)	None	24-in. dia., ASME SA-312 GR. TP 304L, 10	Non-Class 1E

TABLE 11.5.2-1 (SHEET 4 OF 4)

Monitor	Monitor Type	Safety Classification	Mechanical Code	Seismic Qualification	Number of Channels per Unit	Pump	Process Flow (ft^3/min)	Automatic Control Function	Duct Size/Pipe Diameter Material, Schedule	Power Supply
Steam Generator Blowdown										
RE-0021	Effluent, offline	NNS	B 31.1	No	1		360	Closes steam generator blowdown processing system isolation valves and discharge lines	4-in. dia., ASTM A-106 GR. B, 40	Non-Class 1E
Turbine Building Drain										
RE-0848	Effluent, offline	NNS	B 31.1	No	1		50-640	Stops flow from turbine and control bldg drains system to the waste water retention basin	6-in. dia., ASTM A-106 GR. B, 40	Non-Class 1E diesel backed
Auxiliary Component Cooling Water										
RE-1950	Process, offline	NNS	ASME III-3	Yes	1		5400 6600	None	16-in. dia., ASME SA-106 GR. B, 80	Non-Class1E
Chemical and Volume Control System Letdown										
RE-48000	Process, offline	NNS	ASME III-3	Yes	1		75-120	None	3-in. dia., ASME SA-376 TP 304 or 316SS Schedule 160	Non-Class1E

a. NSS - Nonnuclear safety.
b. ASTM - American Society of Testing Materials.
c. ASME - American Society of Mechanical Engineers.
d. Structural integrity only.
e. WGPS - Waste gas processing system.

TABLE 11.5.2-2 (SHEET 1 OF 3)

DETECTOR REQUIREMENTS FOR PROCESS AND EFFLUENT RADIATION MONITORS

Monitor	Location	Detector Type	Radiation Zone (mR/h)	Major Isotopes	Detectable Range ($\mu Ci/cm^3$)
Gas					
Plant Vent Effluent (Low Range)					
RE-12442A (particulate)	Plant vent	Beta scintillation	0.25-2.5	I-131, I-133, Cs-134, Cs-137, Co-58, Co-60, Sr-90	3.5E-12 to 2.3E-7
RE-12442B (iodine)	Plant vent	Gamma scintillation	0.25-2.5	I-131, I-133, I-135	1.1E-10 to 2.3E-5
RE-12442C (radiogas)	Plant vent	Beta scintillation	0.25-2.5	Xe-133, Xe-135, Kr-85, Ar-41	5.0E-7 to 5.0E-2
Containment Vent Effluent					
RE-2565A (particulate)	Containment vent	Beta scintillation	0.25-2.5	I-131, I-133, Cs-134, Cs-137, Co-58, Co-60	3.5E-12 to 2.3E-7
RE-2565B (iodine)	Containment vent	Gamma scintillation	0.25-2.5	I-131, I-133, I-135	1.1E-10 to 2.3E-5
RE-2565C (noble gas)	Containment vent	Beta scintillation	0.25-2.5	Xe-133, Xe-135, Kr-85, Ar-41	1.0E-6 to 1.0E-1
Containment Atmosphere					
RE-2562A (particulate)	Containment bldg	Beta scintillation	0.25-2.5	I-131, Cs-137, Co-58, Co-60	3.5E-12 to 2.3E-7
RE-2562B (iodine)	Containment bldg	Passive iodine cartridge	0.25-2.5	I-131, I-133	NA
RE-2562C (radiogas)	Containment bldg	Beta scintillation	0.25-2.5	Xe-133, Xe-135, Kr-85, Ar-41	5.0E-7 to 5.0E-2

TABLE 11.5.2-2 (SHEET 2 OF 3)

Monitor	Location	Detector Type	Radiation Zone (mR/h)	Major Isotopes	Detectable Range ($\mu Ci/cm^3$)
Control Room Air Intake RE-12116, RE-12117	Control room	Thin walled G-M tubes	<0.25	Xe-133, Xe-135, Kr-85, I-131, I-133, Co-58, Co-60	1.0E-6 to 1.0E-1
Waste Gas Processing System RE-0013	Aux bldg	G-M tubes	2.5-15	Xe-133, Xe-135, Kr-85	9.9E-1 to 2.5E+3
Waste Gas Processing System Effluent ARE-0014	Aux bldg	G-M tube	0.25-2.5	Xe-133, Xe-135, Kr-85	1.0E-1 to 1.0E+4
Waste Gas Decay Tank RE-0039A, B, and C	Aux bldg	Thin walled G-M tube	0.25-2.5 2.5-15 2.5-15	Xe-133, Xe-135, Kr-85	1.0E-6 to 1.0E-1
Fuel Handling Building Effluent ARE-2532A and B ARE-2533A and B	Fuel handling bldg	Thin walled G-M tube	0.25-2.5	Xe-133, Xe-135, Kr-85	1.0E-6 to 1.0E-1
Primary-to-Secondary N16 Leak Detection RE-724	Turbine Building	Gamma Scintillation	<0.25	N-16	2.0E-8 to $2x10^{-3}$
Primary-to-Secondary Noble Gas Leak Detection RE-810	Turbine Building	Beta Scintillation	<0.25	Xe-133, Xe-135, Kr-85 Kr-88	3.74E-8 to 9.09E-2
Radwaste Processing Facility Vent ARE-16980 (particulate)	Radwaste Processing Facility	Beta Scintillation	<2.5	Co-58, Co-60, Cs-134, Cs-137	1.0E-13 to 1.0E-7

TABLE 11.5.2-2 (SHEET 3 OF 3)

Monitor	Location	Detector Type	Radiation Zone (mR/h)	Major Isotopes	Detectable Range ($\mu Ci/cm^3$)
Liquid					
RE-0017A and B (component water)	Aux bldg	Gamma scintillation	0.25	Co-58, Co-60, I-131, I-133, Cs-134, Cs-137	4.0E-7 to 4.0E-2
RE-0018 (waste liquid effluent)	Aux bldg	Gamma scintillation	2.5-15	Co-58, Co-60, I-131, I-133, Cs-134, Cs-137	1.0E-6 to 1.0E-1
RE-0019 (steam generator liquid)	Aux bldg	Gamma scintillation	2.5-15	Co-58, Co-60, I-131, I-133, Cs-134, Cs-137	4.0E-7 to 4.0E-2
RE-0020A and B (nuclear service water)	Aux bldg	Gamma scintillation	0.25-2.5	Co-58, Co-60, I-131, I-133, Cs-134, Cs-137	4.0E-7 to 4.0E-2
RE-0021 (steam generator blowdown)	Aux bldg	Gamma scintillation	0.25-2.5	Co-58, Co-60, I-131, I-133, Cs-134, Cs-137	4.0E-7 to 4.0E-2
RE-0848 (turbine bldg drain)	Turbine bldg	Gamma scintillation	0.25-2.5	Co-58, Co-60, I-131, I-133, Cs-134, Cs-137	4.0E-7 to 4.0E-2
RE-1950 (aux component cooling water)	Aux bldg	Gamma scintillation	0.25-2.5	Co-58, Co-60, I-131, I-133, Cs-134, Cs-137	4.0E-7 to 4.0E-2
RE-48000 (CVCS letdown)	Aux bldg	Gamma scintillation	100-250	Co-58, Co-60, I-131, I-133, Cs-134, Cs-137	2.0E-4 to 4.0E+1

TABLE 11.5.2-3 (SHEET 1 OF 2)

CONDITIONS OF SERVICE FOR PROCESS AND EFFLUENT RADIATION MONITORS

	PROCESS CONDITIONS			MONITOR LOCATION CONDITIONS			
Monitor	Sample Fluid	Operating Temperature (°F)	Operating Pressure (psig)	Temp (°F)	Pressure (psig)	Relative Humidity (%)	Building Location/ Elevation
Gas							
Plant vent effluent RE-12442A	Gas	40-104	0	Unit 1 32-120 Unit 2 40-120	0	35-90	Equip bldg 220 ft
Plant vent effluent RE-12442B	Gas	40-104	0	Unit 1 32-120 Unit 2 40-120	0	35-90	
Plant vent effluent RE-12442C	Gas	40-104	0	Unit 1 32-120 Unit 2 40-120	0	35-90	
Containment vent effluent RE-2565A	Gas	60-120	0	Unit 1 32-120 Unit 2 40-125	0	35-90	Equip bldg 220 ft
Containment vent effluent RE-2565B	Gas	60-120	0	Unit 1 32-120 Unit 2 40-125	0	35-90	
Containment vent effluent RE-2565C	Gas	60-120	0	Unit 1 32-120 Unit 2 40-125	0	35-90	
Containment atmosphere RE-2562A	Gas	60-120	(-1.5)-3	Unit 1 40-104 Unit 2 40-120	0	20-95	Aux bldg 170 ft 6 in.
Containment atmosphere RE-2562B	Gas	60-120	(-1.5)-3	Unit 1 40-104 Unit 2 40-120	0	20-95	
Containment atmosphere RE-2562C	Gas	60-120	(-1.5)-3	Unit 1 40-104 Unit 2 40-120	0	20-95	
Control room intake RE-12116, RE-12117	Gas	5-104	0	65-100	0	20-60	Control bldg 260 ft
Waste gas processing RE-0013	Gas	70-140	0.5-2	40-104	0	20-95	Aux bldg 188 ft 3 in.
Waste gas processing system RE-0014	Gas	70-140	0	40-104	0	20-95	Aux bldg 178 ft
Waste gas decay tank RE-0039A, RE-0039B, RE-0039C	Gas	40-104	0	40-104	0	20-95	Aux bldg 189 ft Aux bldg 190 ft 3 in.

TABLE 11.5.2-3 (SHEET 2 OF 2)

	PROCESS CONDITIONS			MONITOR LOCATION CONDITIONS			
Monitor	Sample Fluid	Operating Temperature (°F)	Operating Pressure (psig)	Temp (°F)	Pressure (psig)	Relative Humidity (%)	Building Location/ Elevation
Fuel handling building effluent ARE-2532A and B	Gas	40-104	0	40-104	(-0.25)-0 (in. WG)	20-95	Fuel handling bldg 253 ft
Fuel handling building effluent ARE-2533A and B	Gas	40-104	0	40-104	-0.25)-0 (in. WG)	20-95	Fuel handling bldg 283 ft
Radwaste processing facility vent effluent ARE-16980	Gas	50-100	0	40-104	0	20-95	Radwaste processing facility
Liquid							
Component cooling water RE-0017A and B	Liquid	108	24	40-104	0	20-95	Aux bldg 240 ft
Waste liquid effluent RE-0018	Liquid	100	125	40-104	0	20-95	Aux bldg 119 ft 3 in.
Steam generator liquid RE-0019	Liquid	130	300	40-104	0	20-95	Aux bldg 170 ft 6 in.
Nuclear service water RE-0020A and B	Liquid	108	90	40-104	0	20-95	Aux bldg 195 ft
Steam generator blowdown RE-0021	Liquid	130	300	40-104	0	20-95	Aux bldg 195 ft
Turbine bldg drain RE-0848	Liquid	104	20	40-104	0	40-85	Turbine bldg 195 ft
Aux component cooling water RE-1950	Liquid	105	26	40-104	0	20-95	Aux bldg 220 ft
Chemical and volume control system letdown RE-48000	Liquid	115	350	40-104	0	20-95	Aux bldg 195 ft

TABLE 11.5.2-4

DELETED

TABLE 11.5.2-5 (SHEET 1 OF 2)

FLOWRATE MONITORING FOR LIQUID AND GASEOUS NORMAL RELEASE PATHWAYS

Pathway	Equipment Number	Range	Control Room Indication	Local Indication
Liquid				
Radwaste effluent (batch)	FE-018	0 - 225 gal/min	Yes	Yes
Steam generator blowdown effluent (batch)	FE-021	125-1250 gal/min	Yes	Yes
Turbine building drain effluent	(a)	(a)	(a)	(a)
Control building sump effluent	(a)	(a)	(a)	(a)
Gaseous				
Plant vent (includes discharges from the containment purge system, gaseous radwaste system, fuel handling building HVAC, and auxiliary building HVAC)	1FE-12442A & 12442B	3,000-190,000 sf^3/min	Yes	Yes
	2FE-12442A & 12442B	1200-120,000 sf^3/min	Yes	Yes
	1FE-12835	3200-160,000 sf^3/min	Yes	No
	2FE-12835	2,000-100,000 sf^3/min	Yes	No
Condenser air ejector and steam packing exhaust	FE-12839	0-6500 sf^3/min	Yes	Yes
Containment purge system[(a)]	FE-2565	0-20,000 sf^3/min	Yes	Yes

TABLE 11.5.2-5 (SHEET 2 OF 2)

Pathway	Equipment Number	Range	Control Room Indication	Local Indication
Waste gas processing effluent[b]	AFE-0014	0.2-50.0 sf^3/min	Yes	Yes
Radwaste processing facility vent effluent	AFE-16980	480-18,280 sf^3/min	Yes	Yes

a. These pathways are not equipped with a release flowrate measuring device due to the fact that radioactivity exceeding preset limits is not sent to the environs but instead to the respective waste processing systems.

b. These pathways are also monitored downstream by the plant vent monitor.

TABLE 11.5.3-1 (SHEET 1 OF 2)

AIRBORNE RADIOLOGICAL SAMPLING CAPABILITY FOR PERMS

	Sample Identification			
Monitor	Fluid	Location	Type	Purpose
RE-12442A (gaseous release pathway)[(a)]	Ventilation air effluent	At monitor	Particulate	Provide quantitative activity release data. Calibration of continuous monitors.
RE-12442B (gaseous release pathway)[(a)]	Ventilation air effluent	At monitor	Iodine	
RE-12442C (gaseous release pathway)[(a)]	Ventilation air effluent	At monitor	Radiogas	
			Vapor sample	
RE-2565A	Containment ventilation air	At monitor	Particulate	Provide diagnostic backup data to RE-12442A, B, and C and RE-2562A, B, and C.
RE-2565B	Containment ventilation air	At monitor	Iodine	Determine effectiveness of containment vent filters.
RE-2565C	Containment ventilation air	At monitor	Radiogas	
			Vapor sample	
RE-2562A	Containment atmosphere	At monitor	Particulate	Determine personnel access time limit inside containment. Calibration of monitor.
RE-2562B	Containment atmosphere	At monitor	Passive iodine cartridge	Determine personnel access time limit inside containment.
RE-2562C	Containment atmosphere	At monitor	Radiogas	
			Vapor sample	
RE-12116 and RE-12117	Control room air intakes	Control room intake duct	Radiogas iodine	Determine isotopic activity in control room following accident.
RE-0013	Waste gas	At monitor	Radiogas	Determine activity inventory waste gas system.
RE-12839A (gaseous release pathway)[(a)]	Condenser air ejector effluent	At monitor	Passive particulate cartridge	Provide quantitative activity release data.

TABLE 11.5.3-1 (SHEET 2 OF 2)

AIRBORNE RADIOLOGICAL SAMPLING CAPABILITY FOR PERMS

	Sample Identification			
Monitor	Fluid	Location	Type	Purpose
RE-12839B (gaseous release pathway)[(a)]	Condenser air ejector effluent	At monitor	Passive iodine cartridge	
RE-12839C (gaseous release pathway)[(a)]	Condenser air ejector effluent	At monitor release data.	Radiogas Calibration	Provide quantitative activity of continuous monitors.
ARE-0014	Waste gas system effluent (batch)	Use system sample at decay tank and at monitor	Radiogas iodine	Determine identify and quantity in effluent.
RE-0039A, B, and C	Decay tank and compressor ventilation air	At monitor	Radiogas iodine	Provide quantitative isotopic activity data.
ARE-2532A and B ARE-2533A and B	Fuel handling building ventilation air	At monitor	Radiogas iodine vapor sample	Provide quantitative isotopic activity data.
RE-12444A RE-12444B RE-12444C (gaseous release pathway)	Ventilation air effluent	At monitor	Passive particulate cartridge Passive iodine cartridge Radiogas Vapor sample	Provide quantitative activity release data. Calibration of continuous monitors.
ARE-16980 (gaseous release pathway)[(a)]	Radwaste processing facility vent effluent	At monitor	Particulate	Provide quantitative activity release data. Calibration of continuous monitors.

a. These monitors require sampling in accordance with Regulatory Guide 1.21.

TABLE 11.5.3-2

LIQUID RADIOLOGICAL SAMPLING CAPABILITY FOR PERMS

Monitor	Sample Identification			Purpose
	Fluid	Location	Type	
RE-0017A and B	Component cooling water liquid	At monitor	Liquid sample	Detect activity leakage into component cooling system. Calibration of continuous monitor.
RE-0018[a][b]	Waste liquid effluent (batch release)	Use system sample at waste monitor tank and at monitor.	Liquid sample	Provide quantitative activity release data. Calibration of continuous monitors.
RE-0019	Steam generator liquid	At monitor	Liquid sample	Detect primary-secondary system leaks; continuous monitor. Locate leaking steam generator. Determine magnitude of primary to secondary leakage.
RE-0020A and B[a]	Nuclear service water liquid effluent (continuous release)	At monitor	Liquid sample	Provide quantitative activity release data. Calibration of continuous monitors.
RE-0021[a]	Steam generator blowdown system liquid process and/or effluent	At monitor	Liquid sample	Provide quantitative activity release data. Calibration of continuous monitors.
RE-0848[a][b]	Turbine bldg liquid effluent (continuous)	Turbine bldg drain tanks and sumps, and at monitor	Liquid sample	Provide quantitative activity release data. Calibration of continuous monitors.
RE-1950	Aux component cooling water liquid	At monitor	Liquid sample	Detect activity leakage into aux component cooling water. Calibration of continuous monitor.
RE-48000	Letdown liquid	At monitor	Liquid sample	To detect abnormal activity levels in the reactor coolant system.

a. These monitors require sampling in accordance with Regulatory Guide 1.21.

b. Sampling also required to provide process concentration following activation of control function (See also subsection 7.3.6.)

TABLE 11.5.5-1 (SHEET 1 OF 2)

DESIGN PARAMETERS FOR POST-ACCIDENT RADIATION MONITORS

Monitor	Monitor Type	Safety Classification	Mechanical Code	Seismic Qualification	Number of Monitors per Unit	Moving Filter	Charcoal Cartridge Particulate Filter	Pump	Process Flow (ft^3/ min)	Automatic Control Function	Duct Size/Pipe Diameter Material, Schedule	Flow Instrumentation	
												Instrument No.	Range (sf/min)
Plant Vent Effluent													
RE-12444A (particulate)	Effluent passive particulate filter	Cat. 2	B 31.1	Yes[e]	1		X	X	150,000 (9000)[b]	None	120 in. x 60 in. duct (Unit 1) 60 in. x 78 in. duct (Unit 2) ASTM 526, 12 gauge	FE-12835	3200-160000
RE-12444B (iodine)	Effluent passive iodine cartridge	Cat. 2	B 31.1	Yes[e]	1		X	X	150,000 (9000)[b]	None			
RE-12444C,D,E (radiogas)	Effluent Beta Scintillation G-M, G-M	Cat. 2	B 31.1	Yes[e]	1			X	150,000 (9000)[b]	None			
									NA	None	Pipe 28-0-5 in. dia. 2.5 in. WT 3.0 in. INS	FT-542/543 FT-512/513 FT-522/523 FT-532/533	0-5 million lb/h
Main Steam Line													
RE-13119	Strap on	SC-3/IE	NA	Yes[a]	1								
RE-13120					1								
RE-13121					1								
RE-13122					1								
Condenser Air Ejector and Steam Packing Exhauster													
RE-12839A	Effluent passive particulate filter	NNS[d]	B 31.1	No	1		X	X	2100	None	18-in. dia. ASTM A-106-GRB std wt	FE-12839	0-6500
RE-12839B	Effluent passive iodine cartridge	NNS	B 31.1	No	1		X	X	2100	None	See ARE-0016 specs		

TABLE 11.5.5-1 (SHEET 2 OF 2)

Monitor	Monitor Type	Safety Classification	Mechanical Code	Seismic Qualification	Number of Monitors per Unit	Moving Filter	Charcoal Cartridge Particulate Filter	Pump	Process Flow (ft^3/ min)	Automatic Control Function	Duct Size/Pipe Diameter Material, Schedule	Flow Instrumentation	
												Instrument No.	Range (sf/min)
RE-12839C,D,E	Effluent Beta Scintillation G-M, G-M	NNS	B 31.1	No	1			X	2100	Yes; diverts air ejector and steam packing exhauster discharge through charcoal filtration	18-in. dia. ASTM A-106-GRB std wt		

a. These monitors are qualified to perform their function following design basis seismic event.

b. Accident condition flow for Unit 1. Flow for Unit 2 is 90,000 ft/min for normal operation and 3200 ft/min for accident conditions.

c. ASTM - American Society of Testing Materials.

d. NNS - Nonnuclear safety.

e. Seismic qualification is limited to the isokinetic nozzles, flow transmitter, monitoring skid, and associated data processing module.

TABLE 11.5.5-2

DETECTOR REQUIREMENTS FOR POST-ACCIDENT RADIATION MONITORS

Monitor	Location	Detector Type	Radiation Zone (MR/h)	Major Isotopes	Detectable Range ($\mu Ci/cm^3$)
Plant Vent Effluent					
RE-12444A (particulate)	Plant vent	Passive particulate	0.25-2.5 (10-100 r/h)[a]	I-131, I-133, Cs-134, Cs-137, Co-58, Co-60, Sr-90	NA
RE-12444B (iodine)		Passive iodine cartridge	0.25-2.5 (10-100 r/h)	I-131, I-133	NA
RE-12444C, (radiogas)		Beta scintillation	0.25-2.5 (10-100 r/h)	Xe-133, Xe-135, Kr-85	$1.0E^{-7}$ to $3.4E^{-3}$ for Xe-133
RE-12444D	Plant vent	G-M tube	0.25 - 2.5 (10 - 100 r/h)	Xe-133, Xe-135, Kr-85, Kr-88	$3.4E^{-3}$ to $4.0E^{+1}$
RE-12444E	Plant vent	G-M tube	0.25 - 2.5 (10 - 100 r/h)	Xe-133, Xe-135, Kr-85, Kr-88	$4.0E^{+1}$ to $5.8E^{+4}$
Main Steam Line RE-13119 RE-13120 RE-13121 RE-13122	MSIV areas	Strap on gamma	0.25-2.5 (10-100 r/h)	Xe-133, Xe-135, Kr-85m, Kr-88	$1.0E^{-1}$ to $1.0E^{+3}$
Condenser Air Ejector and Steam Packing Exhauster					
RE-12839A (particulate)	Turbine bldg.	Passive particulate filter	<0.25 (10-100 r/h)	Cs-134, Cs-137, C0-58, Co-60	NA
RE-12839B (iodine)	Turbine bldg.	Passive iodine cartridge	<0.25 (10-100 r/h)	I-131, I-133	NA
RE-12839C (radiogas)	Turbine bldg	Beta scintillation	<0.25 (10-100 r/h)	Xe-133, Xe-135, Kr-85 Kr-88	$1.0E^{-7}$ to $3.4E^{-3}$ for Xe-133
RE-12839D	Turbine bldg.	G-M tube	<0.25 (10 - 100 r/h)	Xe-133, Xe-135, Kr-85, Kr-88	$3.4E^{-3}$ to $4.0E^{+1}$
RE-12839E	Turbine bldg.	G-M tube	<0.25 (10 - 100 r/h)	Xe-133, Xe-135, Kr-85, Kr-88	$4.0E^{+1}$ to $5.8E^{+4}$

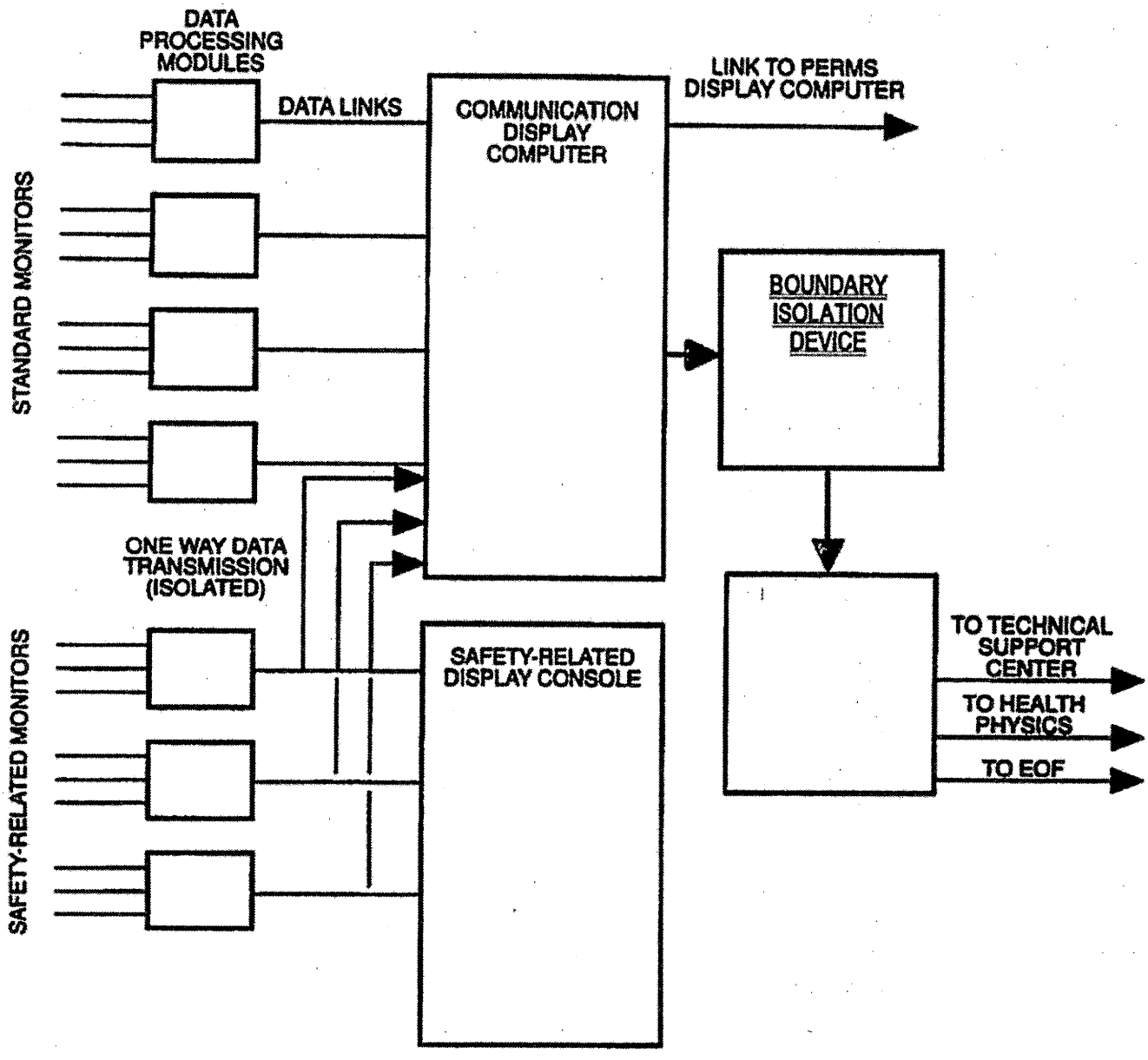
a. Values in parentheses for accident conditions.

TABLE 11.5.5-3

CONDITIONS OF SERVICE FOR POST-ACCIDENT RADIATION MONITORS

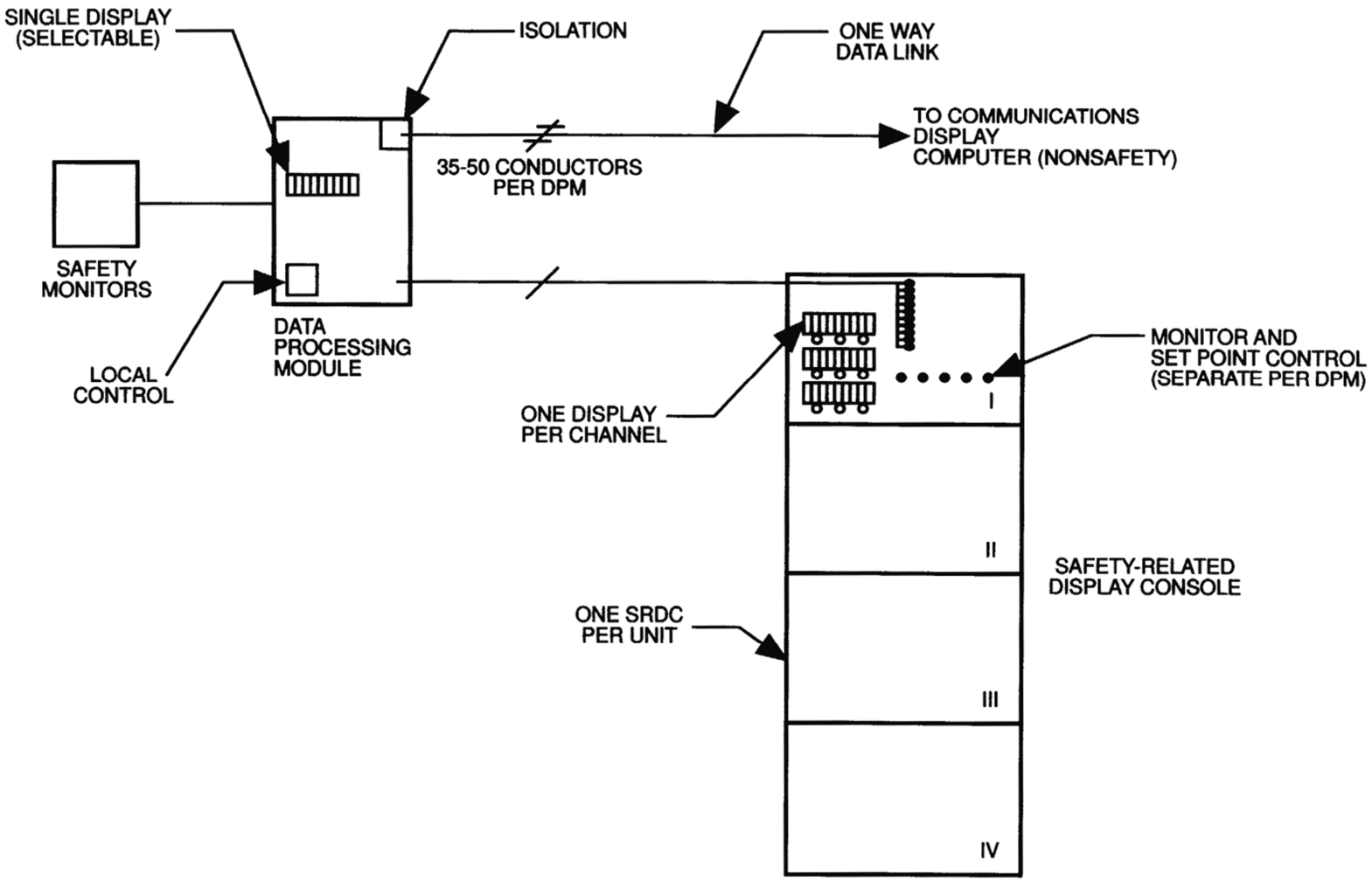
	PROCESS CONDITIONS			MONITOR LOCATION CONDITIONS			
Monitor	Sample Fluid	Operating Temperature (°F)	Operating Pressure (psig)	Temperature (°F)	Relative Pressure (psig)	Building Humidity (%)	Location/ Elevation
Plant Vent Effluent							
RE-12444A	Gas	40-104 (40-120)[a]	0 (0)[a]	32-120 (32-120)[a]	0 (0)[a]	35-90 (20-100)[a]	Equipment bldg 220 ft
RE-12444B	Gas	40-104 (40-120)	0 (0)	32-120 (32-120)	0 (0)	35-90 (20-100)	
RE-12444C,D,E	Gas	40-104 (40-120)	0 (0)	32-120 (32-120)	0 (0)	35-90 (20-100)	
Main Steam Line RE-13119 RE-13120 RE-13121 RE-13122	NA	NA	NA	47-115 (5-115)	0 (0)	10-100 (10-100)	Aux bldg (MSIV area) Control bldg (MSIV area)
Condenser Air Ejector and Steam Packing Exhauster							
RE-12839A	Gas	131-211[b]	0 (0)	40-104 (40-120)	0 (0)	40-85 (20-95)	Turbine bldg 270 ft
RE-12839B	Gas	131-211[b]	0 (0)	40-104 (40-120)	0 (0)	40-85 (20-95)	Turbine bldg 270 ft
RE-12839C,D,E	Gas	131-211[b]	0 (0)	40-104 (40-120)	0 (0)	40-85 (20-95)	Turbine bldg 270 ft

a. Values in parentheses for accident conditions.
b. Main process stream temperature upstream of heat exchanger on monitor skid.



REV 20 9/16

SOUTHERN COMPANY *Energy to Serve Your World®* VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2	RADIATION MONITORING SYSTEM BLOCK DIAGRAM
	FIGURE 11.5.2–1



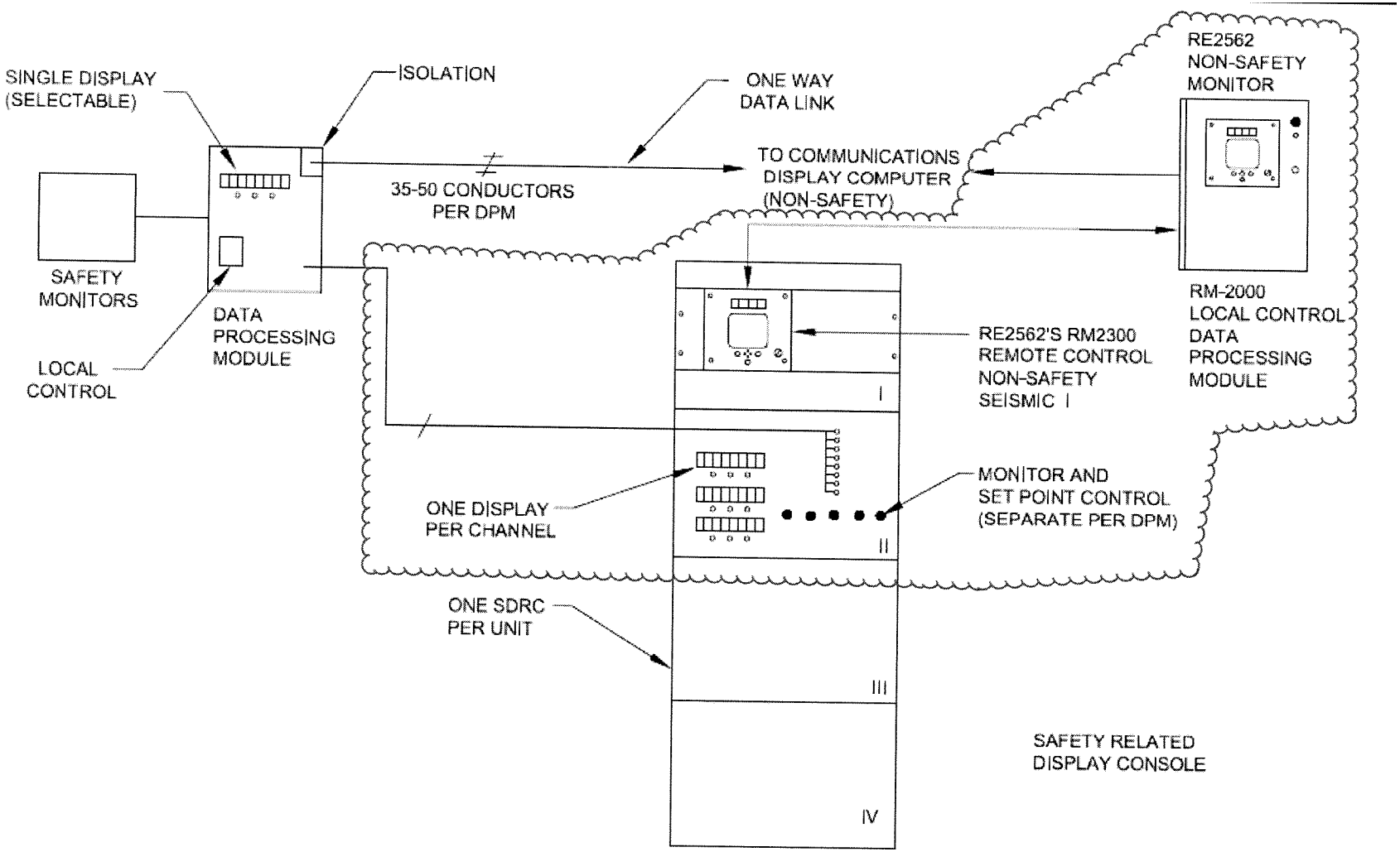
REV 19 4/15

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VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1

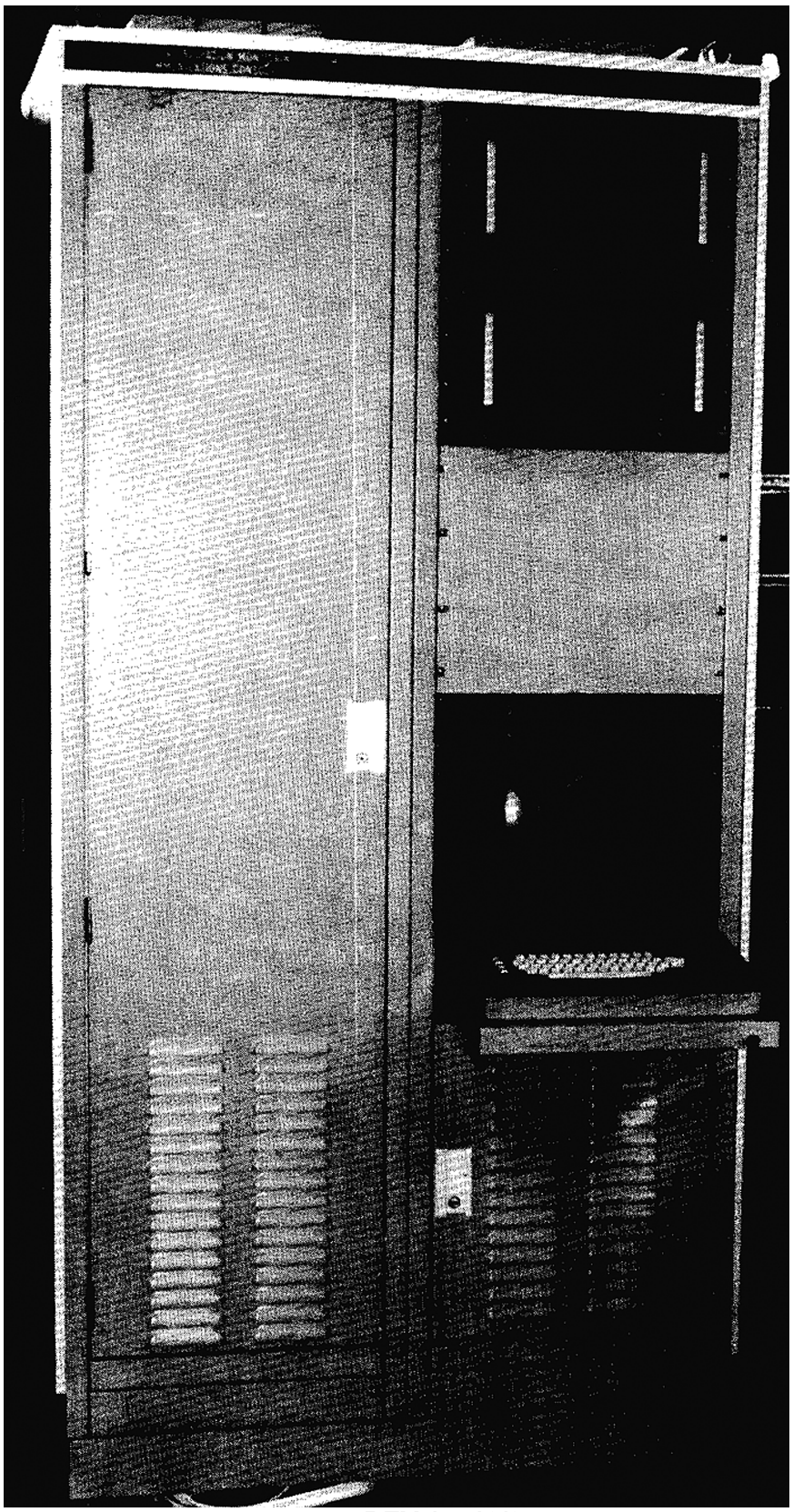
SAFETY SUBSYSTEM BLOCK DIAGRAM

FIGURE 11.5.2-2a



REV 19 4/15

SOUTHERN COMPANY *Energy to Serve Your World®*	**VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 2**	SAFETY SUBSYSTEM BLOCK DIAGRAM FIGURE 11.5.2-2b



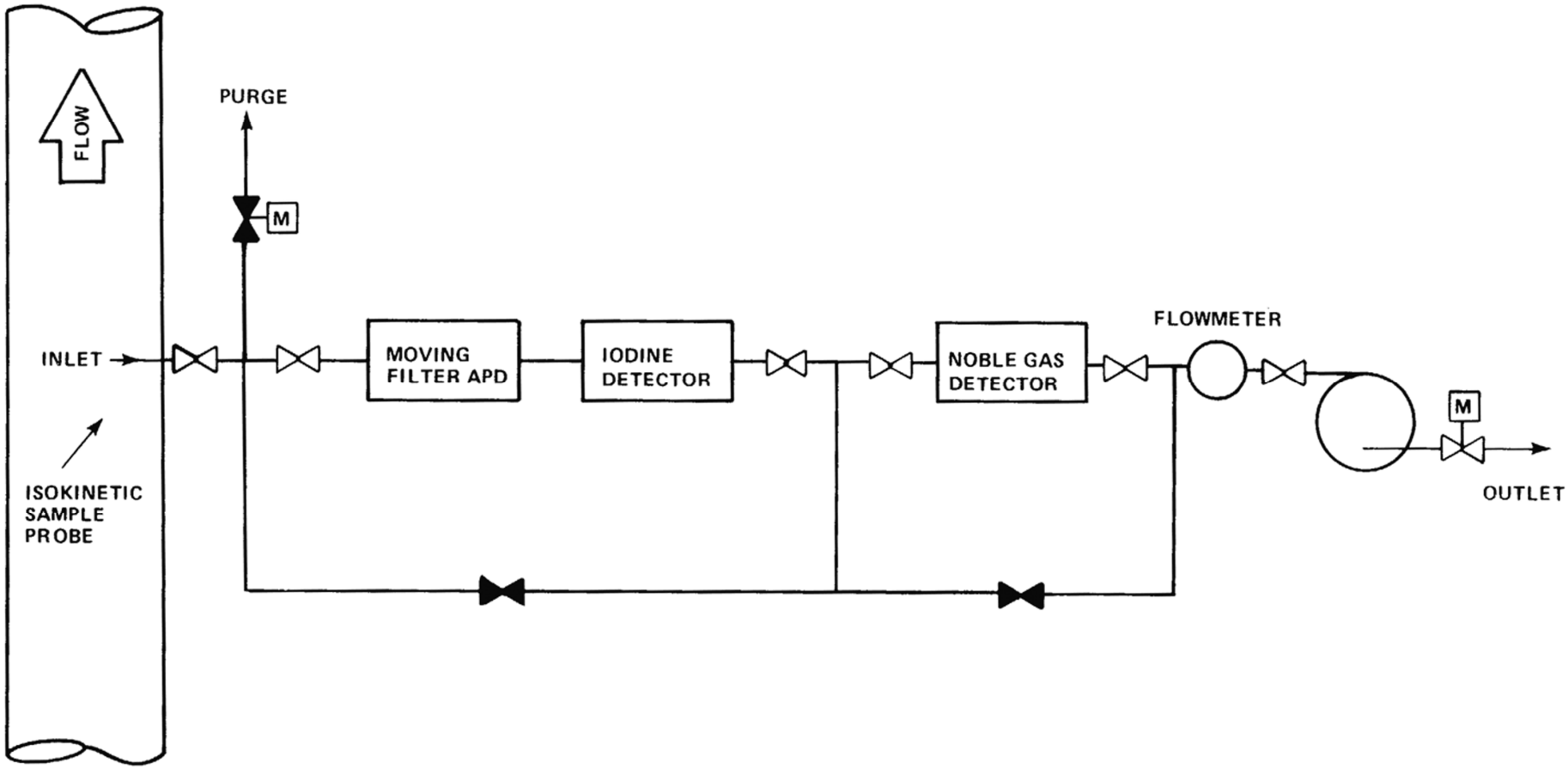
REV 14 10/07

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ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

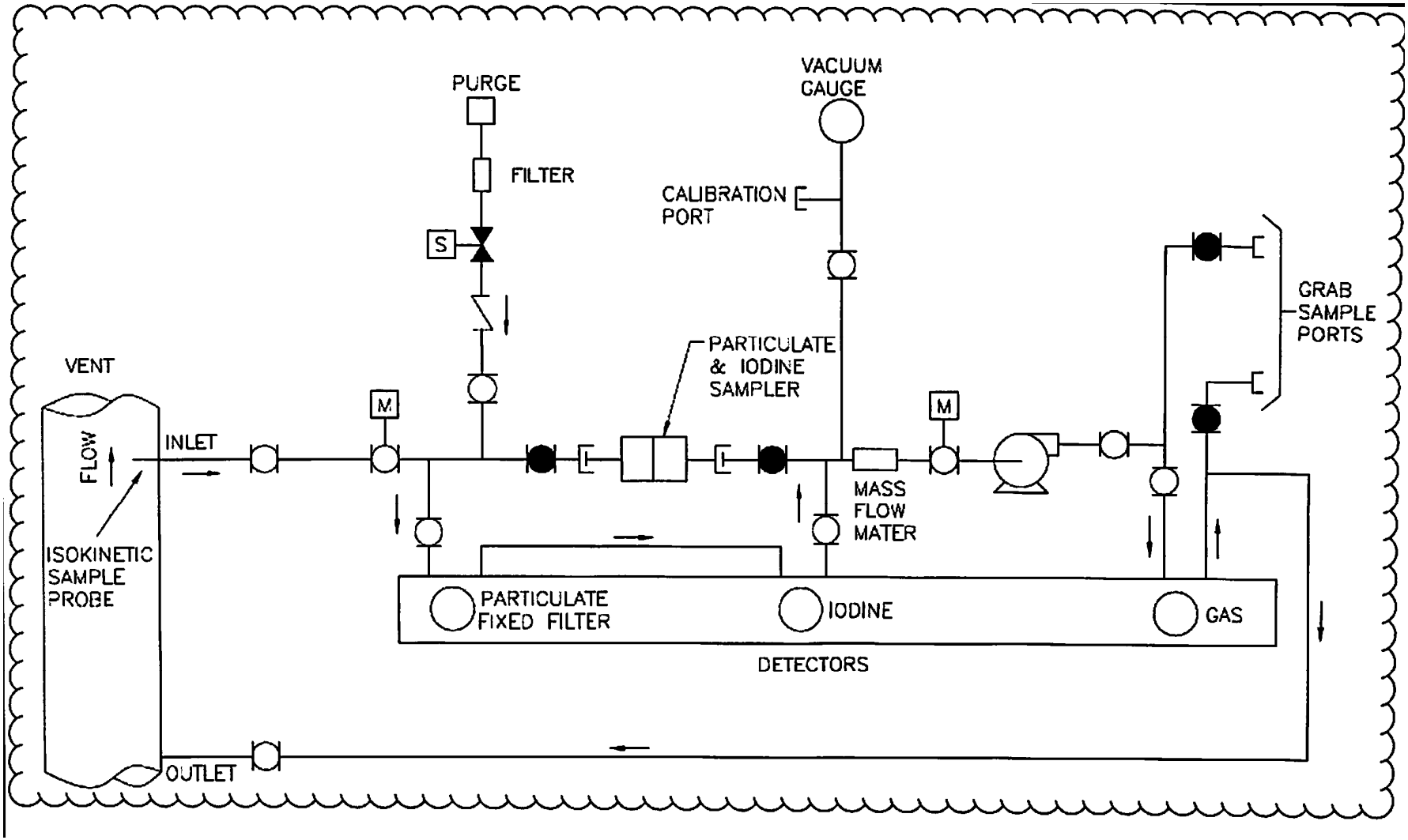
DRMS COMMUNICATION CONSOLE

FIGURE 11.5.2–3



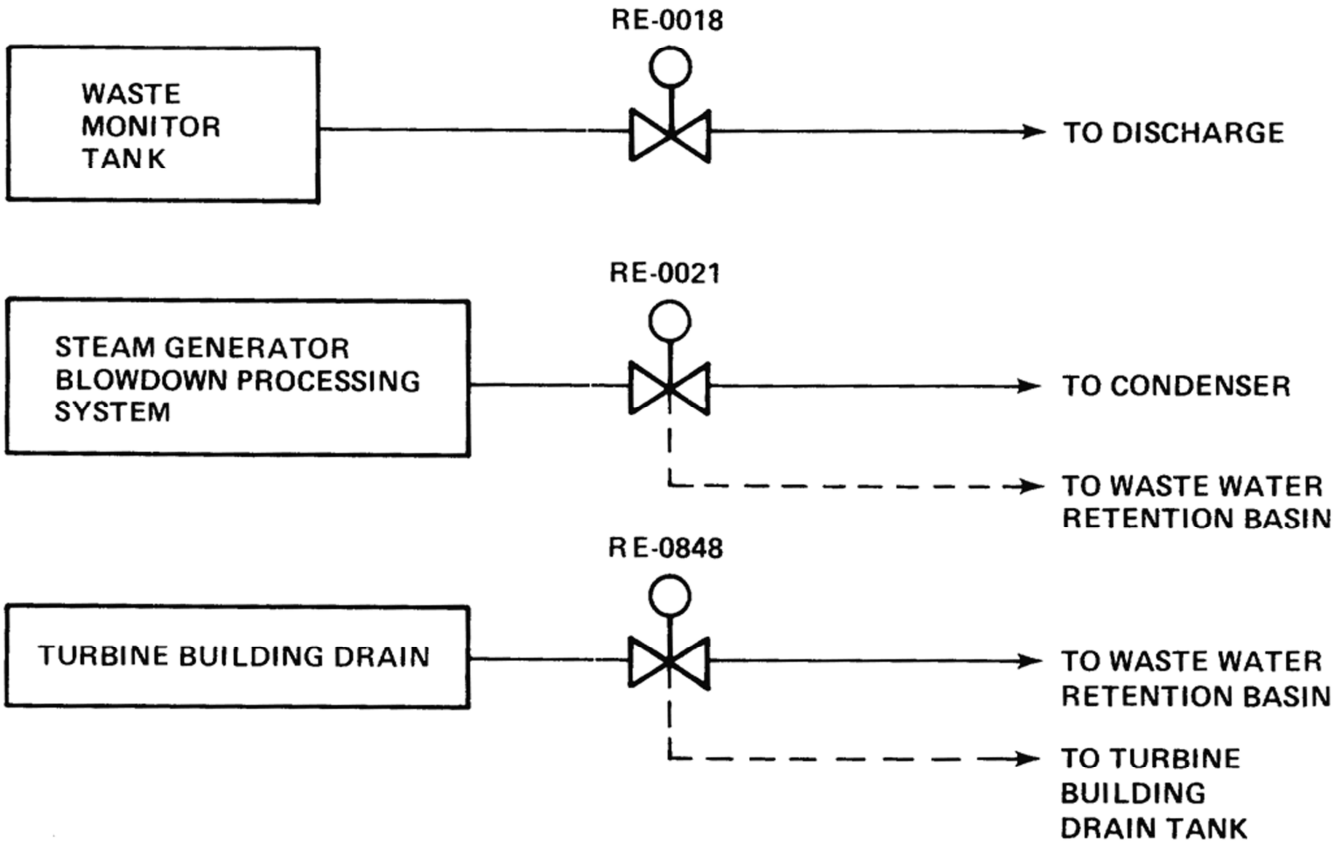
REV 19 4/15

SOUTHERN COMPANY *Energy to Serve Your World®*	**VOGTLE ELECTRIC GENERATING PLANT UNIT 1**	THREE-CHANNEL AIRBORNE MONITORING ASSEMBLY
		FIGURE 11.5.2-4a



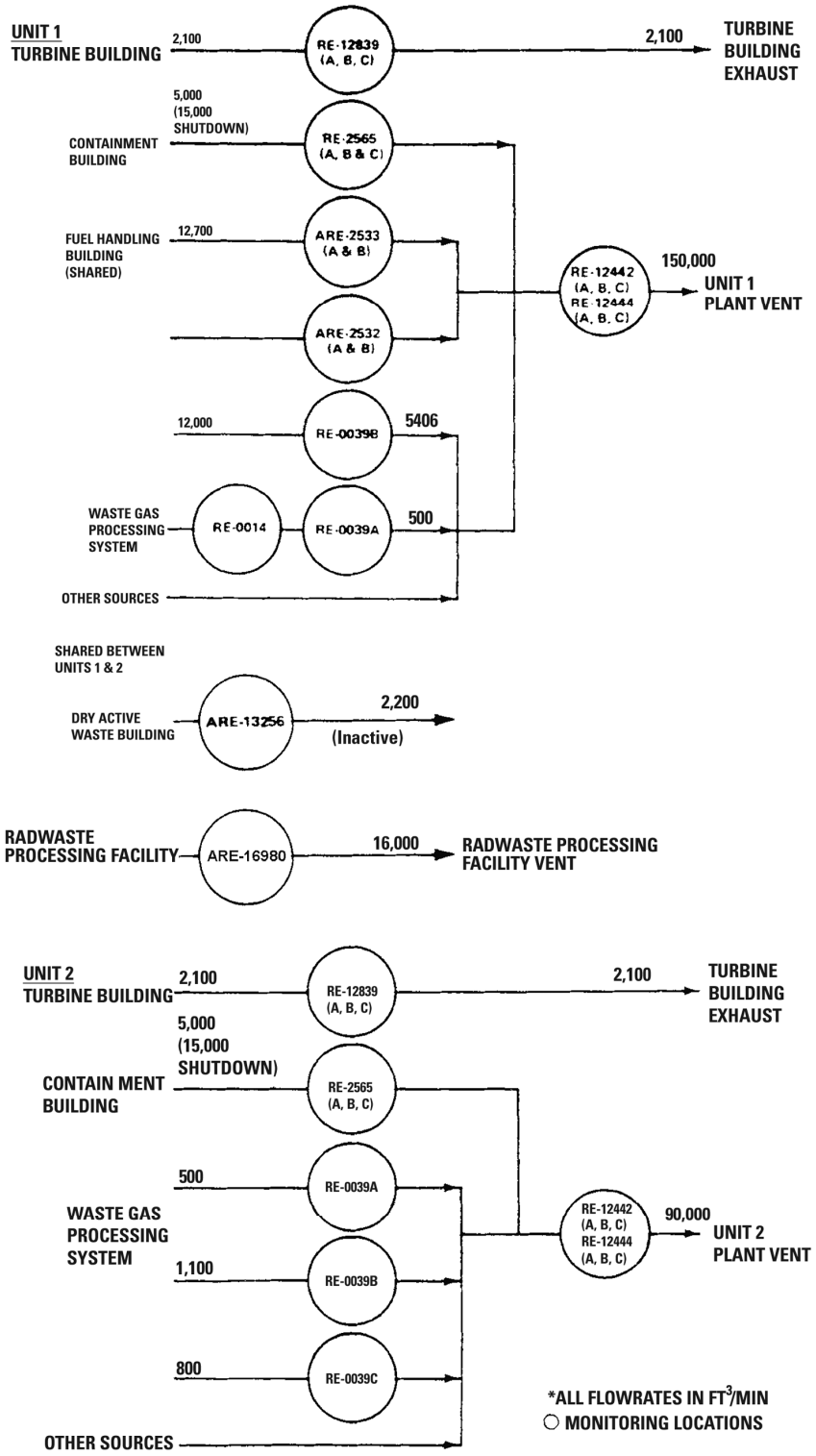
REV 19 4/15

SOUTHERN COMPANY *Energy to Serve Your World®*	**VOGTLE** **ELECTRIC GENERATING PLANT** **UNIT 2**	THREE-CHANNEL AIRBORNE MONITORING ASSEMBLY
		FIGURE 11.5.2-4b



REV 14 10/07

SOUTHERN COMPANY *Energy to Serve Your World®* VOGTLE ELECTRIC GENERATING PLANT UNIT 1 AND UNIT 2	MONITORING LOCATIONS FOR LIQUID RELEASE PATHWAYS
	FIGURE 11.5.2–5



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VOGTLE
ELECTRIC GENERATING PLANT
UNIT 1 AND UNIT 2

PERMS NORMAL GASEOUS EFFLUENT RELEASE PATHWAY

FIGURE 11.5.2–6